MECHANISMS OF PHARMACOLOGICAL ACTIVATION OF BK CHANNELS

A Dissertation
Submitted to
The Temple University Graduate Board

In Partial Fulfillment
Of the Requirements for the Degree
DOCTOR OF PHILOSOPHY

---

by
Michael Elliott Rockman
April, 2020

Thesis Advisor
Dr. Brad. S. Rothberg, Ph.D., Medical Genetics and Molecular Biochemistry

Examining Committee:
Dr. Kathleen M. Giangiacomo, Ph.D., Medical Genetics and Molecular Biochemistry
Dr. Steven R. Houser, Ph.D., Physiology
Dr. Jonathan Soboloff, Medical Genetics and Molecular Biochemistry

External Evaluator
Dr. Crina M. Nimigean, Ph.D., Weill Cornell Medical College, Department of Anesthesiology

©
Copyright
2020

by

Michael Elliott Rockman
All Rights Reserved

ABSTRACT

Large conductance Ca^{2+}-activated K^+ channels (BK channels) are activated by cytosolic calcium and depolarized membrane potential under physiological conditions. These channels control electrical excitability in neurons, smooth muscle, and other tissues by gating K^+ efflux and hyperpolarizing the membrane in response to Ca^{2+} signaling. Mutations to BK channels have been associated with human diseases such as epilepsy and movement disorders. Because activation of BK channels results in relaxation of smooth muscle and reduced neuronal excitability, small-molecule BK channel activating drugs have the potential to treat conditions such as hypertension, ischemia-reperfusion injury, erectile dysfunction, as well as certain forms of epilepsy and movement disorders.

In this work, I examine mechanisms underlying BK channel activation by the small-molecule NS11021. A biarylthiourea derivative, NS11021 is structurally distinct from other BK channel activators, and may thus work by a distinct mechanism of action. I found that increasing NS11021 shifts the voltage required for half-maximal activation ($V_{1/2}$) for BK channels toward more hyperpolarized voltages, in both the presence and nominal absence of Ca^{2+}, suggesting that NS11021 facilitates BK channel activation primarily by a mechanism that is distinct from Ca^{2+} activation. 30 μM NS11021 slows the time course of BK channel deactivation at -200 mV by up to 10-fold compared to 0 μM NS11021, while having little effect on the time course of activation. This action occurs at negative voltages, at which the BK channel voltage sensors are largely at rest. Single-channel kinetic analysis further shows that 30 μM NS11021 increases Po by 62-fold, and increases mean open time from 0.15 ms to 0.52 ms in nominally 0 Ca^{2+} at voltages <-60 mV, conditions where BK voltage sensors are largely in the resting state. The major activating effects of NS11021 can be described by a scheme in which the drug primarily shifts the pore-gate equilibrium toward the open state.

I next examined whether NS11021 could ameliorate changes in gating caused by a disease-linked loss-of-function BK mutation, P805L. The P805L mutation was identified in a patient with severe speech and developmental delays, intellectual disability, and apraxia. My findings indicate that BK channels with the P805L mutation activate with a $V_{1/2}$ shifted towards depolarized voltages, and with a slowed time-course of activation compared to WT. Although P805 is located in the Ca^{2+}-sensing domain of the channel, these effects persist in the nominal absence of Ca^{2+}, indicating that P805L induced changes cannot be limited to Ca^{2+} sensitivity alone and must alter either voltage-sensing or intrinsic opening of the pore. Addition of NS11021 partially mitigates the effect of the mutation by eliciting a left-shift in the $V_{1/2}$, such that 10 μM NS11021 results in P805L channels activating with a $V_{1/2}$ similar to WT channels in 10 μM Ca^{2+}, although the time course of both activation and deactivation are slowed by approximately 4-fold compared to WT. These results may provide a framework for analysis of therapeutic potential of BK channel modulators in treatment of patients with BK channelopathies.

ACKNOWLEDGEMENTS

I would like to thank my mentor Dr. Brad Rothberg for welcoming me into his lab and helping me grow as a scientist. I greatly appreciate the guidance and mentorship he provided throughout my training and for teaching me how to quantitatively approach a research question. Thank you for always being an advocate and for inspiring me to do better. I would also like to thank Alex Vouga of the Rothberg Lab for all the help and support he provided. Whether teaching me how to patch-clamp and perform experiments or talking about our latest accomplishments and setbacks, I greatly value our friendship.

Thank you to committee members Dr. Kathleen Giangiacomo, Dr. Steve Houser, and Dr. Jonathan Soboloff for feedback and advice throughout this process. I would also like to thank external examiner Dr. Crina Nimigean for agreeing to be a part of my defense committee.

A special thanks to all others who helped me develop and maintain a love of science, especially previous research advisors Dr. Bob Lefkowitz and Dr. Justin Hines, and teachers Bert Wartski and Ruben Giral.

Lastly, I would like to thank my family and friends for their love and support over the years. To my parents Howard and Claire Rockman, thank you for the unconditional love and support you have always provided and for emphasizing the importance of education and hard work. To my partner Monica Wentz, thank you for sharing this journey with me and for the encouragement and support every step of the way.

TABLE OF CONTENTS

ABSTRACT iii

ACKNOWLEDGEMENTS v

LIST OF ABBREVIATIONS viii

LIST OF FIGURES x

LIST OF TABLES xii

1. INTRODUCTION 1

Part I: Ion Channels 1
Historical Perspective 1
Properties of ion channels 4
K^+ channels 5

Part II: BK Channels 8
Domain Architecture 9
Dual-allosteric model of activation 11
Role in smooth muscle, brain, and kidney 14
Role in hypertension and neurological disorders 16

Part III: Pharmacology of BK channel modulators 18
Pore Blockers 19
Gating Modulators 21
Distinct mechanisms of altered gating 21
Inhibitors 21
Activators 22

Part IV: Hypothesis 27

2. CHARACTERISTICS OF BK CHANNEL ACTIVATION BY NS11021 29

Introduction 29

Materials and Methods 30
Electrophysiology 30
Data Analysis 32

Results 35
Effects of NS11021 on BK channel activation 35
NS11021 slows channel deactivation time course with little effect on activation 38
NS11021 activation does not require the CSD 41
NS11021 activation may occur primarily through the PGD 43
Concentration-dependence of NS11021 action 46

Discussion 48
NS11021 alters gating kinetics 48
Mechanism of NS11021 activation is unlikely to be highly cooperative 51

3. QUANTITATIVE DESCRIPTION OF NS11021 ACTIVATION 52
Introduction 52
Materials and Methods 53
Description of steady-state channel activity 53
Description of activation and deactivation rates 54
Results 55
NS11021 acts by stabilizing the PGD in the open state 55
Potential NS11021 actions at other gating modules 70
Discussion 80
Mechanism of NS11021 activation 80
Physical mechanism of NS11021 activation 81
Limitations of the proposed mechanism 83
Conclusions 84
4. EFFECTS OF P805L, A LOSS-OF-FUNCTION MUTATION LINKED TO HUMAN DISEASE 86
Introduction 86
Materials and Methods 87
Electrophysiology 87
Data Analysis 89
Kinetic Modeling 90
Results 92
P805 is a highly conserved residue in RCK2 92
P805L alters voltage-dependent gating in the presence of 10 μM Ca^{2+} 92
P805L channels exhibit altered gating over a range of $[Ca^{2+}]$ 95
P805L mutation does not substantially alter single-channel conductance 95
Changes in Ca^{2+}-sensitivity alone cannot describe behavior of P805L channels 97
NS11021 activates P805L mutant channels 111
Discussion 113
Structural changes caused by the P805L mutation 113
Gating mechanism of BK channels with the P805L mutation 113
Other loss-of-function BK channelopathies 115
Pharmacological activation of a loss-of-function BK channelopathy 115
Conclusions 116
5. DISCUSSION 118
Mechanisms of activation similar to NS11021 118
Loss-of-function BK channelopathies 119
Challenges to treating BK channelopathies 123
Outlook for treating BK channelopathies 125
REFERENCES CITED 127

LIST OF ABBREVIATIONS

Ca^{2+}	Calcium Ion
Cl^{-}	Chloride Ion
Cryo EM	Cryogenic electron microscopy
CSD	Calcium-sensing domain
CTx	Charybdotoxin
DHA	Docosahexaenoic Acid
GABA	y-aminobutyric acid
GOF	Gain-of-function mutation
G-V-Ca^{2+}	Conductance-Voltage relations over a range of [Ca^{2+}]
K^{+}	Potassium Ion
KcsA	K+ channel of *Streptomyces lividans*
Kv	Voltage-gated K^{+} channel
LNCaP	Lymph node carcinoma of the prostate
LOF	Loss-of-function mutation
Na^{+}	Sodium Ion
NCA	Negatively charged activator
PNKD	Paroxysmal Nonkinesigenic Dyskinesia
Po	Channel Open Probability
PGD	Pore-forming/Gate Domain
ROMK	Renal outer medullary K^{+} channel
TEA	Tetraethylammonium
VSD	Voltage-sensing Domain

ABBREVIATIONS OF KINETIC MODEL PARAMETERS

Dual-Allosteric Model (**Scheme 1** - 50 state model)

C - O	Closed and Open states of the pore
R - A	Resting and Active states of the voltage-sensor
X - X_{Ca2+}	Unliganded and Ca^{2+}-bound states of the Ca^{2+}-sensor
J	R - A equilibrium constant of the voltage-sensor
K_D	Ca^{2+} dissociation constant of the Ca^{2+}-sensor
L	C - O equilibrium constant of the pore
z_J	Gating valence of R - A equilibrium constant
z_L	Gating valence of C - O equilibrium constant
C	Allosteric coupling constant, between channel opening and Ca^{2+}-sensor activation
D	Allosteric coupling constant, between channel opening and voltage-sensor activation
E	Allosteric coupling constant, between Ca^{2+}-sensor activation and voltage-sensor activation

Voltage-Dependent Model (**Scheme 2** - 10 state model)

α	Voltage-sensor activation rate constant
β	Voltage-sensor deactivation rate constant
ɣ	Pore closing rate constant
δ	Pore opening rate constant
z_α, z_β, z_γ, z_δ	Gating valence associated with each rate constant

LIST OF FIGURES

Figure 1-1: Domain architecture of BK channels 10

Figure 1-2: Physiological Role of BK Channels in Smooth Muscle 15

Figure 1-3: Diversity of chemical structures among BK channel activators 24

Figure 2-1: BK channel activation and deactivation time course 34

Figure 2-2: Effects of NS11021 on BK channel gating 36

Figure 2-3: NS11021 activates BK channels over a range of $[Ca^{2+}]$ 40

Figure 2-4: Effects of NS11021 on truncated BK channels 42

Figure 2-5: NS11021 increases open probability at negative voltages in nominally 0 Ca^{2+} 44

Figure 2-6: Effects of NS11021 on open and closed dwell time distributions 45

Figure 2-7: Estimate of apparent stoichiometry of NS11021 action 49

Figure 3-1: G-V-Ca^{2+} relations with predictions from Scheme 1 using parameters from Fit A 58

Figure 3-2: G-V-Ca^{2+} relations with predictions from Scheme 1 using parameters from Fit B 59

Figure 3-3: G-V-Ca^{2+} relations with predictions from Scheme 1 using parameters from Fit A, with adjustable parameter J_0 61

Figure 3-4: G-V-Ca^{2+} relations with predictions from Scheme 1 using parameters from Fit A, with adjustable parameter D 62

Figure 3-5: G-V-Ca^{2+} relations with predictions from Scheme 1 using parameters from Fit B, with adjustable parameter J_0 63

Figure 3-6: G-V-Ca^{2+} relations with predictions from Scheme 1 using parameters from Fit B, with adjustable parameter D 64

Figure 3-7: Description of BK channel activity at nominally 0 Ca^{2+} using Schemes 1 and 2 67

Figure 3-8: G-V-Ca^{2+} relations from Scheme 1 using parameters from Fit B to describe changing parameter J_0 in addition to L_0. 71

Figure 3-9: G-V-Ca^{2+} relations from Scheme 1 using parameters from Fit B to describe changing parameter D in addition to L_0. 72

Figure 3-10: Additional fits of τ vs. voltage using scheme 2 76

Figure 3-11: Effect of NS11021 on $V_{1/2}$ over a range of [Ca^{2+}]..... 77

Figure 3-12: G-V-Ca^{2+} relations from Scheme 1 using parameters from Fit A to describe changing parameter C in addition to L_0. 78

Figure 3-13: G-V-Ca^{2+} relations from Scheme 1 using parameters from Fit B to describe changing parameter C in addition to L_0. 79

Figure 4-1: BK channel gating was unaffected by the deletion of the N-terminal myc tag and addition of the C-terminal 1D4 tag. 88

Figure 4-2: P805 is a highly conserved residue in RCK2..... 91

Figure 4-3: P805L mutation slows activation 93

Figure 4-4: P805L mutation alters BK channel activity over a range of [Ca^{2+}]. 94

Figure 4-5: P805L mutation does not alter unitary current amplitude. 96

Figure 4-6: G-V-Ca^{2+} relations for P805L channels changing a single parameter from Fit A and Fit B. 99

Figure 4-7: $V_{1/2}$ predicted by changing individual parameters in Scheme 1. 100

Figure 4-8: Description of macroscopic time constants in the absence of Ca^{2+} using Scheme 2 106

Figure 4-9: Altering Ca^{2+}-sensitive parameters in addition to z_L in Scheme 1 to account for changes in gating of P805L channels 110

Figure 4-10: NS11021 activates BK channels with the P805L mutation..... 112

LIST OF TABLES

Table 1-1: Distribution of ions in the cell 6

Table 1-2: Scheme 1 – Dual-allosteric model of BK channel activation..... 12

Table 2-1: Mean values of $V_{1/2}$ and z determined from Boltzmann fits of individual G-V relations..... 37

Table 3-1: Parameters for Scheme 1, fitted with data at 0 μM NS11021..... 57

Table 3-2: Results of changing individual parameters in Scheme 1 to account for effects of NS11021 60

Table 3-3: Fitted parameters for Scheme 2 constrained by time constants acquired with nominally 0 Ca^{2+}, and either 0 or 30 μM NS11021 69

Table 3-4: Results of changing L_0 plus a second parameter in Scheme 1 to account for effect of NS11021 74

Table 3-5: Additional sets of fitted parameters for Scheme 2 constrained by time constants acquired with nominally 0 Ca^{2+}, and either 0 or 30 μM NS11021..... 75

Table 4-1: Results of changing individual parameters in Scheme 1 to account for altered gating in P805L channels..... 98

Table 4-2: Results of changing J_0 (equilibrium of voltage-sensor activation) plus a second parameter in Scheme 1 to account for effects of P805L mutation..... 102

Table 4-3: Results of changing L_0 (equilibrium of pore opening) plus a second parameter in Scheme 1 to account for effects of P805L mutation..... 103

Table 4-4: Results of changing D (coupling between voltage sensors and pore opening) plus a second parameter in Scheme 1 to account for effects of P805L mutation..... 104

Table 4-5: Results of changing individual parameters in Scheme 2 to account for altered gating in P805L mutant channels..... 105

Table 4-6: Results of changing multiple parameters in Scheme 2 to account for altered gating in P805 channels..... 108

Table 4-7: Results of changing z_L plus a second parameter in Scheme 1 to account for effects of P805L mutation 109

CHAPTER 1

INTRODUCTION

Part I: Ion Channels

Historical Perspective

The ability for life to exist in a wide range of environmental conditions is due to the plasma membrane providing a relatively stable environment inside of cells. Sometimes referred to as the “boundary of life”, the plasma membrane can provide protection from extremes in temperature, pH, pressure, radiation, and salinity (Merino et al., 2019). Furthermore, the plasma membrane is a dynamic structure which constantly adjusts to changes in the environment (Singer and Nicolson, 1972), and is “semi-permeable” to allow passage of certain molecules and ions. (Loeb, 1906).

The permeability of the plasma membrane for specific ions is not constant. This phenomena was observed in 1952 by Hodgkin and Huxley describing the action potential propagation in giant squid axon (Hodgkin and Huxley, 1952). We now know that “pores” exist in the plasma membrane by which small ions or molecules can rapidly pass through (Hodgkin and Keynes, 1955). The term “ion channels” is used to describe the aqueous pores formed by protein that can open and close to selectively allow passage of certain ions (Hille et al., 1999). Over the next few decades, many fundamental properties of ion channels were determined by Clay Armstrong, Bertil Hille, and others (Hille and Schwarz, 1978; Hille, 1977, 1978; Armstrong, 1971; Armstrong and Bezanilla, 1974, 1973, 1977; Armstrong, 2007; Hille, 1967). Investigations into the function of ion channels have been greatly aided by the patch-clamp electrophysiology technique (Hille et al., 1999).

When an ion permeates through an ion channel, this movement of charge generates a current. Quantifying electrical properties of cells is therefore a direct way to

study the function of ion channels. Quantitative study of the electrical properties of membranes began in the late 1930s when Kenneth Cole and Howard Curtis demonstrated a rapid fall in the membrane potential during an action potential using extracellular electrodes on the axons of giant squid (Curtis and Cole, 1940). Later, intracellular electrodes were developed and the first direct measurements of transmembrane current were performed (Hodgkin and Huxley, 1939; Curtis and Cole, 1940). The largest breakthrough came in 1976 when Erwin Neher and Bert Sakmann used a smooth electrode tip to electrically isolate a patch of membrane from the remainder of the cell, greatly reducing intrinsic noise of the system (Neher and Sakmann, 1976; Neher and Lux, 1969). This technical breakthrough allowed for the openings and closings of an individual ion channel to be measured in a native cell membrane, a technique now referred to as "single-channel recording" (Neher and Sakmann, 1976; Hamill et al., 1981).

Many of the early experiments to quantify the electrical properties of cell membranes were performed on the giant axons of squid *Loligo pealei*. Because of the large diameter of their axons, the squid made an excellent model system for measuring membrane electrical properties (Adelman and Gilbert, 1990). Until the 1960's, studies of ion channels were centered around the squid giant axon, which often required scientists to work day and night while they had fresh tissue from caught squid. Due to the short time of year where squid could be consistently caught by fisherman, notable scientists would sometimes split time between labs based in Woods Hole, Massachusetts and Valparaíso, Chile (Bezanilla, 2018).

Understanding the function of ion channels at the molecular level began in the mid 1980s with cloning of the first Na^+ channel, followed by the first K^+ channel in *Drosophila melanogaster* (Noda et al., 1984; Tempel et al., 1987). This lead to the identification of the P (pore) region and the K^+ channel signature sequence (MacKinnon and Miller, 1988;

Heginbotham et al., 1992). While mutagenesis experiments utilizing the pore-blocker tetraethylammonium (TEA) could being to uncover information about channel structure (Yellen, 1987; Yellen et al., 1991; Heginbotham and MacKinnon, 1992), the molecular structure of the pore was not yet clear. It took the first X-ray crystal structure of a ion channel determined by Rod MacKinnon in 1998 to learn how K^+ ions are coordinated in the pore during ion permeation (Doyle, 1998).

By determining the structure of the KcsA K^+ channel from *Streptomyces lividans* to 3.4 Å resolution, the specific chemical environment that allowed for charged K^+ ions to diffuse across a hydrophobic membrane was observed for the first time. This structure showed a tetrameric channel pore with two transmembrane alpha helices per subunit, forming an "inverted teepee" with the narrow zone near the cytosolic face of the membrane. The structure also showed details of the selectivity filter for the first time, the highly-conserved structure of K^+ channels that allows for selective passage of K^+ and not Na^+ or Ca^{2+}. The structure of the eukaryotic voltage-gated K^+ channel $K_v1.2$ was determined a few years later to even higher resolution (Long, 2005). While membrane proteins such as ion channels have always been notoriously difficult to crystallize, the rate of structure determination has increased rapidly over the recent years due to the improved resolution from cryogenic electron microscopy (Cryo EM) (Bai et al., 2015; Kühlbrandt, 2014). Structure determination using Cryo EM has led to a revolution in structural biology, improving our understanding about many ion channels that play significant roles in human health (Tao et al., 2016; Wang and MacKinnon, 2017; Sun and MacKinnon, 2017; Liu et al., 2017; Martin et al., 2017; Tempel et al., 1987; Noda et al., 1984; Park et al., 2017; Hartmann et al., 1991; Paulsen et al., 2015; Guy et al., 1991; Liao et al., 2013).

Properties of ion channels

The ***driving force*** for ions to flow through channels relates to the chemical and electrical potentials across the membrane. The equilibrium potential, or the net membrane potential across the membrane, can be described by the Nernst Equation:

$$V_e = \frac{RT}{zF}\, ln(\frac{[ion]_{out}}{[ion]_{in}})$$

(Eq. 1.1)

where V_e is the equilibrium potential, R is the universal gas constant (J K^{-1} mol^{-1}), T is the temperature (˚K), z is the valence of the permeant ion, F is the Faraday constant (C mol^{-1}), and $[ion]_{out}$ and $[ion]_{in}$ represent the concentration of the permeant ion outside and inside of the cell, respectively.

The ***rate*** that ions can permeate is dependent on several factors including pore diameter, relative difference between the radius of each ion and the pore diameter (radius of capture), and pore occupancy (Andersen, 1983; Läuger, 1976; Brelidze and Magleby, 2005; Naranjo et al., 2016; Piskorowski and Aldrich, 2006). For certain channels, the conductance rate can be as high as 10^8 ions per second, approaching the diffusion limit (Hille, 2001).

The opening and closing of ion channels, a process called ***gating***, can be tightly regulated by an assortment of stimuli including membrane potential, intracellular ions, hormones, neurotransmitters, cyclic nucleotides, G-proteins, metabolic intermediates, pressure, light, and temperature (Armstrong and Hille, 1998; Kaupp and Seifert, 2002; Wickman and Clapham, 1995; Delmas and Coste, 2013; Nagel et al., 2003; Yao et al., 2011; Blatt and Thiel, 1993; Madden, 2002). The presence of a stimulus alters the open probability (Po) of the channel by changing the thermodynamic equilibrium associated with certain kinetic states (Horn and Lange, 1983). For example, a stimulus can activate a

channel by making an open state more thermodynamically stable or by making a closed state less thermodynamically stable – both resulting in a similar effect of driving a channel toward the open state.

The physiological gradients of common ions are shown in **Table 1-1** (Hille, 2001). In excitable cells, ion channels are responsible for setting and modulating the resting membrane potential (Hodgkin and Huxley, 1952). The net inward flow of positive charge (Na^+ or Ca^{2+} channels) leads to "depolarization" of the membrane and typically has an excitatory function. Conversely, the outward flow of positive charge (K^+ or Cl^- channels) leads to "hyperpolarization" of the membrane and is typically inhibitory. In non-excitable cells, cytosolic Ca^{2+} plays many important roles in signaling and regulation of transcription (Clapham, 2007). In summary, ion channels are pores in the cell membrane that can open and close to rapidly allow passage of ions to participate in a wide range of functions.

K^+ channels

Potassium channels represent the largest and most widely distributed superfamily of ion channels and are encoded by approximately 80 different genes (Maljevic and Lerche, 2013). The highly-conserved selectivity filter allows K^+ ions to permeate near the rate of diffusion while being up to 10,000 times more selective for K^+ than Na^+ (Blatz and Magleby, 1984; Sansom et al., 2002; Lu et al., 2001). Most K^+ channels fall into one of four major classes: voltage-gated (K_v), inward-rectifying (K_{IR}), two-pore domain (K_2P), and calcium-activated (K_{Ca}).

Voltage-gated K^+ channels are activated by a depolarized membrane potential. The opening of K_v channels during an action potential generates an outward current that drives the membrane potential to more hyperpolarized voltages (Hille, 2001). In contrast, the physiological activation of inward-rectifying K+ channels typically results in an inward

Table 1-1: Distribution of ions in the cell

Ion	Extracellular (mM)	Intracellular (mM)
Na^+	145	12
K^+	4	155
Ca^{2+}	1.5	1×10^{-4}
Cl^-	123	4.2

Values obtained from (Hille, 2001).

current due to voltage-dependent blockade by polyamines and certain ions at depolarized voltages (Vergara et al., 1998; Matsuda et al., 1987; Lopatin et al., 1994; John et al., 2004). Inward-rectifying K^+ channels play a key role in maintaining the resting membrane potential (Reimann and Ashcroft, 1999). Two-pore domain K+ channels, unlike all other K^+ channels that form tetramers, contain only two subunits assembling to form a functioning channel (Lesage et al., 1996). Two-pore domain K^+ channels are weakly voltage dependent and contribute to both frequency and duration of action potentials (Donner et al., 2011; Zheng and Trudeau, 2015; Schewe et al., 2016). The calcium-activated BK, IK, & SK channels open in response to cytosolic Ca^{2+} and are named after their characteristic single channel conductance, which is either big (BK channels), intermediate (IK channels), or small (SK channels).

The major focus of the remainder of this dissertation will be large-conductance Ca^{2+}-activated K^+ channels (BK channels). Synergistically activated by cytosolic Ca^{2+} and a depolarized membrane potential, BK channels are widely expressed in most tissues and participate in many physiological processes (Marty, 1981; Blatz and Magleby, 1984; Latorre et al., 2016). IK and SK channels, by contrast, open with a smaller single-channel conductance than BK channels and activate in response to Ca^{2+}-binding to calmodulin but not by voltage (Adelman, 2016; Li et al., 2009; Schumacher et al., 2001; Joiner et al., 1997; Köhler et al., 1996; Bond et al., 2005; Lee and Mackinnon, 2018; Bond et al., 2004; Ishii et al., 1997; Silverman et al., 1997; Logsdon et al., 1997; Kshatri et al., 2018). In summary, calcium-activated K^+ channels link Ca^{2+} and electrical signaling to participate in important physiological processes.

Part II: BK Channels

Large conductance Ca^{2+}-activated K^+ channels (BK channels, Slo1, $K_{Ca}1.1$, Maxi-K) are activated by cytosolic Ca^{2+} and depolarized membrane potential to control electrical excitability in neurons and smooth muscle cells (Van Goor et al., 2001). Activation of BK channels leads to an outward K^+ current which drives the membrane potential towards more hyperpolarization voltages (Blatz and Magleby, 1984). Altered BK channel function has been linked to epilepsy, dyskinesia, and other neurological deficits in humans, making these channels a key target for drug therapies (Bailey et al., 2019; Bentzen et al., 2014).

The first indication about the existence of a Ca^{2+}-dependent K^+ current came in the early 1970's, when it was observed that Ca^{2+} injected into neurons increased the K^+ conductance (Meech, 1972; Meech and Standen, 1975). It was subsequently proposed that a Ca^{2+}-activated K^+ current may act as a vital link between cell metabolism and membrane conductance (Meech, 1978). Over the next decade, single-channel recordings illuminated a channel with a large unitary conductance (~200 pS) which led to the name "BK channel," short for "Big-conductance K^+ channel" (Pallotta et al., 1981; Marty, 1983).

Identification of the gene encoding the BK channel began in the mid 1980s. The *slowpoke* (*Slo*) gene, named for a *Drosophila* mutant that was slow to recover movement after being flicked from the wall of it's glass container, was found to mediate a Ca^{2+}-dependent K^+ current, and revealed a similar transmembrane architecture to that of previously known K_v channels (Atkinson et al., 1991). In mouse, the sequence of the *Slo* ortholog was cloned from brain and skeletal muscle tissues to reveal that (as observed in *Drosophila*) although a large number of variant cDNA sequences exist, BK channels in all tissues are encoded by a single gene (Butler et al., 1993). Using the current nomenclature, the KCNMA1 gene encodes for the human BK channel and has been mapped to chromosome 10q23.1 (Dworetzky et al., 1994; McCobb et al., 1995).

Domain Architecture

BK channels assemble to form tetramers of KCNMA1 encoded α-subunits (Shen et al., 1994). Each subunit consists of an N-terminal voltage-sensing domain (VSD; transmembrane helices S0-S4), a pore/gate domain (PGD; transmembrane helices S5-P-S6), and a Ca^{2+}-sensing domain (CSD) comprised of two tandem cytosolic RCK domains (abbreviation for Regulator of the Conductance of K^+, a term coined by the MacKinnon laboratory after identification of the structure of this domain; **Figure 1-1**). The transmembrane domains comprising the voltage-sensor and pore have been described as the "core" of the channel, and are structurally similar to K_v channels (Salkoff and Jegla, 1997; Salkoff et al., 1992). In addition to the six transmembrane helices found in K_v channels, BK channels contain an additional helix, S0, that is thought to modulate voltage sensitivity of the channel (Koval et al., 2007; Liu et al., 2008; Pantazis et al., 2010).

The Ca^{2+}-sensing RCK domains contain three Ca^{2+} binding sites per subunit, two of which bind with near micromolar apparent affinity, and one formed at the interface between the VSD and RCK domains with lower (millimolar) apparent affinity (Jiang et al., 2002a; Lingle, 2007; Giraldez and Rothberg, 2017; Schreiber and Salkoff, 1997). The eight total RCK domains per channel (2 RCK domains from each subunit) assemble to form what has been called the cytosolic "gating ring" (Schreiber and Salkoff, 1997; Bao et al., 2002; Ye et al., 2006; Wu et al., 2010). The two higher-affinity binding sites are relatively Ca^{2+}-selective, although other divalent cations, including Cd^{2+} and Ba^{2+}, can bind with varying affinities (Zeng et al., 2005; Zhou et al., 2012; Horrigan and Ma, 2008). The full length structure of the BK channel was determined in 2017, showing for the first time how structural changes caused by Ca^{2+}-binding lead to opening of the pore (Tao et al., 2016; Hite et al., 2016). Based on structural data, it is thought that Ca^{2+} binding causes large conformational changes that result in pulling of the S6 helix to open the gate of the

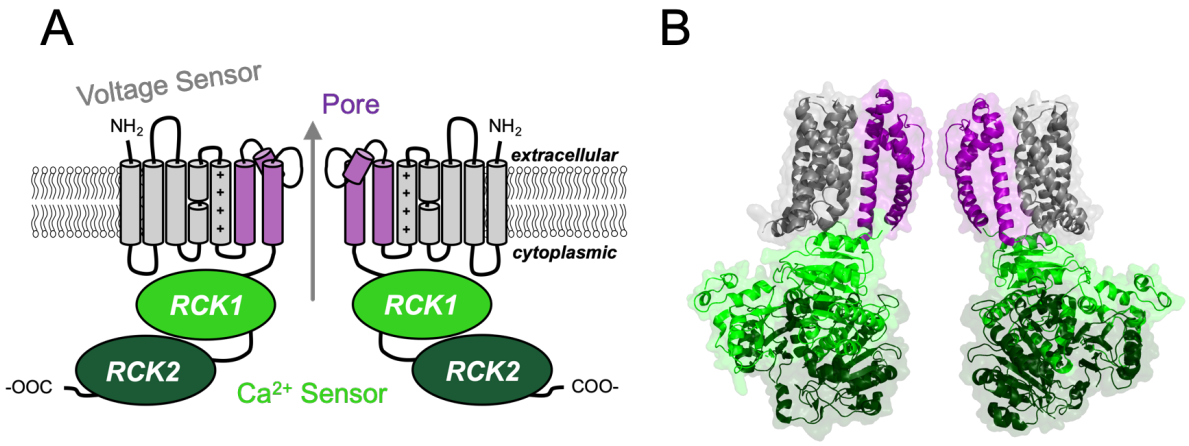


Figure 1-1: Domain architecture of BK channels

A, Cartoon of BK channel architecture showing two of the four subunits from the side to visualize the permeation pathway (arrow). Each subunit contains voltage sensors (S0-S4, gray), a pore forming domain (S5-S6, purple), and two Ca^{2+}-sensing RCK domains (green). **B,** Structure of BK channel (pdb = 5TJ6), adapted from Tao *et al.,* 2017.

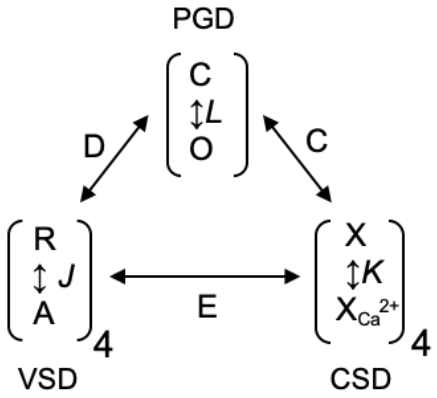
channel, and interactions with the S4-S5 linker region to facilitate voltage-sensor movement (Hite et al., 2016; Fan et al., 2020; Giraldez and Rothberg, 2017).

Dual-allosteric model of activation

The mechanism by which BK channels activate in response to depolarization and increased cytosolic Ca^{2+} can be described by a dual allosteric scheme (Cui et al., 1997; Cui and Aldrich, 2000; Horrigan and Aldrich, 2002; Horrigan et al., 1999; Rothberg and Magleby, 2000). Although activation of BK channels near physiological conditions requires a combination of both cytosolic Ca^{2+} and depolarization, either depolarized voltage (>100 mV) or high [Ca^{2+}] on their own can be sufficient to activate channels experimentally (Rothberg and Magleby, 2000; Horrigan et al., 1999). The "dual-allosteric model" of BK channels accounts for many of the observed properties, and provides a framework in which to quantitatively describe gating of BK channels (**Scheme 1, Table 1-2**) (Horrigan and Aldrich, 2002).

In the dual-allosteric model, BK channels are defined as containing four voltage sensing domains (VSD) and four Ca^{2+} sensing domains (CSD) which contribute to activation of a single pore/gate domain (PGD) (**Scheme 1**). Intrinsic transitions of the pore between the closed (C) and open (O) state are described by the equilibrium constant L. Each of the four voltage-sensors can be in a resting (R) or active (A) state, with equilibrium constant J. Each Ca^{2+}-sensor can likewise be unbound (X) or bound (X_{Ca2+}), with Ca^{2+} dissociation constant K_D. Allosteric coupling constants C, D, and E describe the coupling between the Ca^{2+}-sensor and pore, voltage-sensor and pore, and coupling between the voltage-sensors and Ca^{2+} sensors, respectively. The dual-allosteric model contains multiple simplifying assumptions, such as assuming a single Ca^{2+}-binding site per subunit, but nonetheless provides a quantitative method by which to describe BK channel

Table 1-2: Scheme 1 – Dual-allosteric model of BK channel activation



J R – A Equilibrium Constant
$J = J_0 \exp(-z_J V/k_B T)$
J_0 is the zero-voltage value of J
z_J is the partial charge

K Equilibrium Constant for Ca^{2+} Binding
$K = [Ca^{2+}] / K_c$
K_c is the Ca^{2+} dissociation of the closed channel

L C – O Equilibrium Constant
$L = L_0 \exp(-z_L V/k_B T)$
L_0 is the zero-voltage value of L
z_L is the partial charge

C Allosteric Coupling Constant between Ca^{2+} binding and channel opening
$C = K_c/K_o$
K_C is the Ca^{2+} dissociation of the closed channel

D Allosteric Coupling Constant between voltage sensor activation and channel opening
$D = \exp(-z_J[V_{1/2,\,o} – V_{1/2,\,c}] / k_B T)$
$V_{1/2,\,o}$ and $V_{1/2,\,c}$ are the half-activating voltage of Q_c and Q_o
Q_o and Q_c are the steady–state gating charge distribution for closed and open channels

E Allosteric Coupling Constant between Ca^{2+} binding and voltage sensor activation

activation over a range of voltages and [Ca^{2+}]. A value for each equilibrium constant (J, K, and L) and allosteric coupling constant (C, D, and E) can be determined from experimental data, and this quantitative model can be used to predict BK channel open probability over a rage of voltages and [Ca^{2+}] (Horrigan and Aldrich, 2002; Latorre et al., 2016). In addition to describing activity of wild-type BK channels, alterations to BK channel gating caused by addition of a drug, mutation, or auxiliary subunit can be described in the context of the dual-allosteric model (Koval et al., 2007; Wang et al., 2006, 2009; Ma et al., 2006; Gessner et al., 2012; Zhou et al., 2018; Horrigan and Aldrich, 1999; Hu et al., 2005; Sweet and Cox, 2009).

Conditions which activate BK channels can be different in certain cell types based on assembly with auxiliary (β or ɣ) subunits (Li and Yan, 2016). For example, the β1 subunit expressed primarily in smooth muscle enables channels to activate at less depolarized voltages by modulating the apparent Ca^{2+} sensitivity (Qian and Magleby, 2003; Morrow et al., 2006; Tseng-Crank et al., 1994; Bao and Cox, 2005; Nimigean and Magleby, 1999, 2000; Qian et al., 2002). In contrast, association with the β3 subunit that is expressed in chromaffin cells results in rapid inactivation (Lingle et al., 2001; Zeng et al., 2001). Auxiliary ɣ subunits were discovered in lymph node carcinoma of the prostate (LNCaP) cells (Gessner et al., 2005). Association with ɣ subunits causes a substantial leftward shift (~140 mV with the $ɣ_1$ subunit) in the voltage-dependence of activation, which allows BK channels to activate near normal cell resting potentials even without increased of cytosolic Ca^{2+}(Yan and Aldrich, 2010, 2012; Yang et al., 2017; Gonzalez-Perez et al., 2014, 2018). Expression of the $ɣ_1$ subunit has been identified in secretory epithelium, arterial and bronchial smooth muscle cells, and in malignant cells (Evanson et al., 2014; Noda et al., 2019; Yang et al., 2017; Huang and Jan, 2014; Miyagawa et al., 2018; Yoshihara et al., 2011). In summary, the ability for BK channels to associate with a unique

combination of auxiliary subunits in specific tissues is a mechanism to generate diversity in the modulation of electrical signaling by BK channels in these tissues.

Role in smooth muscle, brain, and kidney

Remarkably, BK channels are expressed in nearly every tissue of the human body including epithelial cells (Furuya et al., 1989), smooth muscle (McCobb et al., 1995; Vogalis et al., 1996), brain (Ha et al., 2000), kidney (Hunter et al., 1984), and ear (Jones et al., 1999) to participate in important physiological processes. In smooth muscle cells, regulation of tone is mediated in part by BK channels (**Figure 1-2**). During initiation of smooth muscle contraction, Ca^{2+} influx through plasma membrane voltage-gated Ca^{2+} channels produces a large Ca^{2+} release from the sarcoendoplasmic reticulum in a process called "Ca^{2+}-induced Ca^{2+} release" (Berridge, 1993). The [Ca^{2+}] in local transients, known as "calcium sparks", can reach concentrations up to 100 μM while only increasing the global calcium concentration by less than 2 nM (Nelson et al., 1995; Jaggar et al., 1998, 2000). The large increase in [Ca^{2+}] leads to an increased open probability of BK channels, resulting in an outward hyperpolarizing current. Increased BK channel current in smooth muscle cells therefore leads to relaxation, whereas a reduction in BK channel current leads to contraction.

In the brain, BK channels regulate neuronal firing rate by mediating the repolarization and afterhyperpolarization phases of the action potential (Gu et al., 2007; Faber and Sah, 2002; Shao et al., 1999). Paradoxically, BK channels can either increase or decrease neuronal firing rates, in part due to expression in both excitatory and inhibitory neurons and association with the β4 subunit, which has the effect of slowing channel activation as well as deactivation kinetics (Montgomery and Meredith, 2012; Contet et al., 2016; Brenner et al., 2005). The onset and duration of BK current also exhibits substantial

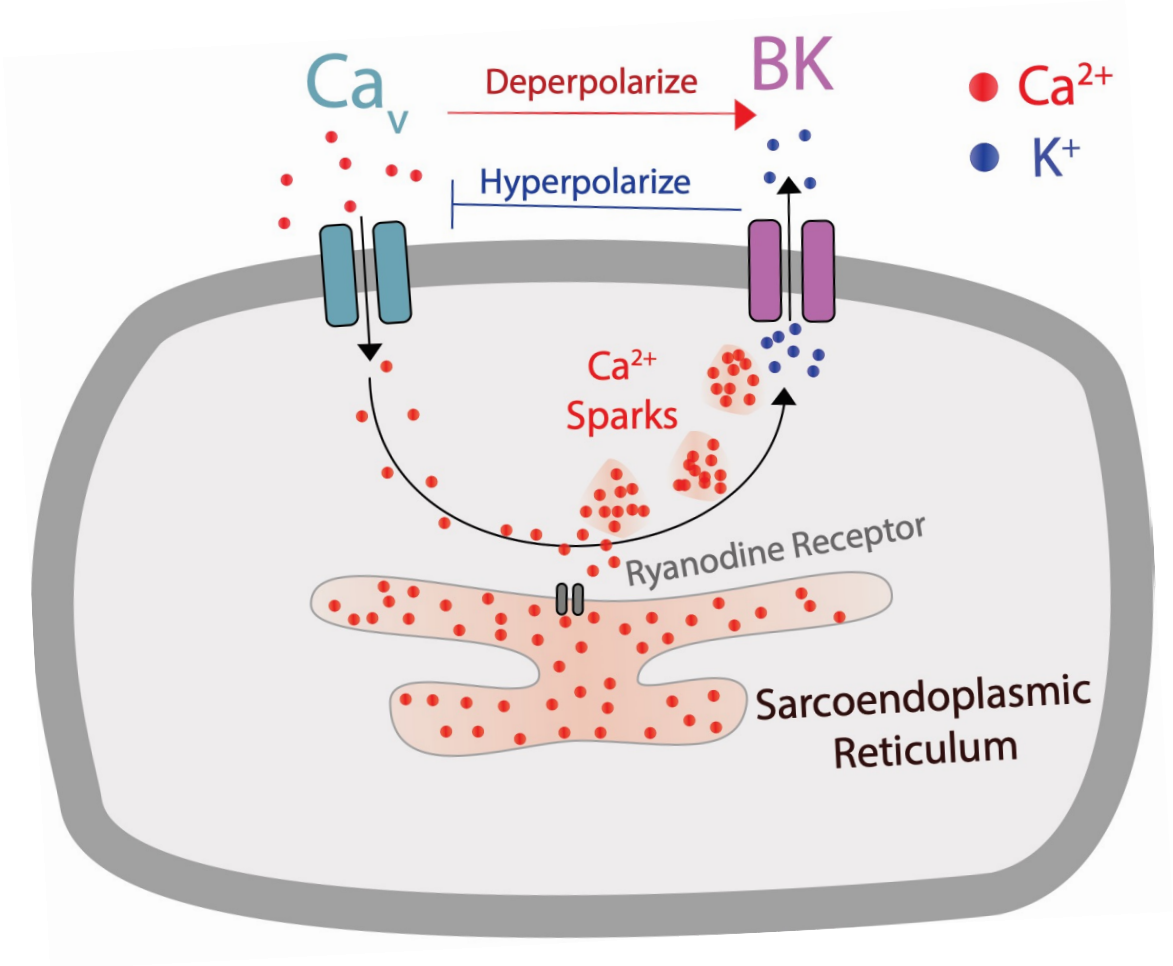


Figure 1-2: Physiological Role of BK Channels in Smooth Muscle

In smooth muscle cells, entry of Ca^{2+} (red circles) via plasma membrane voltage-gated Ca^{2+} channels stimulates release of Ca^{2+} from the sarcoendoplasmic reticulum by activating ryanodine receptors. BK channels produce a hyperpolarizing current in response to increased Ca^{2+}, resulting in smooth muscle relaxation.

variation depending on cell type and the composition of ion channels expressed in close proximity to BK channels. (Hu et al., 2001; Pattillo et al., 2001). Furthermore, in a separate role, BK channels have been proposed to act as an “emergency brake” in the nervous system to activate under excitotoxic conditions where cytosolic [Ca^{2+}] rises above physiological levels (Hu et al., 2003).

Although its roles in all segments of the nephron are not fully understood, BK channels are expressed in most renal cell types including in podocytes, glomerular mesangial cells, proximal convoluted tubule, distal convoluted tubule, and the cortical collecting duct (Meera et al., 2000; Morton et al., 2004; Stockand and Sansom, 1994). While the majority of K^+ secretion in the distal kidney is typically conducted by the Renal Outer Medullary K^+ Channel (ROMK) (Ho et al., 1993; Grimm and Sansom, 2010), *in vivo* data suggest that BK channels are upregulated in response to a high K^+ diet, likely acting as a mechanism to ensure a stable K^+ concentration even under conditions of K^+ excess (Rieg et al., 2007; Pluznick and Sansom, 2006; Carrisoza-Gaytan et al., 2020). Different renal cell types have additionally been shown to assemble with different auxiliary subunits including β1, β2, and β4 (Grimm and Sansom, 2007), further indicating the diversity of BK channel function in the kidney.

Role in hypertension and neurological disorders

BK channels have been implicated in the pathogenesis of numerous diseases including hypertension (Morrow et al., 2006; Zeng et al., 2001; Fernández-Fernández et al., 2004), ischemia-reperfusion injury (Bentzen et al., 2009), erectile dysfunction (Bentzen et al., 2009), bladder incontinence (Meredith et al., 2004), seizure (Brenner et al., 2005; Sheehan et al., 2009) and movement disorders (Du et al., 2005). The specific alteration to BK channel function in the pathogenesis of disease encompasses a wide range of

possible abnormalities including changes in expression (Pacheco Otalora et al., 2008), posttranslational modifications (Pietrzykowski et al., 2008), and auxiliary subunit composition (Fernández-Fernández et al., 2004). KCNMA1 mutations and polymorphisms have been identified in human patients presenting with a wide range of symptoms particularly in the neurological system (Bailey et al., 2019; Plante et al., 2019a).

Activation of BK channels in smooth muscle leads to hyperpolarization of the cell and subsequent relaxation. For vascular smooth muscle, the net result of BK channel activation is a reduction of blood pressure (Vang et al., 2010). For this reason, activators of BK channels have been an attractive drug target for the potential treatment of hypertension (Bentzen et al., 2014; Grimm and Sansom, 2010). Furthermore, BK channels in smooth muscle are co-assembled with the β1 subunit (Nelson and Bonev, 2004). A mouse knockout of the β1 subunit has also been shown to increase arterial tone and blood pressure with a reduced Ca^{2+} sensitivity (Brenner et al., 2000). It has also been speculated that renal loss of the β1 subunit in these mice may contribute to hypertension, since these mice exhibited volume expansion along with an increase in aldosterone concentration (Grimm et al., 2009). In summary, BK channels appear to participate in important feedback mechanisms by which blood pressure is regulated. A reduction of baseline BK channel function is associated with hypertension while activation of BK channels by selective small-molecule drugs has shown the potential to reduce blood pressure.

At least 21 specific KCNMA1 mutations have been associated with human disease with varying severity (Bailey et al., 2019). There is no distinct phenotype that defines each point mutation, but rather an assortment of symptoms ranging from movement disorder, developmental delay, epilepsy, and vision/hearing deficits (Bailey et al., 2019). Both gain-of-function (GOF) and loss-of-function (LOF) mutations have been identified, with either familial inheritance or occurring de-novo (Du et al., 2005; Liang et al., 2019). The GOF mutations D434G and N999S have also been observed to cause paroxysmal

nonkinesigenic dyskinesia (PNKD). Briefly, patients have sudden attacks characterized by limited control of muscle movement. These events range in duration from a few seconds to minutes, are not triggered by movement, and can occur dozens of times each day (Bailey et al., 2019). The direct causal link between increased BK channel function in certain cell types and the development of symptoms remains incompletely understood.

The mechanisms of action of certain mutations have been uncovered. The GOF mutation D434G has been observed to increase the open probability of BK channels by increasing the apparent Ca^{2+} sensitivity (Wang et al., 2009; Yang et al., 2010; Díez-Sampedro et al., 2006). Meanwhile, the GOF mutation N999S increases the open probability by causing activation to occur more rapidly and deactivation to occur more slowly than WT (Moldenhauer et al., 2019; Plante et al., 2019b). Identifying the specific mechanism by which each mutation alters gating will be an important step in the development of future therapeutics. The observed effect of a drug as well its potential to treat human disease will therefore greatly depend on its specific mechanism of action.

Part III: Pharmacology of BK channel modulators

The function of BK channels can be altered by addition of certain peptide toxins, biologically active metabolites, ions, and synthetic drugs (Nardi and Olesen, 2008). Certain substances physically block the permeation of ions (pore blockers), to reduce ion channel function, while others alter the gating of BK channels directly by modulating either Ca^{2+}-sensitivity, voltage-sensitivity, opening of the pore directly, or coupling between these domains. In this section, I will describe the effects of several substances which are known to alter the functioning of BK channels.

Pore Blockers

Certain substances inhibit BK channel function by blocking the permeation of ions through the channel pore. Working as “biological plugs”, pore blockers are thought to work by binding to a receptor site in the channel pore to occlude K^+ entry into the conduction pore of the channel (Anderson et al., 1988; MacKinnon and Miller, 1988; Park and Miller, 1992; Giangiacomo et al., 1992; Miller, 1995). Quaternary ammonium salts like tetraethylammonium (TEA) were initially identified to selectively inhibit K^+ currents in axons of giant squid (Tasaki and Hagiwara, 1957; Armstrong and Binstock, 1965; Armstrong, 1971; Latorre and Miller, 1983; French and Shoukimas, 1981). TEA blocks most K^+ channels by binding to the to the intracellular cavity of the channel (Carrisoza-Gaytan et al., 2020; Lenaeus et al., 2005). However, in addition to exhibiting high selectivity for K^+ channels over Na^+ channels, TEA binds BK channels on the extracellular side of the channel with a much higher affinity than other K^+ channels due to interactions with Y293 near the mouth of the pore (Shen et al., 1994; Lagrutta et al., 1998; Blatz and Magleby, 1984; Yellen, 1984a; b). Use of TEA in early experiments aided the identification of Ca^{2+}-activated K^+ currents in many different tissue types and facilitated the characterization of BK channels (Meech and Standen, 1975; Meech, 1978; Blatz and Magleby, 1984; Marty and Neher, 1985; Lang and Ritchie, 1990; Schwartzkroin and Prince, 1980; McAfee and Yarowsky, 1979).

Similarly, ions such as Ba^{2+} have been shown to have an inhibitory effect on BK channels (Miller, 1987; Neyton and Pelleschi, 1991; Neyton and Miller, 1988a; b; Demo and Yellen, 1992). A single Ba^{2+} ion can enter the pore and obstruct permeation, an effect that can be observed as an increase in the frequency of long-duration close intervals (Neyton, 1996). Although Ba^{2+} has been a useful tool to study BK channel gating and ion permeation mechanisms in experimental conditions, physiologically Ba^{2+} (often used as a

contrast agent in medical imaging) is bound tightly to sulfate ions, and thus is not toxic to humans.

Due to the physiological importance of ion channels, certain plants and animals produce potent peptide toxins that selectively block certain ion channels (Quintero-Hernández et al., 2013). Venomous species of scorpions produce peptide toxins that inhibit certain ion channels with very high affinity to cause symptoms such as pain, arrhythmia, respiratory failure, neurological abnormalities, and hemodyamic shock (Sofer and Gueron, 1988; Campos et al., 1980; Carbone et al., 1982; Rogers et al., 1996; Catterall, 1977). Identified from the venom of *Leiurus quinquestriatis*, charybdotoxin (CTx) was one of the first scorpion toxins identified with inhibitory effects on BK channels (Carbone et al., 1982; Miller et al., 1985; Miller, 1995). Since CTx blocks from the external solution and increasing intracellular K^+ results in faster toxin dissociation, it was reasoned that CTx likely binds at the external end of the pore to act like a physical plug to reduce current (MacKinnon and Miller, 1988; Anderson et al., 1988). The use of pore-blocking toxins was helpful in identifying which residues of K^+ channels contribute to formation of the pore (MacKinnon and Miller, 1989; Giangiacomo et al., 2008). Subsequently, iberiotoxin was identified to block using a similar mechanism, while binding with higher affinity and exhibiting much greater selectivity for BK channels (Galvez et al., 1990; Giangiacomo et al., 1992; Candia et al., 1992). Noxiustoxin was also found to inhibit BK and K_v channels with lower affinity (Valdivia et al., 1988; Gurrola and Possani, 1995). In summary, pore blockers have been useful tools to selectively probe BK channel function to aid the characterization of these channels.

Gating Modulators

Distinct mechanisms of altered gating

Drugs can also alter the function of BK channels by modulating specific aspects of channel gating. For example, a drug has the potential to modulate the Ca^{2+}-sensor (K in **Scheme 1**), the voltage-sensor (J), the intrinsic gating of the pore (L), as well as the allosteric coupling between these domains (C, D, and E). Assessing channel activity over a range of extreme conditions is a way to experimentally determine mechanisms of action. For example, if a drug works on BK channels solely by modulating Ca^{2+}-sensing, one would not expect to observe significant changes in open probability in the absence of Ca^{2+}. In contrast, if a drug primarily modulates voltage-sensing, one would not expect substantial activation at negative voltages at which voltage sensors are primarily in the resting state. Finally, if drug activity persists at negative voltages ***and*** in the nominal absence of Ca^{2+}, this would imply action on the intrinsic gating of the pore directly. The mechanisms of action of some BK channel modulators like NS1619, Cym04, and GoSlo-SR-5-6 have been investigated previously (Bentzen et al., 2014; Hoshi and Heinemann, 2016). Learning how specific chemical entities may modulate channel gating, along with their sites of action, will be critical to rational drug design and development of therapeutics to treat specific BK channelopathies.

Inhibitors

Certain compounds can inhibit BK channels by altering aspects of channel gating. The fungal alkaloid paxilline, a member of the indole diterpene family of compounds, was identified using a competition binding assay with [^{125}I]-CTX (Knaus et al., 1994). In the initial study, eight compounds in the indole diterpene family were shown to all inhibit BK channels with low nanomolar affinity (Knaus et al., 1994). The mechanism of paxilline inhibition was initially hypothesized to be related to Ca^{2+} concentration, since paxilline

blockade could be reduced at higher concentrations of Ca^{2+} (Imlach et al., 2011). It was later determined that paxilline inhibition is instead related to open probability, where paxilline binds to the closed conformation with >500x greater affinity than the open conformation (Zhou and Lingle, 2014). Structurally, paxilline interacts near the interface between S6 and the pore-forming helix in a mechanism that does not modulate voltage sensor activation (Zhou and Lingle, 2014; Zhou et al., 2010, 2020). Additionally, unintended BK channel inhibition can be a side effect of certain FDA approved drugs, such as Ca^{2+}-channel blockers like verapamil (Harper et al., 2001), general aesthetics like ketamine (Denson and Eaton, 1994; Denson et al., 1994), antifungals like clotrimazole (Wu et al., 1999), and μ-opioid agonists like loperamide (Vouga et al., 2019).

Activators

Endogenous substances

Certain biomolecules and natural products have been shown to have an activating effect on BK channels (Tian et al., 2019). The signaling molecule PIP_2 can activate BK channels by interacting with the S6-RCK linker, in a mechanism that is enhanced by the β1 subunit (Vaithianathan et al., 2008). While the complete mechanism of PIP_2 activation of BK channels appears to be multifaceted, PIP_2 activation may exhibit some Ca^{2+}-dependencies (Tang et al., 2014). Ca^{2+}-binding to RCK1 has been proposed to dampen the activating effects of PIP_2 based on mutagenesis experiments and docking simulations (Tang et al., 2014).

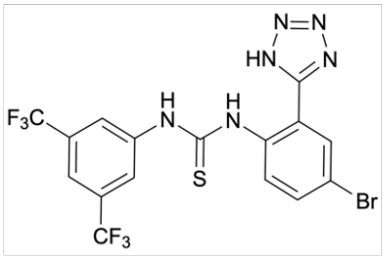
Natural Products

Additionally, Docosahexaenoic acid (DHA, **Figure 1-3**), a long-chain fatty acid found in fish oil, has been shown to reduce blood pressure by activating BK channels

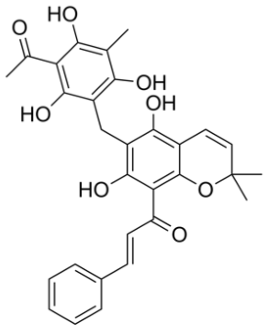
(Horrocks and Yeo, 1999; Hoshi et al., 2013). DHA was observed to increase currents by up to 20-fold in a mechanism that does not depend on the Ca^{2+}- or voltage-sensors (Hoshi et al., 2013). Administration of DHA results in more rapid activation while not significantly impacting deactivation kinetics (Hoshi et al., 2013).

Dehydrosoyasaponin-I (**Figure 1-3)**, a triterpene glycoside found in the medicinal herm *Desmodium adscendens,* was found to activate BK channels in a highly cooperative manner (McManus et al., 1993; Giangiacomo et al., 1998). Kinetic modeling suggests that Dehydrosoyasaponin-I binds to the open state with 10-20x higher affinity than the closed state, and may require binding of three or four molecules to maximally activate the channels (Giangiacomo et al., 1998). The β1 subunit is required for Dehydrosoyasaponin-I activation of BK channels (McManus et al., 1993).

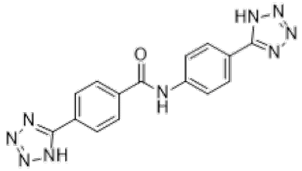
Originally identified from the *M. phillippinenis* tree in India, Kamala is red powder that has been used as a herbal medicine to treat tapeworm and skin conditions such as psoriasis (Maioli and Valacchi, 2010; Semper, 1910). It has been proposed that the usefulness of kamala to treat tapeworm comes from the paralyzing effect on motor nerves and muscle (Zakharov et al., 2005; Semper, 1910). An active ingredient of Kamala, mallotoxin (**Figure 1-3)**, has been shown to activate BK channels by a mechanism that is distinct from Ca^{2+} activation (Zakharov et al., 2005). Assembly with the β1 and β2 subunits did not affect mallotoxin activation of BK channels, however assembly with ɣ1, ɣ2, or ɣ3 subunits dampened the effect of mallotoxin on shifting the $V_{1/2}$ to more hyperpolarized voltages (Guan et al., 2017; Zakharov et al., 2005). The ability for mallotoxin activation to persist with expression of certain auxiliary β subunits while activity is decreased with expression of other auxiliary subunits underscores the importance of understanding the mechanisms of action of all known drugs as well as specific pharmacology in channels expressed with different subunits.



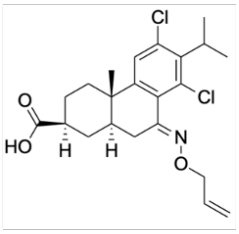
NS11021



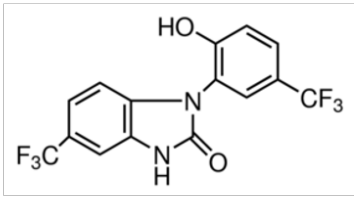
Mallotoxin



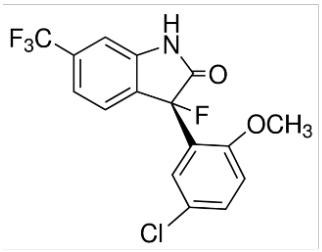
Andolast



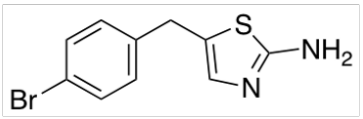
Cym04



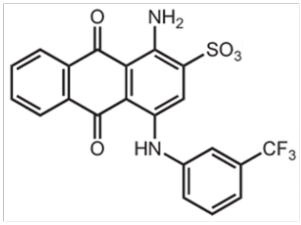
NS1619



BMS204352



NS19504



GoSlo-SR-5-6

Figure 1-3: Diversity of chemical structures among BK channel activators

Activators of BK channels are structurally diverse and likely contain multiple distinct mechanisms of action.

Ions

In addition to Ca^{2+}, other cations such as Mg^{2+} can have activating effects on BK channels. The low-affinity Ca^{2+} binding site located at the interface between the VSD and RCK domains, does not select strongly for Ca^{2+} over $Mg^{2+,}$ and is thought to be constitutively occupied by Mg^{2+}, which is present in the cytosol at 0.5 to 2 mM (Yang et al., 2007, 2008; Chen et al., 2011; Shi and Cui, 2001). Mg^{2+} appears to enhance coupling between the voltage sensor and pore in a mechanism distinct from Ca^{2+} activation (Horrigan and Ma, 2008). It was observed that 10 mM Mg^{2+} could alter G-V relations similar to 2 μM Ca^{2+}, yet had no detectible effect on open probability at negative voltages when the majority of voltage sensors were in the resting state (Horrigan and Ma, 2008).

While BK channels are able to selectively allow passage of K^+ over other physiologically relevant ions such as Na^+ and Ca^{2+}, certain less-common ions such as Tl^+, Cs^+, and Rb^+ have been experimentally observed to permeate through the pore (Piskorowski and Aldrich, 2006; Blatz and Magleby, 1984). Interestingly, substitution of external K^+ with Rb^+ decreases the outward current while simultaneously allowing channels to activate at less depolarized voltages (Demo and Yellen, 1992). This activation is accompanied by a slowing of deactivation (Mienville and Clay, 1996; Swenson and Armstrong, 1981; Thompson and Begenisich, 2012; Piskorowski and Aldrich, 2006; Demo and Yellen, 1992). The mechanism of activation for Rb^+ and other permeant ions stabilizing the open state to increase open probability has been described as a 'foot in the door' activator (Swenson and Armstrong, 1981).

Synthetic drugs

Small molecule synthetic activators of BK channels comprising a wide range of chemical structures are currently known (**Figure 1-3**). The company NeuroSearch has

identified multiple compounds with BK activating effects including the benzimidazolones NS004 and NS1619, the biarylthiourea NS11021, and the thiazolidine NS19504 (Olesen et al., 1994a; b; Bentzen et al., 2014, 2007; Nausch et al., 2014; Calderone et al., 2008). Furthermore, other small molecules include the GoSlo family of openers, the droabietic acid derivative Cym04, the aryloxindole BMS-204352, and the tetrazolyl-benzamido derivative Andolast (CR-2039) (Roy et al., 2014; Large et al., 2015; Webb et al., 2015; Gessner et al., 2012; Jensen, 2002; Cheney et al., 2001; Malerba et al., 2015). Although these small-molecule activators have shown therapeutic potential in animal models for treatment of a wide range of conditions, only Andolast is currently in development and has been show to improve airflow in asthmatic patients in a dose dependent manner (Bentzen et al., 2014; Malerba et al., 2015).

Although the structures of Cym04 and NS1619 are different, it has been proposed that these drugs work by a similar mechanism of action involving the S6-RCK linker (Gessner et al., 2012). A neuronal BK splice variant (slo1_9a) that contains alternative splicing in the S6-RCK linker is insensitive to Cym04, and activation by Cym04 can be abrogated by a two-residue deletion in this linker. Since Cym04 activation persisted in the nominal absence of Ca^{2+} and at negative voltages when the majority of voltage sensors are at rest, the authors concluded that it Cym04 is unlikely mediated by alterations to the Ca^{2+} sensor and likely is due to facilitating voltage-sensor activation or pore opening directly (Gessner et al., 2012). It has been suggested in a separate structure-activity study that the activity of droabietic derivatives (Cym04) on activating BK channels relates to the activity of a carboxyl group and hydrophobic moiety (Ohwada et al., 2003).

An anthraquinone derivative, GoSlo-SR-5-6, was shown to shift the $V_{1/2}$ around 100 mV towards more hyperpolarized voltages when 1 μM was applied to the cytosolic face of the channel (Roy et al., 2014). Coupled with an EC_{50} value of 251 nM, GoSlo-SR-

5-6 is one of the most potent and efficacious BK channel openers identified to date (Roy et al., 2014). In terms of mechanism, it was shown that mutations to the S4-S5 linker and S6 abolished the activating activity of GoSlo-SR-5-6 (Webb et al., 2015). Furthermore, a truncated construct with complete removal of the Ca^{2+} sensors could still be activated by GoSlo-SR-5-6, suggesting a mechanism that is not dependent on the Ca^{2+}-sensing RCK domains. At negative voltages when the majority of voltage sensors are at rest and in the nominal absence of Ca^{2+}, GoSlo-SR-5-6 increased Po by over 1,000 fold at -140 mV (Webb et al., 2015). By modeling with the dual-allosteric model (Scheme 1), the authors suggest that the principle effect of GoSlo-SR-5-6 is on increasing the pore closed-to-open equilibrium constant (L_0) by 700-fold, but do not rule out smaller impacts to other voltage-dependent parameters of gating (Webb et al., 2015).

Part IV: Hypothesis

Ion channels are the second-most common target class of all FDA-approved drugs (Santos et al., 2016). Meanwhile, K^+ channels represent the largest superfamily of ion channels and participate in a wide-range of physiological processes. BK channels provide an important link between cytosolic Ca^{2+} and membrane depolarization, yet there are currently no FDA-approved drugs that specifically target BK channels. While BK channels are structurally similar to K_v channels, the existence of unique structural elements such as the S0 helix and the Ca^{2+}-binding RCK domains provide the opportunity for relatively specific BK channel activators and inhibitors. Furthermore, currently known activators exhibit large structural diversity.

The hypothesis of this work is that multiple distinct mechanisms of pharmacological activation of BK channels may exist due to the large diversity in chemical structures of small-molecule activators (**Figure 1-3**). It will be important to determine the specific

mechanisms by which BK channel gating is altered by each family of drugs in order to potentially identify novel mechanisms of BK channel activation.

To test this hypothesis, I will further establish the distinct mechanisms by which a drug can activate BK channels in terms of effects on each gating module (Ca^{2+}-sensor, voltage-sensor, and pore), effects on activation and deactivation kinetics, and effects on channel open probability over a range of conditions. I will first characterize the small-molecule activator NS11021 and compare it to currently known mechanisms of BK channel activation. NS11021 has been demonstrated to be a relatively potent vasodilator and been tested in animal models for the treatment ischemia-reperfusion injury (Bentzen et al., 2009), bladder instability (Layne et al., 2010), and erectile dysfunction (Kun et al., 2009; Cui, 2020). Next, I investigate whether NS11021 can mitigate effects on channel gating caused by a loss-of-function BK channel mutation associated with human disease, P805L. To do so, I characterize a loss-of-function BK channel mutation P805L that has been associated with human disease. By initially determining how the gating of a mutant BK channel is altered and by subsequently measuring the extent by which BK currents can be ameliorated using small-molecule activators, I begin to investigate whether currently known activators may have therapeutic potential for patients with a KCNMA1 channelopathy.

CHAPTER 2

CHARACTERISTICS OF BK CHANNEL ACTIVATION BY NS11021

Introduction

As noted above, the role of BK channels in control of vascular tone is well established; thus targeting of BK channels by pharmacological activators may have the potential to treat conditions such as ischemia-reperfusion injury, overactive bladder, and erectile dysfunction. In addition, BK activators may be useful as therapeutics in disease where decreased BK channel function is an underlying factor, and these activators may also serve as research tools to understand and probe molecular mechanisms of channel gating (Bentzen et al., 2014; Kshatri et al., 2018; Hoshi and Heinemann, 2016).

It was recently suggested that the BK channel openers Cym04 and NS1619 may act through a mechanism that involves binding to the S6-RCK linker, since a BK splice variant with alternative splicing in the S6-RCK linker is insensitive to Cym04 (Gessner et al., 2012). In addition to the structurally distinct activators NS1619 and Cym04, several other compounds comprising a diverse range of chemical classes are known BK channel activators, including the natural products mallotoxin and dehydroxysoyasaponin, and synthetic compounds such as BMS-204352 and NS11021. Compounds comprising such a wide array of structures may contain distinct pharmacophores, and may thus have distinct structural or functional actions that may be exploited for therapeutic purposes. It will be important to identify molecular mechanisms for each of these diverse compounds to improve understanding of BK channel gating mechanisms (Horrigan et al., 1999; Horrigan and Aldrich, 2002; Rothberg et al., 1997; Rothberg and Magleby, 1999; Cox et al., 1997; Cui and Aldrich, 2000).

In this chapter, the mechanism of a potent and relatively specific BK channel activator, NS11021, will investigated using patch-clamp electrophysiology. NS11021 has previously been studied in animal models for treatment of ischemia-reperfusion injury and erectile dysfunction through its vasodilatory effects (Kun et al., 2009; Bentzen et al., 2007, 2009; Cui, 2020; Coskun et al., 2020). We found that NS11021 shifts the voltage-dependence of activation primarily through slowing of the overall channel closing rate. This action of NS11021 persists in the nominal absence of Ca^{2+} in hslo1-WT channels, as well as in truncated BK channels from which the CSD has been deleted (slo1c-Kv-minT). Addition of NS11021 leads to an increase BK channel open probability (Po) at negative voltages under conditions where the VSD is largely in the resting state, consistent with the idea that NS11021 is acting at the PGD. Partial results of the presented work have been published in (Rockman et al., 2020).

Materials and Methods

Electrophysiology

Experiments were performed using excised inside-out patches from human embryonic kidney cells (HEK-293T; American Type Culture Collection) stably-transfected with human BK channel α-subunit cDNA (referred to as HF1 cells; Meera et al., 1997), obtained from the laboratory of Dr. R. Aldrich (Univ of Texas, Austin, TX). Slo1c-Kv-minT (in pcDNA3.3; gift of Dr. T. Giraldez, Univ of La Laguna, Spain; Budelli et al., 2013; Webb et al., 2015), was overexpressed in HEK-293T cells following transient transfection by electroporation.

Except where noted, solutions bathing both sides of the membrane contained 160 mM KCl and 10 mM HEPES (pH 7.4). For data in the presence of 1 or 10 μM free Ca^{2+}, solutions at the cytoplasmic face of the patch additionally contained 2 mM N-(2-

hydroxyethyl)-ethylene-diamine-triacetic acid (HEDTA), with 0.394 mM or 1.429 mM $CaCl_2$ added to bring the free [Ca^{2+}] to the indicated levels respectively for pH 7.4, T = 22°C, and ionic strength = 0.16 (Bers et al., 2010). For patches with 100 μM free Ca^{2+}, the solution contained 2 mM nitrilotriacetic acid (NTA) as a calcium buffer, with 0.972 mM $CaCl_2$. For experiments at nominally 0 free Ca^{2+}, the solution contained 5 mM EGTA with no added $CaCl_2$ (free [Ca^{2+}] estimated to be less than 0.3 nM (Bers et al., 2010)).

NS11021 was included in solutions applied to the cytoplasmic face of the patch. NS11021 powder (Tocris Bioscience, Minneapolis, MN) was dissolved in DMSO to make a 100 mM stock solution, which was then aliquotted and further dissolved to yield a series of less concentrated stock solutions, so that the final concentration of DMSO in all experiments was 0.1% (v/v). All experiments performed in the absence of NS11021 also included 0.1% DMSO (v/v, final concentration), which was observed to have no effect on BK channel gating compared to experiments performed in DMSO-free solutions.

Patch clamp experiments were performed at room temperature (22°C), using a Dagan PC-ONE amplifier controlled by pClamp9 software. Solutions at the cytosolic face of the excised inside-out membrane patch were typically changed multiple times during the course of each experiment in a gravity-fed perfusable recording chamber, to measure BK channel activity over a range of [Ca^{2+}] and [NS11021]. To minimize voltage errors arising from series resistance, we analyzed only recordings in which the maximal current was <4 nA. We estimate that the maximal voltage error contributed by series resistance for these recordings was 6 mV, and analyzed data are presented without correction for series resistance.

Data Analysis

Following patch excision, channel activity was assessed by acquisition of currents in response to a series of voltage steps, from a holding voltage of typically -90 mV to test voltages ranging from -80 to 250 mV (100 ms) to yield a range of activity levels, and then back to typically -60 mV to elicit a tail current. Each voltage protocol was typically repeated at least five times for each patch, and current traces were digitized and averaged in real time using pClamp9 software. Tail current amplitudes were measured, plotted as a function of test voltage, and fitted with a Boltzmann equation:

$$G/G_{max} = \frac{1}{1 + \exp\left(\frac{-z(V - V_{1/2})}{k_B T}\right)}$$

(Eq. 2.1)

where G/G_{max} is the normalized tail current amplitude, z is the effective gating valence, $V_{1/2}$ is the voltage at half-maximal activation, k_B is Boltzmann's constant, and T is temperature.

The time constant of current relaxation (τ) was quantified over a range of voltages for both channel activation and deactivation using standard voltage-step protocols (Horrigan et al., 1999). For activation kinetics, patches were initially held at a hyperpolarized voltage at which channels are mainly closed, before stepping to a series of depolarizing voltages; for deactivation kinetics, patches were conversely held at depolarized voltage to maximally activate the channels, followed by steps to a series of hyperpolarizing voltages. Currents were fitted directly with a single exponential function

$$I = Aexp\left(\frac{-t}{\tau}\right) + y_0$$

(Eq. 2.2)

where I is the current, A is the amplitude, t is time, τ is the time constant of relaxation, and y_0 is the level of steady-state current. We have found that under the conditions of our experiments, BK current relaxations were described well by a single exponential over a wide range of voltages, as observed previously (**Figure 2-1**) (Horrigan et al., 1999; Cui et al., 1997).

For single-channel recordings, NPo and dwell times were measured using 50% threshold detection and log-binned at 10 bins per log unit using pClamp9 (Sigworth and Sine, 1987; McManus et al., 1987). Dwell time histograms were fitted with sums (mixtures) of exponential components from a lower limit of twice the system dead time (0.036 ms) using the maximum likelihood method (Sigworth and Sine, 1987; Horn and Lange, 1983; McManus et al., 1987). The number of active channels (N) per patch was estimated by dividing the macroscopic tail current by single channel current amplitude (I/i) (Koval et al., 2007; Wang et al., 2006, 2009). The open probability per channel (Po) was calculated by dividing the measured NPo by N.

The relation between [NS11021] and channel activity was analyzed by plotting G/Gmax as a function of [NS11021] for a given voltage and fitting with a Hill equation,

$$G/G_{max} = min + \frac{1 - min}{1 + \left(\frac{EC_{50}}{[NS11021]}\right)^{n_H}}$$

(Eq. 2.3)

where min = the minimum G/G_{max} at that voltage, EC_{50} is the concentration at half maximal response, and n_H is the Hill coefficient. To obtain reasonably well-constrained parameter

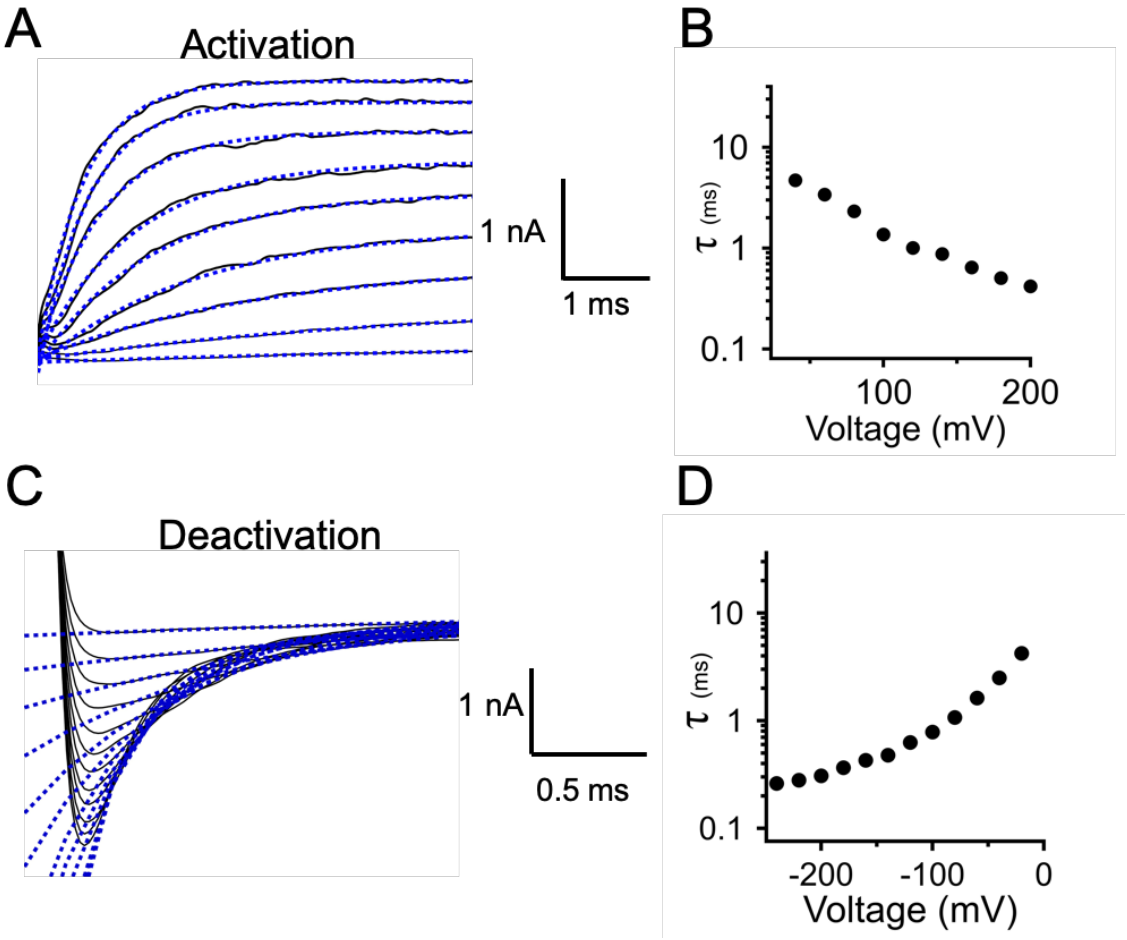


Figure 2-1: BK channel activation and deactivation time course

Representative fits of BK channel activation and deactivation time course with a single exponential for currents with 10 μM Ca^{2+} at the cytosolic side of the patch. **A,** Representative macroscopic currents (black lines) , activated by voltage step from -100 mV to between +40 mV and +200 mV (20 mV increments). Single exponential fits (Eq. 2.2) are shown as blue dashed lines. **B,** Time constants corresponding to the fits in panel A, plotted as a function of voltage. **C,** Representative macroscopic currents (black lines), driven by voltage steps from +100 mV to between -20 mV and -240 mV (20 mV increments). Single exponential fits are shown as blue dashed lines. **D,** Time constants corresponding to the fits in panel C plotted as a function of voltage.

estimates, this analysis was limited to data sets (voltages) in which the minimum G/G_{max} in the absence of NS11021 was less than 0.5, and the highest G/G_{max} in the presence of NS11021 was > 0.5.

Data across different patches are presented as means ± SEM. The experimental data presented represent results from a total of 160 different patches.

Results

Effects of NS11021 on BK channel activation

To quantify the effect of NS11021 on the voltage-dependence of activation, we determined G-V relations from macroscopic recordings of BK currents over a range of $[Ca^{2+}]$ and [NS11021] at the cytosolic side of the membrane, from membrane patches excised from HEK293T cells overexpressing BK channels. **Figure 2-2** illustrates the effect of NS11021 with 10 μM Ca^{2+} at the cytosolic side of the membrane. We found that increasing [NS11021] results in an increasing leftward shift of the G-V relation, for [NS11021] greater than 0.1 μM. Fitting these G-V relations for channels in the presence of 10 μM Ca^{2+} with Boltzmann equations yielded $V_{1/2}$ values (in mV) of: 23.4 ± 2.0 (n=20), 23.1 ± 4.6 (n=7), 7.8 ± 2.3 (n=7), -8.2 ± 2.3 (n=6), and -38.5 ± 3.4 (n=6), for 0, 0.1, 1, 10, and 30 μM NS11021, respectively. These leftward shifts in $V_{1/2}$ were not associated with changes in the effective gating valence (z) of G-V relations. The z values for each [NS11021] at 10 μM Ca^{2+} (in e_0) were: 1.29 ± 0.05, 1.13 ± 0.10, 1.22 ± 0.07, 1.22 ± 0.05, and 1.00 ± 0.05, for 0, 0.1, 1, 10, and 30 μM NS11021, respectively. Because the leftward shifting of $V_{1/2}$ values does not reach a “plateau” at the highest [NS11021] used in these experiments, it seems that 30 μM NS11021 is not sufficient to produce a maximal effect on the channel. However, the limited solubility of the drug in aqueous solutions posed a barrier to measurements at [NS11021]>30 μM. Nevertheless under these conditions,

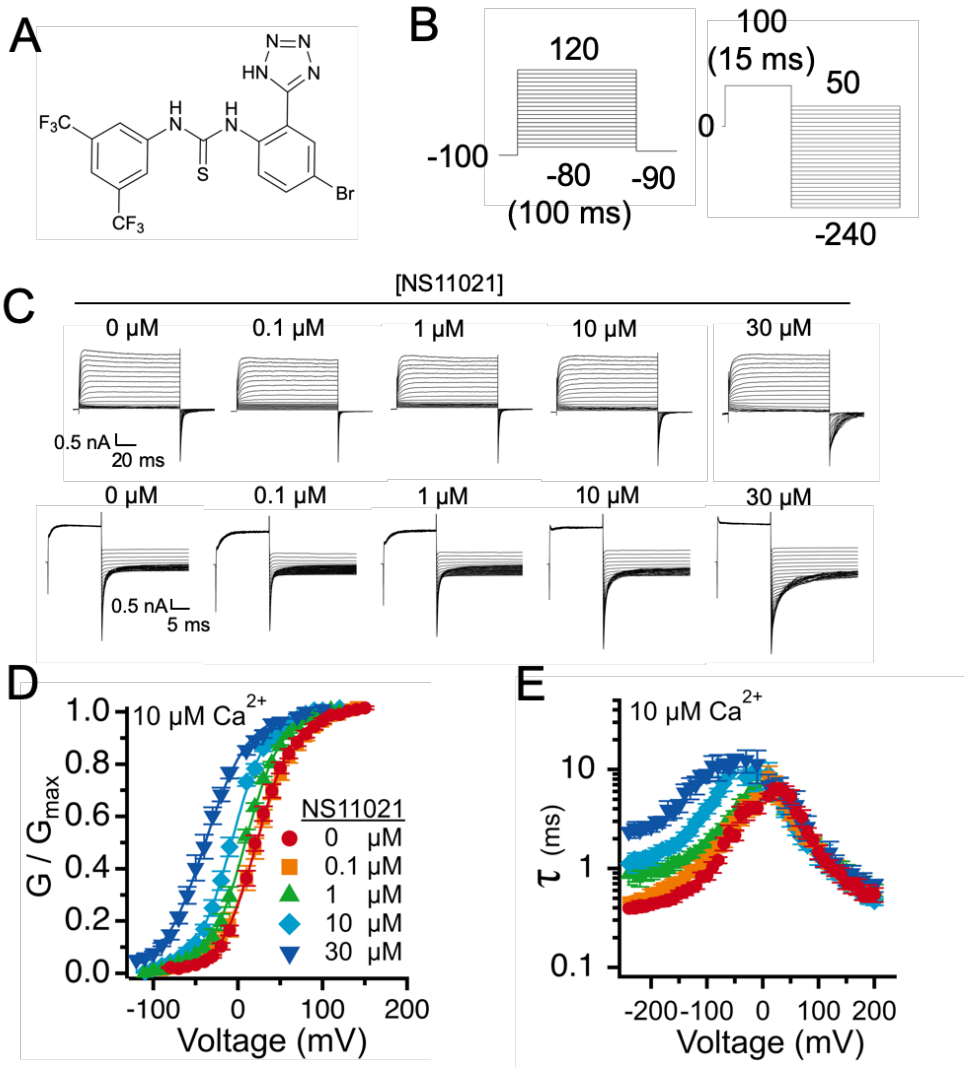


Figure 2-2: Effects of NS11021 on BK channel gating

A, Structure of NS11021. **B,** Representative voltage protocols for channel activation (left) and deactivation (right). Numbers correspond to voltage (in mV) and duration of activating pulse (in ms). **C,** Representative BK currents from an inside-out patch with 10 μM Ca^{2+} and the indicated [NS11021] at the cytosolic side of the membrane. "0 μM NS11021" experiments contained 0.1% DMSO (vehicle) with no added NS11021. **D,** Conductance vs. voltage (G-V) relations from patches with 10 μM Ca^{2+} and the indicated [NS11021]. Solid lines represent fits with a Boltzmann equation. Values for Boltzmann equation parameters are provided in Table 2-1. **E,** Time constant of relaxation (τ) vs. voltage (10 μM Ca^{2+} and the indicated [NS11021]). Time constants were estimated by fitting activation and deactivation traces with a single exponential (Eq. 2-2). Increasing [NS11021] primarily slows channel deactivation.

Table 2-1: Mean values of $V_{1/2}$ and z determined from Boltzmann fits of individual G-V relations

[NS11021] (μM)	0 Ca^{2+} $V_{1/2}$ (in mV), (n)	1 μM Ca^{2+} $V_{1/2}$ (in mV), (n)	10 μM Ca^{2+} $V_{1/2}$ (in mV), (n)	100 μM Ca^{2+} $V_{1/2}$ (in mV), (n)
0	158 ± 2 (54)	110. ± 2 (26)	23 ± 2 (20)	-41 ± 4 (15)
0.1	145 ± 6 (12)	106 ± 3 (14)	23 ± 5 (7)	-55 ± 5 (10)
1	130. ± 4 (4)	91 ± 6 (11)	8 ± 2 (7)	-72 ± 5 (7)
10	89 ± 3 (4)	68 ± 4 (9)	-8 ± 2 (6)	-84 ± 4 (4)
30	73 ± 6 (14)	43 ± 4 (7)	-38 ± 3 (6)	-89 ± 6 (10)
[NS11021] (μM)	0 Ca^{2+} z (e_0)	1 μM Ca^{2+} z (e_0)	10 μM Ca^{2+} z (e_0)	100 μM Ca^{2+} z (e_0)
0	1.4 ± 0.1	2.0 ± 0.1	1.3 ± 0.1	1.3 ± 0.1
0.1	1.5 ± 0.1	2.0 ± 0.1	1.1 ± 0.1	1.0 ± 0.1
1	1.7 ± 0.1	1.8 ± 0.2	1.2 ± 0.1	1.1 ± 0.2
10	1.5 ± 0.3	1.5 ± 0.1	1.2 ± 0.1	1.0 ± 0.2
30	1.3 ± 0.4	1.5 ± 0.1	1.0 ± 0.1	1.2 ± 0.2

increasing [NS11021] substantially activates BK channels. Based on our current understanding of BK channel gating, the leftward shifts in $V_{1/2}$ observed with 10 μM Ca^{2+} could arise through the drug acting at the VSD, CSD, or PGD, or at positions that effect energetic coupling between domains, mechanisms that will be explored below.

NS11021 slows channel deactivation time course with little effect on activation

The leftward shift in G-V relation could be attributed to an action of NS11021 on voltage-dependent BK channel activation, deactivation, or both. To gain further insight toward the underlying mechanism, we analyzed the time course of channel activation and deactivation as a function of voltage over a range of [NS11021], using increasing depolarizing voltage steps applied to channels held in the closed state, and increasing hyperpolarizing steps applied to channels held in the open state, respectively (**Figure 2-2 B,C**). As observed previously, macroscopic BK current kinetics were observed to be essentially monoexponential, and currents were fitted directly with single exponential functions to estimate activation or deactivation time constant (τ) (**Figure 2-1**) (Horrigan et al., 1999; Cui et al., 1997).

At 10 μM Ca^{2+}, we observed that increasing [NS11021] resulted primarily in marked slowing of tail current decays, which was apparent at voltages more negative than the activation $V_{1/2}$ (**Figure 2-2C**, top traces). Time constants of BK current relaxation determined by exponential fitting were plotted as a function of voltage for each [NS11021] (**Figure 2-2E**). From these data it was apparent that increasing [NS11021] elicited a marked slowing of channel deactivation of a range of hyperpolarizing voltages. For channel deactivation, the value for τ (in ms) at -200 mV in 10 μM Ca^{2+} for each [NS11021] was: 0 μM = 0.44 ± 0.04 (n=9), 0.1 μM = 0.55 ± 0.07 (n=7), 1 μM = 0.92 ± 0.15 (n=5), 10 μM = 1.29 ± 0.22 (n=4), 30 μM = 2.66 ± 0.31 (n=5).

In contrast, time constants for channel activation measured at depolarized voltages were affected relatively little with increasing [NS11021]. The value for τ (in ms) at +200 mV in 10 μM Ca^{2+} for each [NS11021] was: 0 μM = 0.55 ± 0.09, 0.1 μM = 0.79 ± 0.23, 1 μM = 0.59 ± 0.10, 10 μM = 0.47 ± 0.04, 30 μM = 0.68 ± 0.10. These results suggest a mechanism in which NS11021 acts on a region of the channel that underlies a distinct subset of transitions in the gating pathway.

The results with 10 μM Ca^{2+} establish benchmarks for further analysis of BK channel activation by NS11021. To determine the relation between the effects of NS11021 and activation of the Ca^{2+}-sensing domain (CSD), we measured G-V and τ vs. V relations for BK channels over a range of [NS11021] at several [Ca^{2+}], ranging from nominally 0 Ca^{2+} (buffered with 5 mM EGTA) to 100 μM Ca^{2+}. **Figure 2-3** illustrates that under each condition of 0 through 100 μM Ca^{2+}, increasing [NS11021] yields qualitatively similar effects, namely a leftward shift in activation $V_{1/2}$, and slowing of time constants for voltage-dependent deactivation, but essentially no NS11021-dependent change in voltage-dependent activation time course. The observation that NS11021 elicits similar dose-dependent shifts in activation $V_{1/2}$ and slowing of deactivation time constants at nominally 0 Ca^{2+} as it does with higher Ca^{2+} is consistent with the ideas that (1) NS11021 does not act primarily through CSD activation, and (2) NS11021 is not Ca^{2+}-mimetic. If one compares the action of NS11021 at any single [Ca^{2+}] vs. the action of Ca^{2+} at any single [NS11021], one can see that increasing [Ca^{2+}] results in hyperpolarizing shifts in $V_{1/2}$ that are associated with *both* speeding up of the activation time course and slowing of the deactivation time course, actions that together are distinct from that of NS11021. In addition, the asymptotic slowing of deactivation kinetics at very hyperpolarized voltages (<-200 mV) over the entire examined range of [Ca^{2+}] with increasing NS11021 is

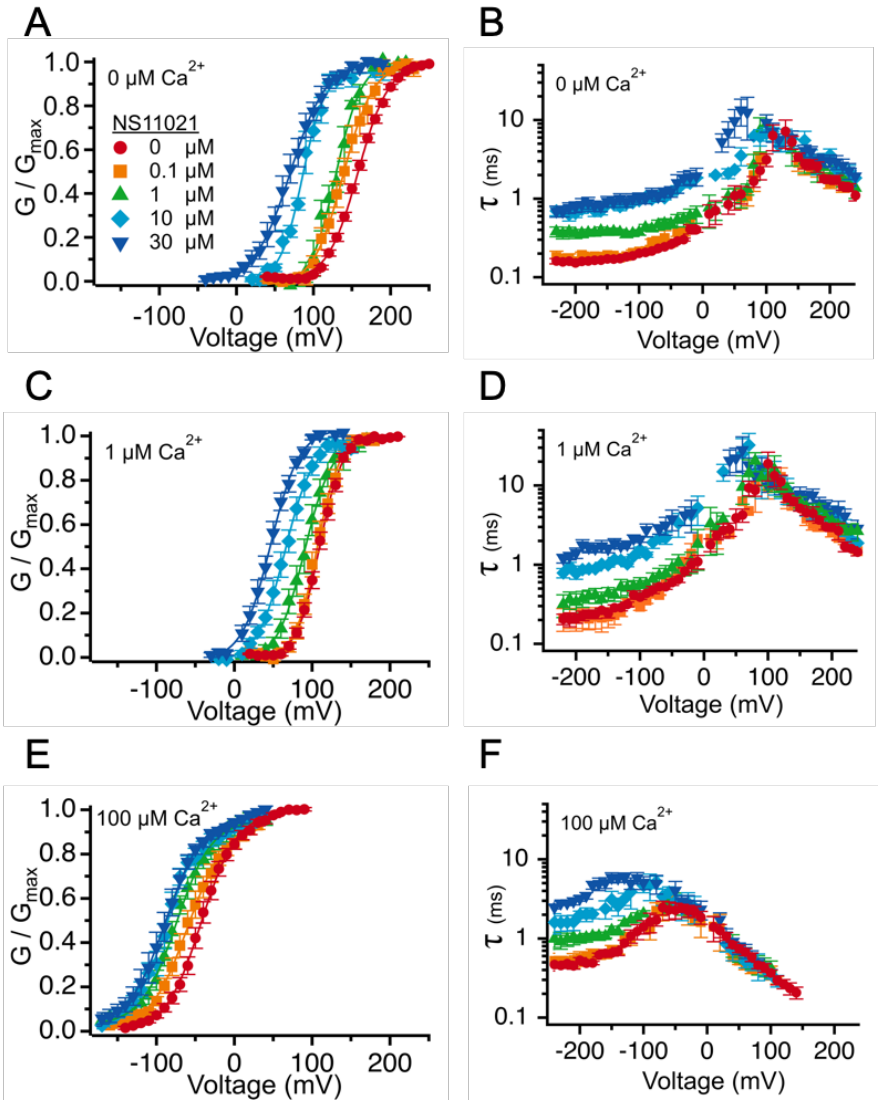


Figure 2-3: NS11021 activates BK channels over a range of [Ca^{2+}]

A, Conductance vs. voltage (G-V) relations from patches with nominally 0 μM Ca^{2+}. In panels A, C, and E, solid lines represent fits with a Boltzmann equation; parameters can be found in Table 2-1. **B,** Time constant (τ) vs. voltage from patches with nominally 0 μM Ca^{2+}. Time constant at -200 mV (τ_{-200}) for each [NS11021]: 0 μM, 0.15 ± 0.01 ms (n=33); 0.1 μM, 0.16 ± 0.02 ms (n=11); 1 μM = 0.36 ± 0.05 ms (n=7); 10 μM, 0.65 ± 0.08 ms (n=5); 30 μM, 0.79 ± 0.12 ms (n=14). **C,** G-V relations from patches with 1 μM Ca^{2+}. **D,** τ vs. voltage from patches with 1 μM Ca^{2+}. τ_{-200} for each [NS11021]: 0 μM, 0.21 ± 0.03 ms (n=21); 0.1 μM, 0.20 ± 0.06 ms (n=8); 1μM, 0.34 ± 0.09 ms (n=10); 10 μM, 0.78 ± 0.12 ms (n=7); 30 μM, 1.41 ± 0.21 ms (n=11). **E,** G-V relations from patches with 100 μM Ca^{2+}. **F,** τ vs. voltage from patches with 100 μM Ca^{2+}. τ_{-200} for each [NS11021]: 0 μM, 0.53 ± 0.04 ms (n=11); 0.1 μM, 0.56 ± 0.12 ms (n=8); 1μM, 0.99 ± 0.11 ms (n=6); 10 μM, 2.01 ± 0.22 ms (n=6); 30 μM = 3.22 ± 0.25 ms (n=9).

consistent with the idea that NS11021 is acting on BK channels when VSD's are in the resting state.

NS11021 activation does not require the CSD

Although the activating effect of NS11021 on BK channels in nominally 0 Ca^{2+} suggests that the action of NS11021 does not require CSD activation, we hypothesized that the CSD could yet play a structural role in NS11021 action, perhaps through binding at the CSD-VSD interface. To test this we assayed the effect of NS11021 in BK channels lacking the CSD, using the construct Slo1C-Kv-minT (**Figure 2-4A**) (Budelli et al., 2013). In this construct, the BK channel is truncated at the end of the C-linker following the PGD (at residue R342) to result in a channel with voltage-activation properties similar to wild-type BK channels in nominally 0 Ca^{2+}. An 11-residue sequence derived from the C-terminal region of the Kv1.4 channel is added to the C-terminus of Slo1C-Kv-minT to enable its trafficking to the plasma membrane. The resulting channel is non-responsive to Ca^{2+} while exhibiting similar voltage-dependence to BK channels. **Figure 2-4** illustrates that in nominally 0 Ca^{2+}, voltage-dependent activation of Slo1C-Kv-minT channels is shifted toward hyperpolarizing voltages by NS11021. We evaluate these Slo1C-Kv-minT recordings with caution, as the requirement for very depolarized activating voltages (≥ 200 mV) combined with relatively low macroscopic current amplitudes (~100 pA) contributed to experimental variability. Nonetheless, we observed that the 75 mV shift elicited by 30 μM NS11021 to Slo1C-Kv-minT channels was not significantly different from the approximately 85 mV hyperpolarizing shift elicited by 30 NS11021 in wild-type BK channels in nominally 0 Ca^{2+} ($p=0.43$, two-tailed t-test), consistent with the idea that NS11021 activation does not require the presence of the CSD. Additionally, we observed

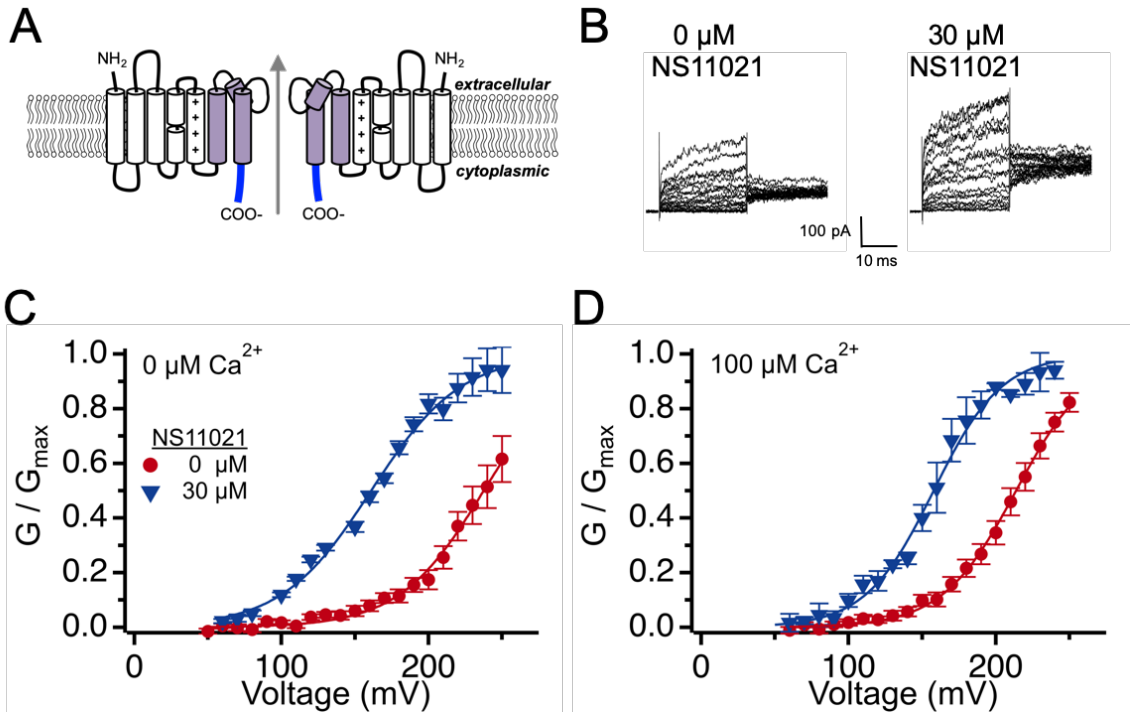


Figure 2-4: Effects of NS11021 on truncated BK channels

A, Topology of the truncated BK channel construct Slo1c-Kv-MinT, showing two of the four domains side-by-side. Arrow illustrates the permeation pathway. The VSD (S0-S4, white), PGD (S5-S6, purple), and 11 residues from the Kv1.4 tail (blue) are sufficient for trafficking of these voltage-gated channels to the plasma membrane. **B,** Representative Slo1c-Kv-MinT currents (nominally 0 Ca^{2+}). Patches were held at 0 mV and stepped to voltages ranging from +50 mV to +250 mV, followed by a step to +160 mV for tail current measurement. **C,** G-V relations from Slo1c-Kv-MinT at nominally 0 Ca^{2+}, with 0 (red) or 30 μM NS11021 (blue). $V_{1/2}$ was shifted from 238 ± 8.1 mV (n=13) to 161 ± 2.6 mV (n=7) with addition of 30 μM NS11021. **D,** G-V relations from Slo1c-Kv-MinT at 100 μM Ca^{2+}, with 0 (red) or 30 μM NS11021 (blue). For these data, $V_{1/2}$ shifted from 214 ± 3.4 mV (n=24) to 158 ± 2.0 mV (n=5) with addition of 30 μM NS11021.

that voltage-activation of Slo1C-Kv-minT channels is not substantially affected by the presence of 100 μM Ca^{2+} at the cytosolic side of the patch, and under these conditions 30 μM NS11021 elicited a similar shift in $V_{1/2}$ to 158 mV. Together these results support the idea that action of NS11021 does not absolutely require CSD activation, and further suggest that presence of the CSD is not a structural requirement for NS11021 binding or activation.

NS11021 activation may occur primarily through the PGD

If NS11021 action does not require the CSD, then we reasoned that it may occur through the VSD, PGD, and/or through an effect on VSD-PGD coupling. To distinguish among these possibilities, we quantified BK channel gating in nominally 0 Ca^{2+} at negative voltages (≤-60 mV), to drive the channel toward open and closed conformations in which both the CSD and VSD are largely at rest. Under these conditions, BK channels gate with very low open probability (Po), such that single-channel openings can be resolved in patches containing many (~20 to 200) active channels (**Figure 2-5A-D**) (Koval et al., 2007; Horrigan and Aldrich, 2002; Wang et al., 2006, 2009). In nominally zero Ca^{2+} and with the VSD at rest, 30 μM NS11021 increased Po by 62 fold, from (3.5 ± 0.8) x 10^{-6} (n=8) at -80 mV without drug, to (2.2 ± 0.6) x 10^{-4} (n=8) with 30 μM NS11021. This result is consistent with the idea that NS11021 action does not require either CSD or VSD activation, and may thus occur largely through an effect at the PGD.

Under these conditions, it was possible to directly measure the durations of BK channel openings; with nominally 0 Ca^{2+} and at -80 mV, the mean open time was 0.15 ± 0.01 ms (n=5) and this increased to 0.52 ± 0.07 ms (n=5) with addition of 30 μM NS11021 (**Figure 2-6 A,C**). This increase in mean open time was associated with a relative increase in the frequency of openings (i.e. a decrease in mean closed time), although mean closed

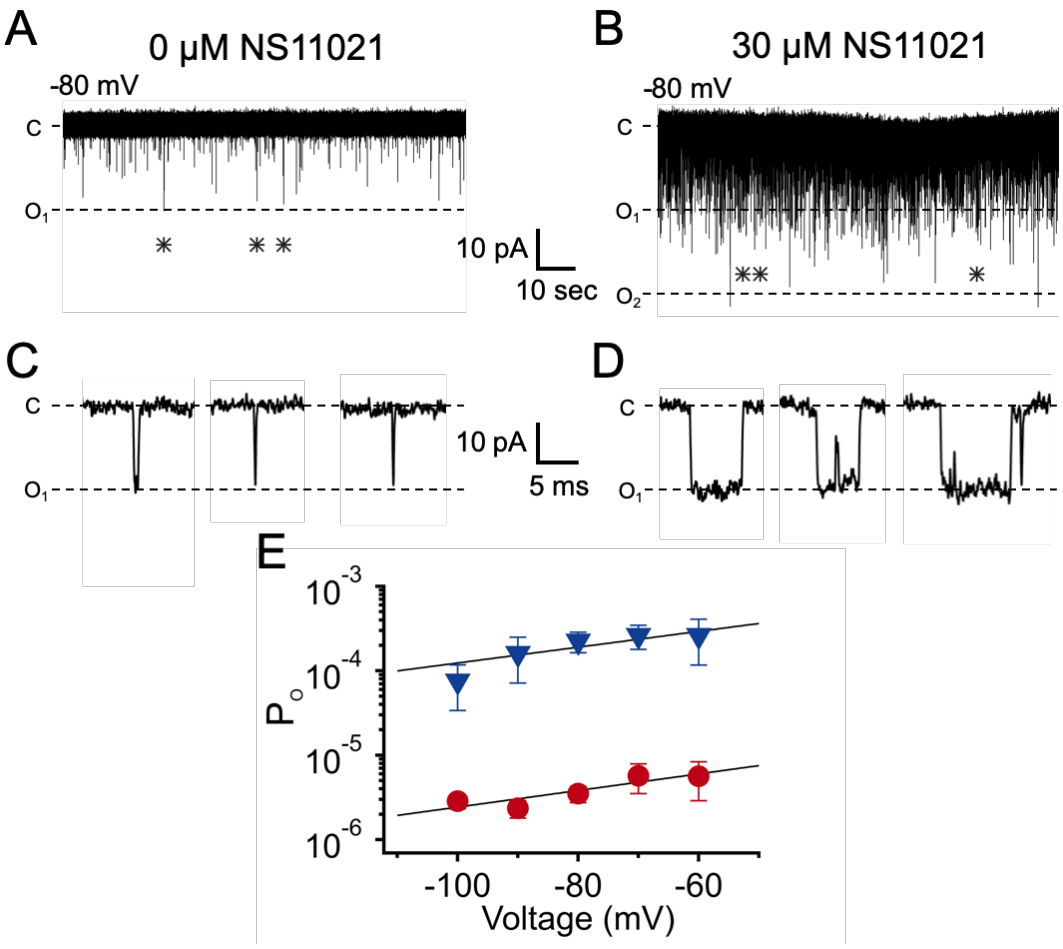


Figure 2-5: NS11021 increases open probability at negative voltages in nominally 0 Ca^{2+}

A, Representative BK currents in nominally 0 Ca^{2+} at -80 mV, with 0 μM NS11021. $NP_o = 2.5 \times 10^{-4}$. **B,** BK currents from the same patch as in panel A, following addition of 30 μM NS11021. $NP_o = 1.5 \times 10^{-2}$. Patch estimated to have 127 active channels, determined by dividing the maximum macroscopic current amplitude by the single channel conductance. **C, D,** Representative channel openings from the traces in panels A and B, respectively (from positions indicated by *), at an expanded time scale. **E,** Po vs. voltage for 0 μM (red circles) and 30 μM NS11021 (blue triangles) activity. Data points represent mean Po ± SEM from 3 to 8 experiments. Lines represent best fit with $P_o = L_0 \exp(-z_L V/k_B T)$. Parameters were: 0 μM NS11021 ($z_L = 0.57\ e_0$, $L_0 = 2.3 \times 10^{-5}$); 30 μM NS11021 ($z_L = 0.55\ e_0$, $L_0 = 1.1 \times 10^{-3}$). Estimated numbers of channels in these multi-channel patches ranged from 22 to 199.

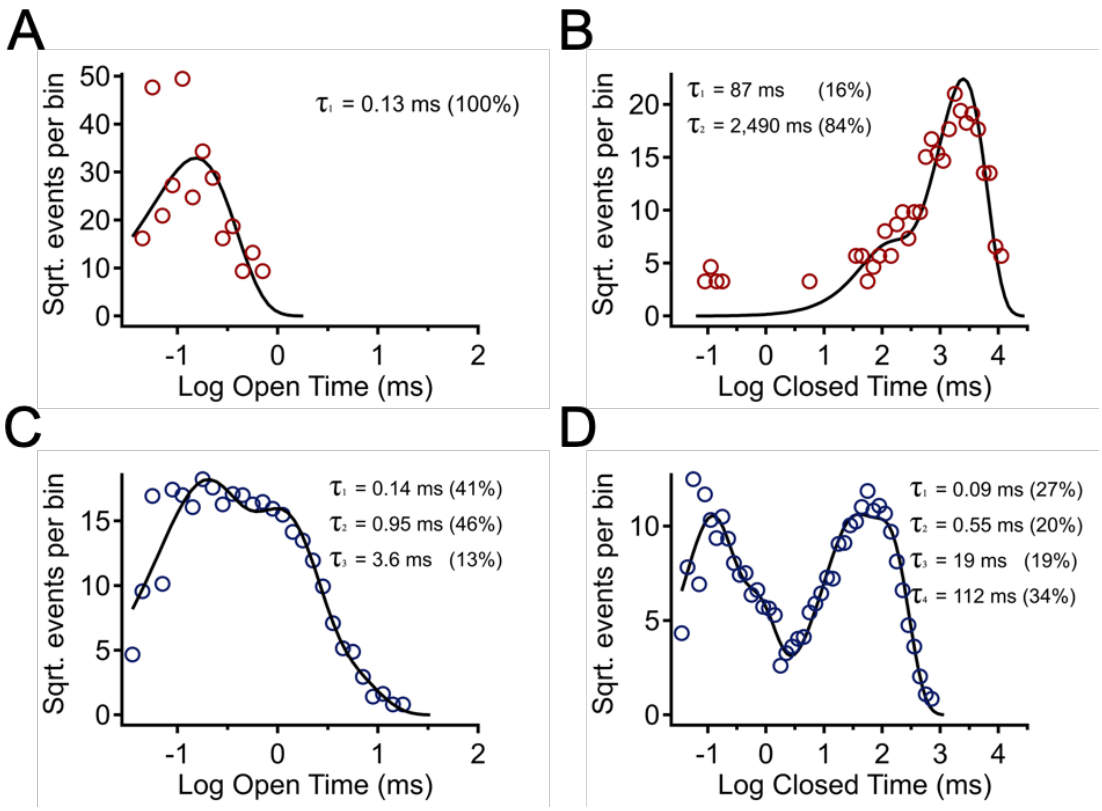


Figure 2-6: Effects of NS11021 on open and closed dwell time distributions

A, Distribution of open times with 0 μM NS11021 (red circles). Line represents fit with a single exponential, τ = 0.13 ms. Distribution contains 396 events; this and subsequent distributions were normalized to contain 100,000 events. **B,** Distribution of closed times with 0 μM NS11021 (red circles). Line represents fit with sum of two exponentials, with τ and % area of each component as indicated. Number of events = 395. **C,** Distribution of open times with 30 μM NS11021 (blue circles) added to the same patch. Line represents fit with sum of three exponentials, with τ and % area as indicated. Number of events = 11,115. **D,** Distribution of closed times with 30 μM NS11021 (blue circles). Line represents fit with sum of four exponentials, with τ and % area as indicated Number of events = 11,145. Distributions in panels A-D show activity in nominally 0 Ca^{2+}, -80 mV.

times are sensitive to the number of channels in the patch. To gain further insight toward NS11021 actions on channel gating, we constructed open dwell time distributions (**Figure 2-6 A,C**). At -80 mV, open times were described by a single exponential component with a time constant of 0.140 ± 0.004 ms (n=5). In the presence of 30 μM NS11021, open times were best described by three components, with τ_1 = 0.09 ± 0.01 ms (area = 0.41 ± 0.02), τ_2 = 0.53 ± 0.09 (area = 0.39 ± 0.02), and τ_3 = 2.5 ± 0.4 ms (area = 0.20 ± 0.02). Thus NS11021 drives the channel to gate among additional long lifetime open states.

To gain further insight toward channel gating under conditions where the CSD and VSD are at rest, we compared closed dwell time distributions with 0 and 30 μM NS11021 (**Figure 2-6 B,D**). Whereas the number of active channels in the patch determines the closed times, **Figure 2-6 B and 2-6 D** show distributions constructed from measurements performed on the same patch; thus effects of NS11021 on these dwell times can be determined from a direct comparison. These show a >20-fold decrease in the time constant for the long closed component (from 2490 ms to 112 ms). In addition, the fraction of closings with lifetimes <1 ms comprised few closings in 0 μM NS11021, whereas this increased to 47% of closings with 30 μM NS11021 (similar results were observed in four additional patches). Because the majority of these brief closings are likely found within bursts of openings, these results support the idea that NS11021 drives BK channel gating from single, brief openings (**Figure 2-5 C**) toward bursts of two or more longer openings (**Figure 2-5 D**) under conditions where the VSD and CSD are presumably at rest.

Concentration-dependence of NS11021 action

NS11021 activates BK channels over a wide range of Ca^{2+} and voltages, through a mechanism that may not involve the CSD or VSD modules. Under the conditions of our experiments, these channels are presumed to be homotetramers. If NS11021 were found

to act at the VSD or CSD, then it might be reasonable to assume a minimal stoichiometry of four NS11021 molecules binding to activate one BK channel. However, an action at the PGD, located at the confluence of the four subunits, might require fewer than four NS11021 molecules; for example, binding of one or two NS11021 molecules might hinder binding of additional molecules. Alternatively, binding of fewer than four NS11021 molecules may be sufficient to produce a maximal effect, or multiple NS11021 molecules may act in a non-cooperative manner on each subunit to stabilize the open state. To gain further insight toward the drug activation mechanism, we analyzed the relation between [NS11021] and channel activity by fitting G/Gmax data at individual voltages over a range of [NS11021] with a Hill equation. Although strictly empirical, this analysis can be used as an indicator of the minimal stoichiometry for drug:receptor interaction, and in the case of multiple binding sites, whether the ligand exhibits either positive or negative cooperativity (i.e. increases or decreases in coupling energetics when multiple sites are occupied).

Figure 2-7 illustrates that over the range of [Ca^{2+}] examined (including at nominally 0 Ca^{2+}), G/Gmax vs. [NS11021] relations could be described with Hill coefficients close to 1. Although these estimates do not confirm that there is an obligatory 1:1 relation for NS11021 to act on the BK channel, they do effectively rule out the idea that multiple NS11021 molecules activate the channel through a highly positive cooperative mechanism, and are consistent with mechanisms in which either a single NS11021 molecule is sufficient to activate the channel, or multiple NS11021 molecules stabilize activated channel subunits effectively independently.

Discussion

NS11021 alters gating kinetics

NS11021 allows BK channels to activate at less depolarized voltages by slowing channel deactivation. In contrast to Ca^{2+}, which alters both activation and deactivation kinetics, NS11021 does not appear to substantially alter activation kinetics. The activatory effects of NS11021 persist over a wide range of Ca^{2+} concentrations and upon truncation of the RCK domains in the Slo1c-Kv-MinT construct, indicating that the mechanism of NS11021 activation is distinct from Ca^{2+} activation.

Macroscopic time constants roughly correspond to the burst duration of single-channel open events (Colquhoun and Hawkes, 1994; Vandenberg and Horn, 1984; Colquhoun and Sakmann, 1981). The increase in mean open time caused by 30 μM NS11021 can be observed as a shift in the distribution of open lifetimes towards longer-duration open states (**Figure 2-6 A/C**). Thus, measurements of single-channel and macroscopic BK channel activity are both consistent with a mechanism where NS11021 is slowing channel deactivation. Likewise, two major effects of 30 μM NS11021 can be observed in the distributions of closed lifetimes (**Figure 2-6 B/D**). First, addition of 30 μM NS11021 increases the frequency of short-duration closed events, likely corresponding to an increase in the number of openings per burst (Colquhoun and Hawkes, 1987). Second, addition of 30 μM NS11021 is associated with the longest-duration closed states occurring more rapidly than without drug, which correspond to NS11021 increasing the intrinsic opening of the pore (Rothberg et al., 1996). In summary, both macroscopic and single channel activity of BK channels suggest a mechanism where NS11021 is primarily slowing channel deactivation kinetics.

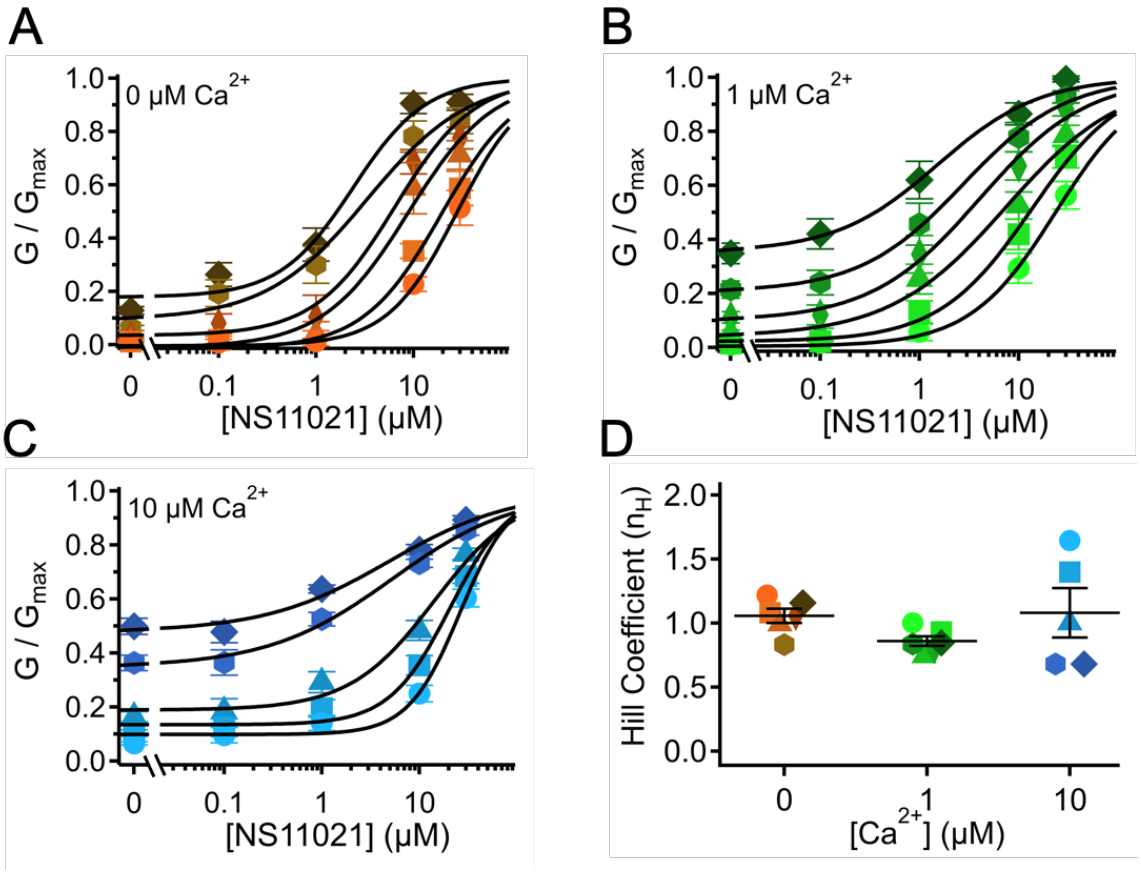


Figure 2-7: Estimate of apparent stoichiometry of NS11021 action

A, Normalized G (G/Gmax) vs. [NS11021] in nominally 0 Ca^{2+}. Data were grouped by voltage and fitted with a Hill equation (Eq. 2.3; black lines) to estimate Hill coefficient (n_H) and EC_{50}. Parameters at each voltage were: 70 mV (●; n_H = 1.2; EC_{50} = 28 μM), 80 mV (■; n_H = 1.1; EC_{50} = 20 μM), 90 mV (▲; n_H = 1.0; EC_{50} = 8.8 μM), 100 mV (♦; n_H = 1.1; EC_{50} = 6.4 μM), 110 mV (⬢; n_H = 0.8; EC_{50} = 3.4 μM), and 120 mV (◆; n_H = 1.2; EC_{50} = 2.4 μM). **B**, G/Gmax vs. [NS11021] with 1 μM Ca^{2+}. 50 mV (●; n_H = 1.0; EC_{50} = 24 μM), 60 mV (■; n_H = 0.9; EC_{50} = 13 μM), 70 mV (▲; n_H = 0.7; EC_{50} = 7.5 μM), 80 mV (♦; n_H = 0.8; EC_{50} = 4.0 μM), 90 mV (⬢; n_H = 0.8; EC_{50} = 2.7 μM), and 100 mV (◆; n_H = 0.8; EC_{50} = 1.5 μM). **C**, G/Gmax vs. [NS11021] with 10 μM Ca^{2+}. -30 mV (●; n_H = 1.6; EC_{50} = 26 μM), -20 mV (■; n_H = 1.4; EC_{50} = 21 μM), -10 mV (▲; n_H = 1.0; EC_{50} = 14 μM), 10 mV (⬢; n_H = 0.7; EC_{50} = 5.3 μM), and 20 mV (◆; n_H = 0.7; EC_{50} = 4.9 μM). **D**, Hill coefficients at each [Ca^{2+}]; parameters at each voltage are shown as shapes; mean at a given [Ca^{2+}] is indicated by horizontal line, and error bars show the SEM. Means for each [Ca^{2+}] are: 0 Ca^{2+}(1.06 ± 0.06; n=6), 1 μM Ca^{2+} (0.86 ± 0.04; n=6), and 10 μM Ca^{2+} (1.08 ± 0.19; n=5).

Similar to NS11021, the activators mallotoxin and Cym04 also preferentially slow channel deactivation while not substantially altering activation kinetics. Addition of 2 μM mallotoxin slows deactivation by approximately 10-fold (Guan et al., 2017), while 10 μM Cym04 slows deactivation by approximately 2-fold (Gessner et al., 2012). Although slowing of deactivation is observed with each drug, the partial valence for τ at very negative voltages (z_N) appears to decrease only with mallotoxin administration and remains largely unchanged by NS11021 and Cym04. Thus, the activatory effects of these drugs may be caused by different mechanisms and will need to be further elucidated using additional approaches.

The observation that NS11021 substantially increases open probability at negative voltages and in the nominal absence of Ca^{2+}, conditions where the majority of the voltage- and Ca^{2+}-sensors are at rest, illustrates that NS11021 alters the intrinsic gating of the pore directly to stabilize the open state (**Figure 2-5**). While this does not eliminate the possibility that NS11021 alters additional aspects of gating, it supports the hypothesis that NS11021 is not acting by exclusively modulating either the voltage or Ca^{2+} sensitivity of the channel. The 60-fold increase in Po observed with NS11021 is greater than the approximately 6-fold increase observed with NS1619 and Cym04 (Gessner et al., 2012), yet substantially less than the >1700-fold increase with GoSlo-SR-5-6 (Webb et al., 2015). The large difference in magnitude of effect on open probability caused by NS11021, Cym04, NS1619, and GoSlo-SR-5-6 when the voltage- and Ca^{2+}-sensors are largely at rest provides further evidence that each drug modulates intrinsic gating of the pore to different extents, and therefore may be activating with different mechanisms of action.

Mechanism of NS11021 activation is unlikely to be highly cooperative

Analysis of G-V relations at individual voltages shows a Hill coefficient (n_H) of approximately 1.0 for NS11021 in the presence of nominally 0, 1, and 10 μM Ca^{2+} (**Figure 2-7**). While this empirical approach represents a minimum stoichiometry for NS11021 molecules binding per channel, it can essentially rule out a highly cooperative mechanism. A similar experimental approach showed the activator dehydrosoyasaponin-I to have a hill-slope of 3.4 to suggest approximately three or four molecules bind to activate each channel (Giangiacomo et al., 1998). Comparably, analysis of Cym04 resulted in a Hill slope of 1.6 (Gessner et al., 2012). While more direct structural measurements would be required to more accurately quantify the stoichiometry of NS11021, these data further suggest that these drugs may all be activating BK channels with different mechanisms of action.

CHAPTER 3

QUANTITATIVE DESCRIPTION OF NS11021 ACTIVATION

Introduction

The major features of BK channel gating by transmembrane voltage and cytosolic Ca^{2+} can be described by a dual allosteric model in which the sequential activation of four voltage-sensors and/or four Ca^{2+}-sensors are energetically coupled to opening of the pore. This scheme (**Scheme 1**), illustrated in Table 1-2, can quantitively describe BK channel activation over a wide range of voltages (from -200 to +200 mV) and Ca^{2+} concentrations (from nominally 0 to 1 mM) (Horrigan and Aldrich, 2002; Rothberg and Magleby, 1999; Rothberg et al., 1997; Cui and Aldrich, 2000). Beyond summarizing our understanding of channel gating transitions and their energetics in “wild-type” BK channels, quantitative models provide a framework for understanding and describing the effects of mutations in terms of how a change in channel structure might alter activation of the channel by Ca^{2+} or voltage, or how a specific gating step might be altered by binding of a pharmacological activator or inhibitor (Koval et al., 2007; Wang et al., 2006, 2009; Ma et al., 2006; Gessner et al., 2012; Zhou et al., 2018; Horrigan and Aldrich, 1999; Hu et al., 2005; Sweet and Cox, 2009; Kshatri et al., 2020).

In this chapter, I provide a quantitative description for how the small molecule NS11021 activates BK channels. NS11021 has been used as a vasodilator in animal models due to selective activation of BK channels (Kun et al., 2009; Bentzen et al., 2007, 2009; Cui, 2020). We can account for the major features of activity of NS11021 through a kinetic scheme in which the drug acts by slowing transition rates of the PGD toward the closed state by approximately 5-fold. Partial results of the presented work have been published in (Rockman et al., 2020)

Materials and Methods

Description of steady-state channel activity

To analyze the mechanism underlying effects of NS11021 on the gating of BK channels, conductance-voltage (G-V) curves in the presence of various concentrations of Ca^{2+} were fitted with the dual allosteric model (**Scheme 1**, Horrigan and Aldrich, 2002):

$$P_o = \frac{1}{1 + \frac{(1 + J + K + JKE)^4}{L(1 + KC + JD + JKCDE)^4}}$$

(Eq. 3.1)

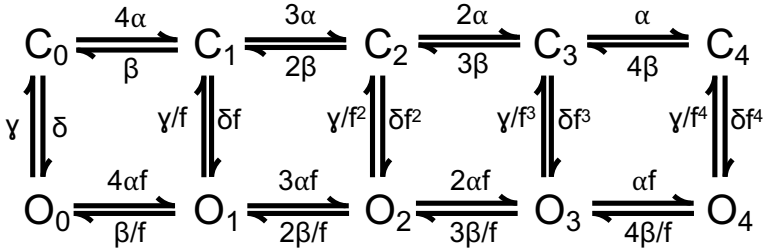
where $J = J_0 \exp(z_J V/k_B T)$, $L = L_0 \exp(-z_L V/k_B T)$, and $K = [Ca^{2+}]/K_D$. For these equations, J_0 represents the equilibrium constant for voltage sensor movement at 0 mV with z_J as the effective valence, V is the membrane potential, k_B is Boltzmann's constant, T is temperature, L_0 is the closed-to-open pore equilibrium constant with z_L as the effective valence, and K_D represents the dissociation constant for Ca^{2+}. The allosteric coupling constants C, D, and E represent the interaction between Ca^{2+} binding and opening, voltage sensor activation and opening, and Ca^{2+} binding and voltage sensor movement, respectively, as illustrated in Scheme 1 (Horrigan et al., 1999; Horrigan and Aldrich, 2002; Ma et al., 2006; Wang et al., 2006; Koval et al., 2007).

For our analysis, sets of values for the model parameters in Eq. 3.1 were first estimated using a range of different initial starting parameters, to minimize the value of χ^2 determined from comparison with mean G-V curves and Po data at nominally 0, 1, 10, and 100 μM Ca^{2+} all at 0 μM NS11021. This was achieved by global fitting using the Levenberg-Marquardt algorithm, implemented in the Global Fit package in IgorPro 8 (WaveMetrics), in which $\chi^2 = \sum_{i=1}^{n} \left(\frac{Y_i - \widehat{Y_i}}{\sigma_i}\right)^2$ where Y_i and $\widehat{Y_i}$ are the ith observed and

corresponding predicted values and σ_i is the observed variance. Based on this approach, two "standard" parameters sets ("Fit A" and "Fit B") were selected, each yielding very similar values of χ^2 and thus statistically equivalent descriptions of the BK channel G-V data with 0 μM NS11021. These two parameter sets were used in fitting of datasets in the presence of 0.1, 1, 10, and 30 μM NS11021 by the following approach: A dataset at a given [NS11021] (at nominally 0, 1, 10, and 100 μM Ca^{2+}) was fitted, again using the Global Fit package in IgorPro 8, by allowing only one parameter to be adjusted (i.e. only J_0, K_D, L_0, z_L, D, *etc.*). Thus estimates of each single parameter were obtained as a function of [NS11021] against the set of standard "Fit A" or "Fit B" parameters.

Description of activation and deactivation rates

The mechanism of NS11021 action on voltage-dependent gating at nominally 0 Ca^{2+} was further analyzed using a 10-state voltage-dependent scheme (Scheme 2), which used microscopic rate constants rather than equilibrium constants, to enable calculation of activation and deactivation time constants for BK channel gating. This is similar to approaches used in previous work (Horrigan et al., 1999), except that Scheme 2 was based on the simplifying constraint that all "closed - open" (C-O) transitions were allosterically coupled to independently-activated VSD's. Scheme 2 is depicted below:



SCHEME 2

In Scheme 2, rate constants were defined by rates for VSD activation and deactivation (α and β, respectively), PGD closing and opening rates (γ and δ), a gating valence for each of these rate constants (z_α, z_β, z_γ, and z_δ), and allosteric coupling between the VSD and PGD (D) using the additional simplifying assumption $D = f^2$ (i.e. VSD-activation coupling to PGD opening is energetically equivalent to VSD-deactivation coupling to PGD closing). Effective rates for each transition were calculated at voltages between -240 mV and +230 mV, in 10 mV increments:

$$rate_{ij}(V) = rate_{ij}(0)\exp(z_{ij}V/k_BT) \qquad \text{(Eq. 3.2)}$$

where V represents the membrane potential, k_B is Boltzmann's constant, T is temperature, $rate_{ij}$ denotes the rate from state "i" to state "j". For example, if α=100 and z_α=1.5, then the effective rate from C_0 to C_1 in Scheme 2 = 4(100)(exp(1.5V/kT)) (Weiss and Magleby, 1992; Zagotta et al., 1994; Rothberg and Magleby, 2000). The principal (i.e. rate-limiting) time constants for voltage-dependent activation were calculated over a range of voltages using Q-matrix methods (Colquhoun and Hawkes, 1995) in IgorPro 8. The rate-limiting rate constants were compared with experimentally-determined time constants by χ^2 as defined above, and model parameters were optimized through subsequent iterations using a direct search approach based on Patternsearch (Colquhoun, 1971).

Results

NS11021 acts by stabilizing the PGD in the open state

To gain further insight toward the molecular mechanism of NS11021 action, we developed a quantitative description of gating in the presence and absence of drug, based on the well-established dual allosteric model for BK channel gating (**Scheme 1**). Although BK channels are known to contain a total of eight high-affinity Ca^{2+} binding sites per

channel, here we have used a simplified gating scheme containing only four Ca^{2+} binding sites per channel, and four voltage sensors that are each activated independent of one another in single transitions (Rothberg and Magleby, 1999; Horrigan and Aldrich, 2002). Our approach was to first identify a set of parameters to describe the steady-state G-V relations over a range of [Ca^{2+}] (G-V-Ca^{2+} relations), which would be constrained by our experimental single-channel and macroscopic current data. Because some parameters in the kinetic scheme are poorly constrained by these data, we developed two sets of parameters, Fit A and Fit B, which yielded nearly equivalent descriptions of the data as evaluated by chi-square statistic (see Methods and **Table 3-1**). Values used for each parameter were similar to values determined previously to describe BK channel activity over a wide range of voltages and [Ca^{2+}] (Koval et al., 2007; Wang et al., 2006, 2009; Ma et al., 2006; Gessner et al., 2012; Zhou et al., 2018; Horrigan and Aldrich, 1999; Hu et al., 2005; Sweet and Cox, 2009; Kshatri et al., 2020). After identifying these sets of parameters, we next evaluated whether the G-V-Ca^{2+} relations in the presence of increasing [NS11021] could be described by changing only a single parameter, while keeping the other parameters at their values determined from fits with data in the absence of NS11021. Thus we attempted fits by varying only J_0, K, L_0, *etc.*, to determine which parameter [NS11021] had the single greatest impact, also evaluated by a chi-square statistic.

Using this approach, we observed that in the case of both parameter sets (Fit A and Fit B), fits with three of the individual parameters, L_0, J_0, and D, provided nearly equivalent descriptions of the experimental G-V-Ca^{2+} relations (**Figures 3-1, 3-2**; **Table 2**). As an example of how much these parameters were affected by the drug, L_0, J_0, and D were altered by 2.6-, 1.5-, and 1.3-fold for Fit A, and 2.2-, 1.6-, and 1.5-fold for Fit B per

Table 3-1: Parameters for Scheme 1, fitted with data at 0 μM NS11021

Parameter	Fit A	Fit B
J_0	0.08	0.06
K_D (μM)	41	17
L_0	2.8×10^{-6}	1.0×10^{-5}
z_J (e_0)	0.30	0.41
z_L (e_0)	0.57	0.58
C	2.5	17
D	29	16
E	20	1.1

Simultaneous (global) fitting was performed using G-V relations over a range of Ca2+ with 0 μM NS11021, as described in Methods. G-V relations predicted by above parameters in Fit A and Fit B can be found in Figures 7A and 8A, respectively. $\chi 2$ values for comparison of experimental data vs. corresponding prediction (based on 90 experimental data points): Fit A, 0.045; Fit B, 0.044.

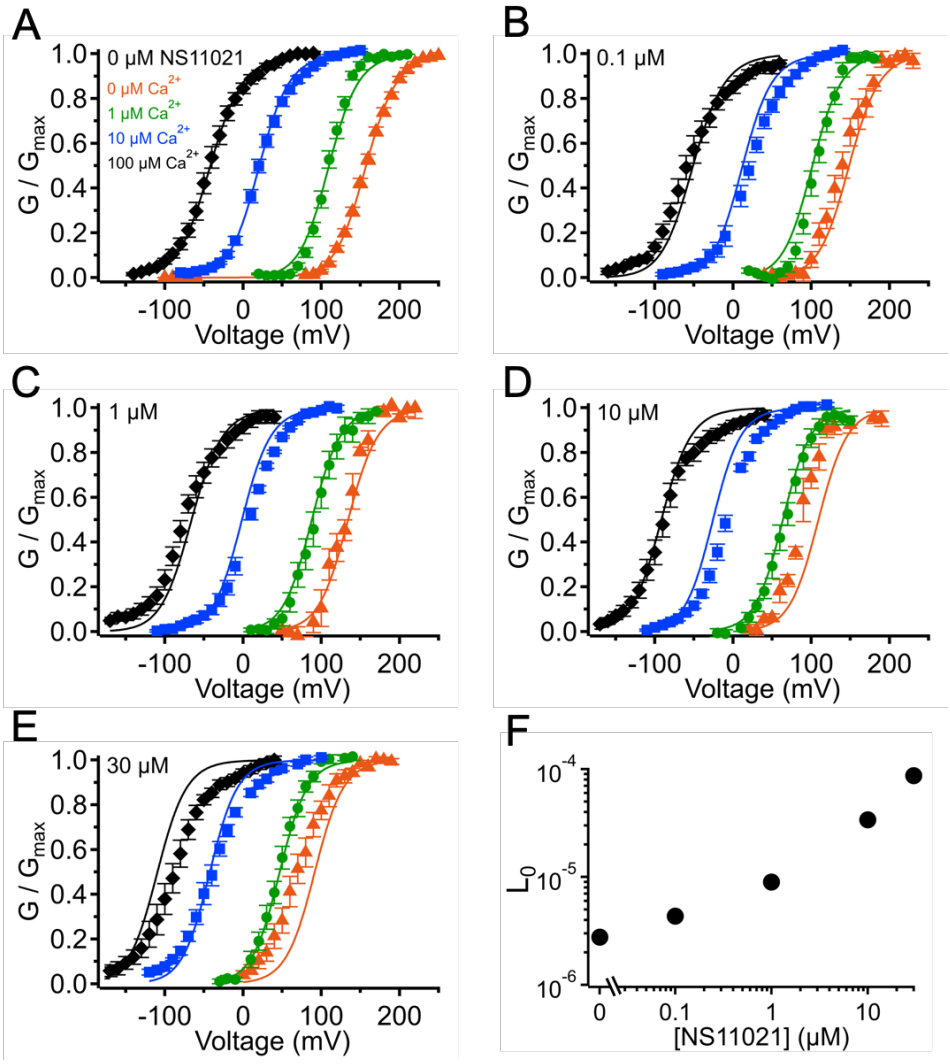


Figure 3-1: G-V-Ca^{2+} relations with predictions from Scheme 1 using parameters from Fit A

A, G-V-Ca^{2+} relations in the absence of NS11021 bathed in 0 μM (orange triangle), 1 μM (green circle), 10 μM (blue square), and 100 μM Ca^{2+} (black diamond). Parameters are listed in Table 1, Fit A. Lines represent the predictions with L_0= $2.8x10^{-6}$. **B-E,** G-V-Ca^{2+} relations in 0.1, 1, 10, and 30 μM NS11021, respectively. Lines represent predictions with L_0= $4.3x10^{-6}$ (0.1 μM), $9.0x10^{-6}$ (1 μM), $3.4x10^{-5}$ (10 μM), and $8.7x10^{-5}$ (30 μM). **F,** Fitted values for L_0 vs. [NS11021].

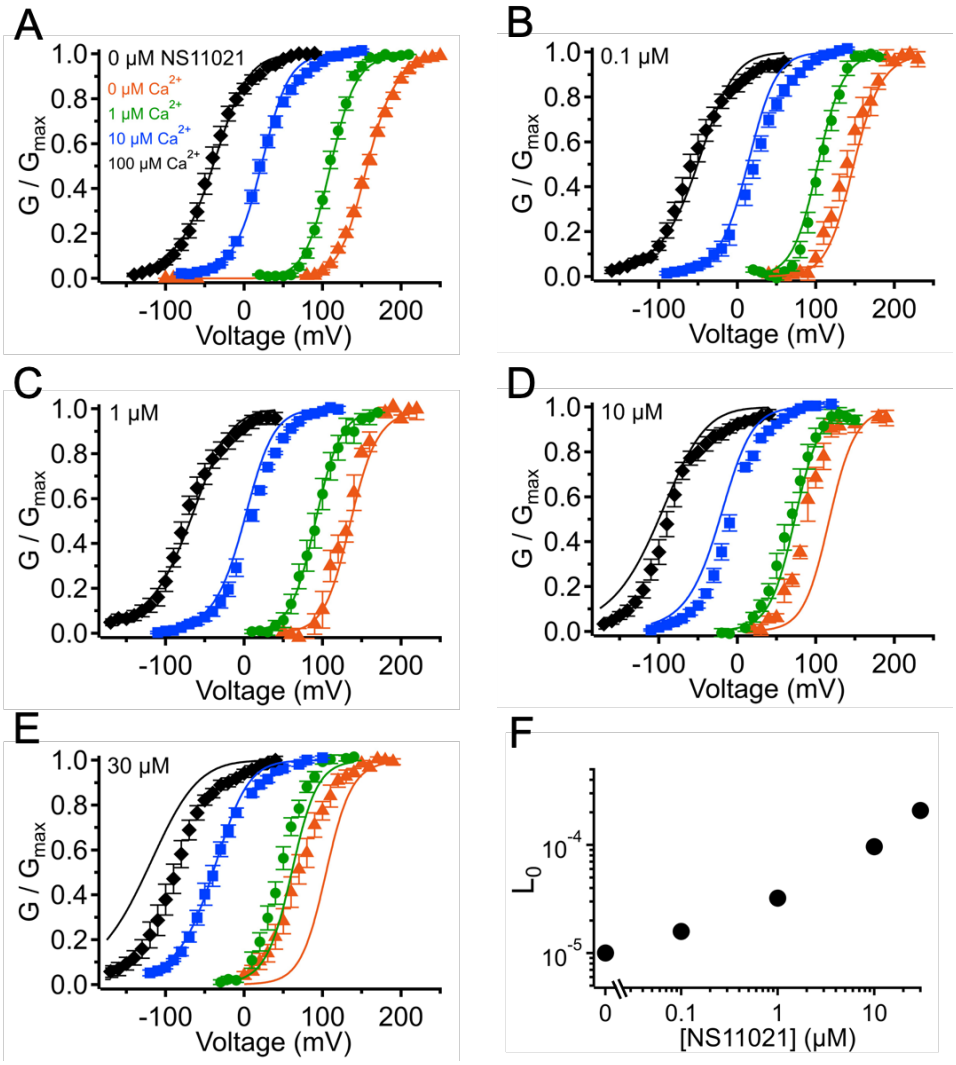


Figure 3-2: G-V-Ca^{2+} relations with predictions from Scheme 1 using parameters from Fit B

A, G-V-Ca^{2+} relations in the absence of NS11021 bathed in 0 μM (orange triangle), 1 μM (green circle), 10 μM (blue square), and 100 μM Ca^{2+} (black diamond). Parameters are listed in Table 1, Fit B. Lines represent the predictions using L_0= $1.0x10^{-5}$. **B-E,** G-V-Ca^{2+} relations in 0.1, 1, 10, and 30 μM NS11021, respectively. Lines represent predictions with L_0= $1.6x10^{-5}$ (0.1 μM), $3.2x10^{-5}$ (1 μM), $9.6x10^{-5}$ (10 μM), and $2.1x10^{-4}$ (30 μM). **F,** Fitted values for L_0 vs. [NS11021].

Table 3-2: Results of changing individual parameters in Scheme 1 to account for effects of NS11021

Parameter	Fit A		Fit B	
	Value at 30 μM NS11021	χ^2	Value at 30 μM NS11021	χ^2
J_0	**0.32**	**1.96**	**0.36**	**2.53**
K_D (μM)	6.8	11.8	2.9	11.29
L_0	**8.7×10^{-5}**	**2.16**	**2.1×10^{-4}**	**3.77**
z_J (e_0)	0.82	11.35	1.3	9.14
z_L (e_0)	1.8	11.52	2.0	12.09
C	6.0	10.40	35	11.66
D	**74**	**2.37**	**61**	**1.56**
E	5.0	9.46	5.1	9.86

Simultaneous (global) fitting was performed using G-V relations acquired over a range of Ca^{2+} for [NS11021] at 0, 0.1, 1, 10, and 30 μM, with parameters shown above adjusted individually, as described in Methods. Adjusted parameter values at 30 μM NS11021 are shown above, along with summed χ^2 values for fitting over the entire range of [NS11021] and [Ca^{2+}] (based on 410 experimental data points). Unadjusted parameter values (corresponding to 0 μM NS11021) can be found in Table 3-1.

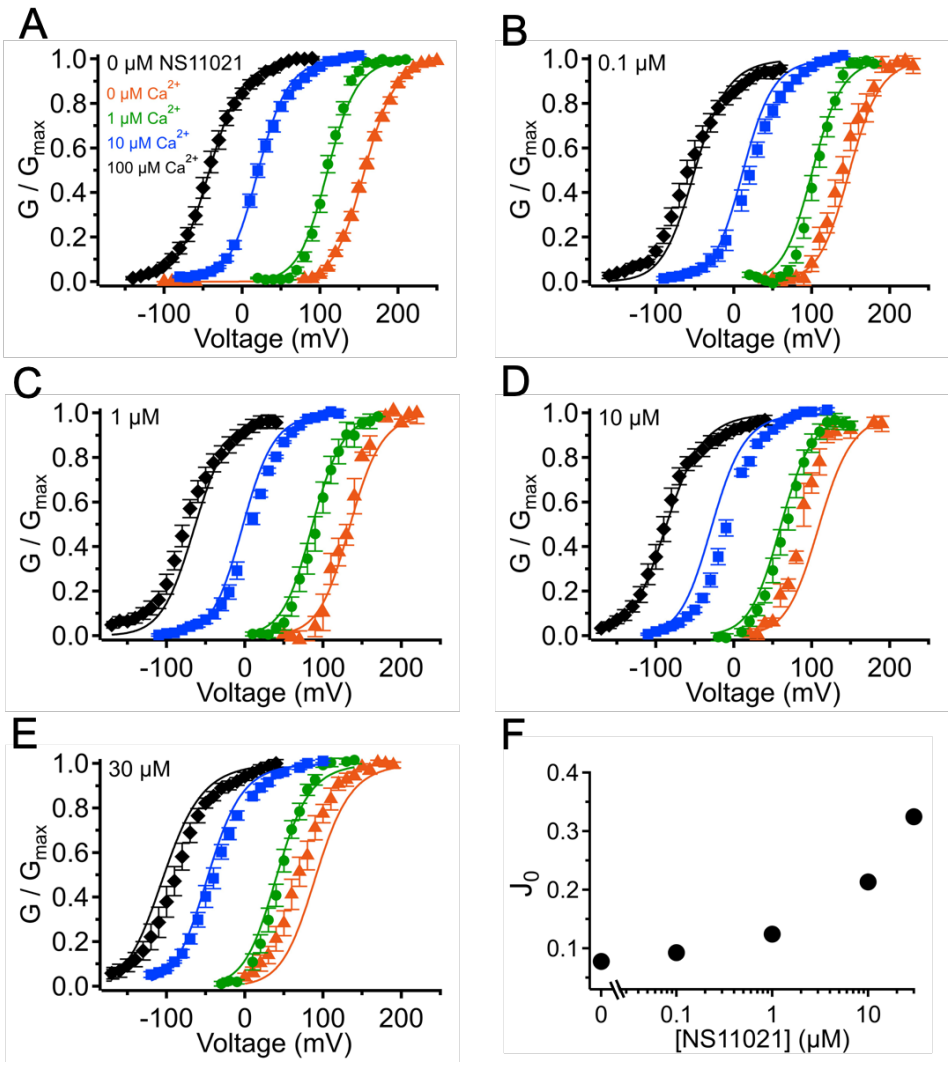


Figure 3-3: G-V-Ca^{2+} relations with predictions from Scheme 1 using parameters from Fit A, with adjustable parameter J_0

A, G-V-Ca^{2+} relations in the absence of NS11021 bathed in 0 μM (orange triangle), 1 μM (green circle), 10 μM (blue square), and 100 μM Ca^{2+} (black diamond). Parameters are listed in Table 1, Fit A. Lines represent the predictions with J_0= 0.08. **B-E,** G-V-Ca^{2+} relations in 0.1, 1, 10, and 30 μM NS11021, respectively. Lines represent predictions with J_0= 0.09 (0.1 μM), 0.12 (1 μM), 0.21 (10 μM), and 0.32 (30 μM). **F,** Fitted values for J_0 vs. [NS11021].

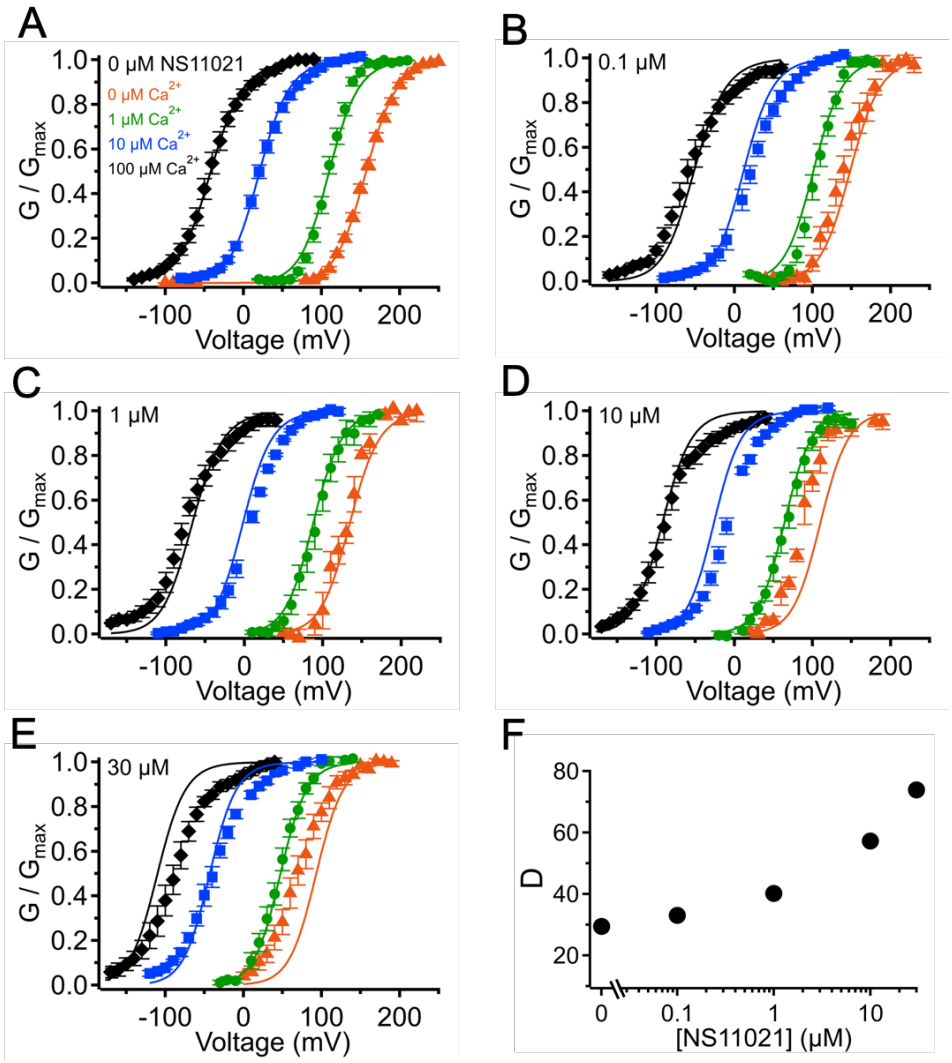


Figure 3-4: G-V-Ca^{2+} relations with predictions from Scheme 1 using parameters from Fit A, with adjustable parameter D

A, G-V-Ca^{2+} relations in the absence of NS11021 bathed in 0 μM (orange triangle), 1 μM (green circle), 10 μM (blue square), and 100 μM Ca^{2+} (black diamond). Parameters are listed in Table 1, Fit A. Lines represent the predictions with D = 29. **B-E,** G-V-Ca^{2+} relations in 0.1, 1, 10, and 30 μM NS11021, respectively. Lines represent predictions with D = 33 (0.1 μM), 40 (1 μM), 57 (10 μM), and 74 (30 μM). **F,** Fitted values for D vs. [NS11021].

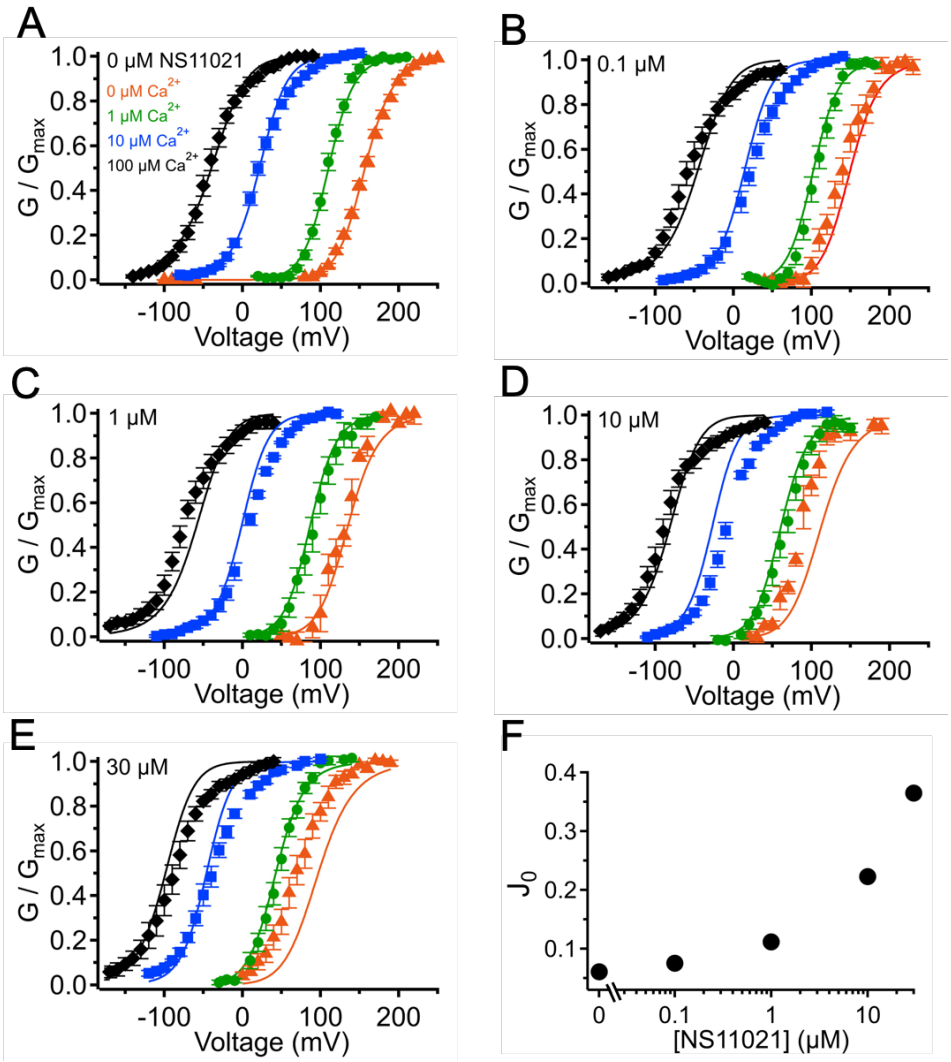


Figure 3-5: G-V-Ca^{2+} relations with predictions from Scheme 1 using parameters from Fit B, with adjustable parameter J_0

A, G-V-Ca^{2+} relations in the absence of NS11021 bathed in 0 μM (orange triangle), 1 μM (green circle), 10 μM (blue square), and 100 μM Ca^{2+} (black diamond). Parameters are listed in Table 1, Fit A. Lines represent the predictions with J_0= 0.08. **B-E,** G-V-Ca^{2+} relations in 0.1, 1, 10, and 30 μM NS11021, respectively. Lines represent predictions with J_0= 0.09 (0.1 μM), 0.12 (1 μM), 0.21 (10 μM), and 0.32 (30 μM). **F,** Fitted values for J_0 vs. [NS11021].

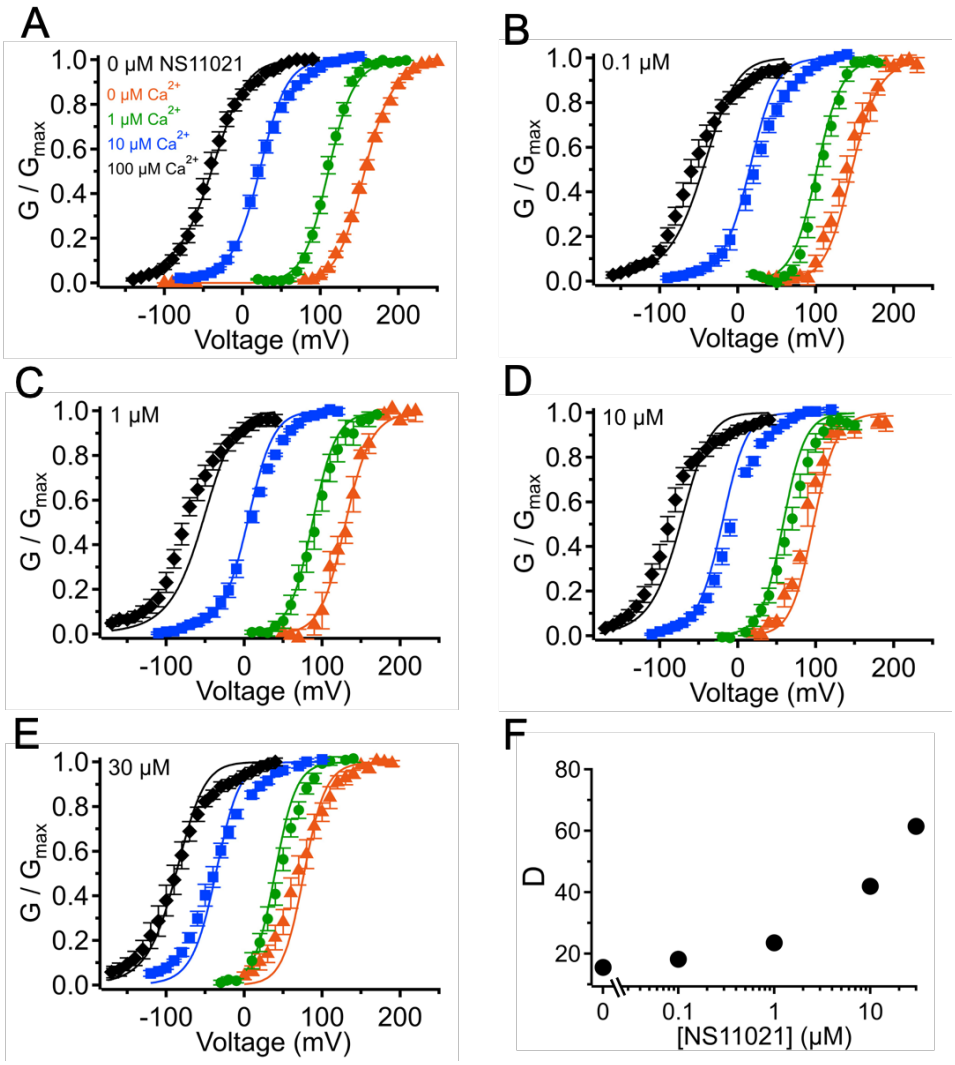


Figure 3-6: G-V-Ca^{2+} relations with predictions from Scheme 1 using parameters from Fit B, with adjustable parameter D

A, G-V-Ca^{2+} relations in the absence of NS11021 bathed in 0 μM (orange triangle), 1 μM (green circle), 10 μM (blue square), and 100 μM Ca^{2+} (black diamond). Parameters are listed in Table 1, Fit B. Lines represent the predictions with D = 16. **B-E,** G-V-Ca^{2+} relations in 0.1, 1, 10, and 30 μM NS11021, respectively. Lines represent predictions with D = 18 (0.1 μM), 23 (1 μM), 42 (10 μM), and 61 (30 μM). **F,** Fitted values for D vs. [NS11021].

threefold increase in [NS11021] between 10 and 30 μM NS11021 (**Figures 3-1 F, 3-2 F; Figures 3-3 – 3-6**). Although these results could be interpreted to mean that effects on L_0, J_0, or D were approximately equally likely to underlie NS11021 action in the channel, we had previously reasoned that the NS11021 activation must, at a minimum, affect L_0 (which governs the open-closed equilibrium at the PGD) because NS11021 altered the gating at very negative voltages at which the VSD is largely at rest. In general, the model described channel activity relatively well over a range of voltages. Model predicted values began to deviate from experimental data at low or saturating [Ca^{2+}] and high [NS11021].

To further explore the validity of the three best single-parameter fits with Scheme 1, we first calculated the Po values predicted by Scheme 1 at nominally 0 Ca^{2+} and Vm ≤ -60 mV, for 0 and 30 μM NS11021, using the values for L_0, J_0, and D shown in Figures 3-1 – 3-6. Here we reasoned that the more valid parameter set should yield a better prediction of these Po data with 30 μM NS11021. Among these, Scheme 1 with L_0 adjusted (from value in Table 3-1 to value in Table 3-2) to account for the effects of increasing [NS11021] yielded the best description of these Po data at which the CSD and VSD are predicted to be largely in the resting state (**Figure 3-7A, B, solid line**).

Although consistent with the idea that an increase in L_0 could describe the effects of NS11021 on BK channel gating, the results determined from predictions of steady-state activation with Scheme 1 are not unequivocal. Thus we reasoned that a more stringent test of model validity would be to use a reduced kinetic scheme, Scheme 2, to describe the voltage-dependent activation and deactivation time constants that had been estimated with nominally 0 Ca^{2+}, and then identify a set of parameters for Scheme 2 that can account for the effects of NS11021 on the time constants (Marks and Jones, 1992; Horrigan et al., 1999). Scheme 2, in principle, should account for the major features of voltage-dependent

BK channel gating in the absence of Ca^{2+}, and so it should be possible to identify which specific rate constants in Scheme 2 best account for the effects of NS11021.

In the 10 state model of voltage-dependent BK channel activation (Scheme 2) horizontal transitions represent voltage-sensor activation, with activation rate constant α and deactivation rate constant β. Likewise, vertical transitions represent pore opening, which can be described by C $\rightarrow$ O rate constant δ and O $\rightarrow$ C rate constant γ. Coupling between voltage sensor activation channel opening was denoted by f, where $D = f^2$. Although pore opening and closing rates exhibit a weak voltage dependence (represented by a small gating valence z_γ and z_δ), the majority of the voltage-dependence of BK channel activation comes from activation of the voltage-sensors (denoted by gating valence z_α and z_β) (Weiss and Magleby, 1992; Horrigan et al., 1999). Thus, when the voltage sensors are mainly at rest, BK channels activate with a weak voltage dependence which is observed as a monoexponential τ – V relationship at very negative (≤ 150 mV) voltages (Horrigan et al., 1999; Cui et al., 1997). As described in the methods, the approach was to use the Q-matrix method to determine values for rate parameters (α, β, γ, δ, z_α, z_β, z_γ, z_δ and D) where the rate-limiting time constant described experimental time constats (Colquhoun and Hawkes, 1981, 1982),

Using this approach, we first identified a set of parameters (rate constants and their corresponding voltage- dependences) to describe the macroscopic time constants of relaxation as a function of voltage (τ vs. V) at 0 Ca^{2+} in the absence of NS11021. We then used these parameters as a starting point to find which rate constants might be altered to best describe the τ vs. V relation with 30 μM NS11021. Using this approach we found that it was possible to describe the τ vs. V relation with 30 μM NS11021 by increasing δ (the rate constant for the C-O transition) by 1.6-fold, and decreasing γ (rate constant for O-C

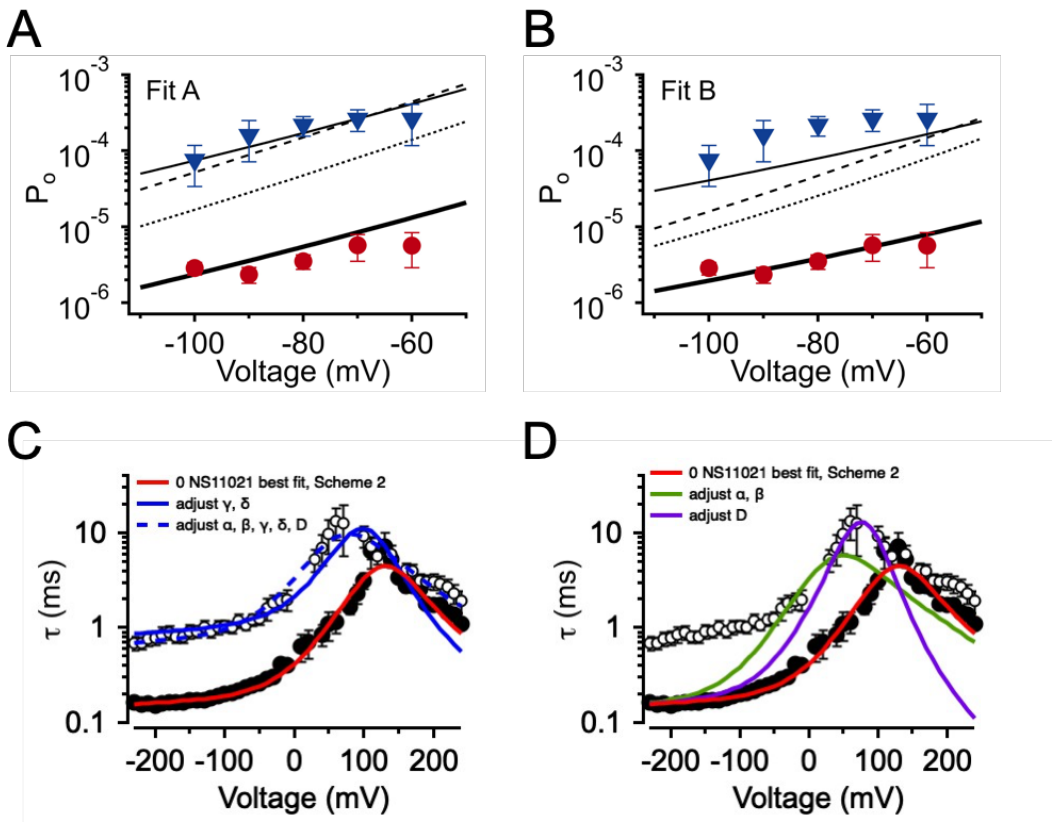


Figure 3-7: Description of BK channel activity at nominally 0 Ca^{2+} using Schemes 1 and 2

A, Po vs. voltage with 0 NS11021 (red circles) and 30 μM NS11021 (blue triangles). Solid line represents Po predicted from Scheme 1 using parameters in Table 1, Fit A. Predicted Po with 30 μM NS11021 was generated by using Fit A substituting L_0 = $8.7x10^{-5}$ (solid thin line), J_0 = 0.32 (dashed line), or D = 74 (dotted line). Using parameters from Fit A in Table 1 resulted in a χ^2 value of 3.85. χ^2 values obtained by changing only one parameter to account for Po at 30 μM NS11021: χ^2 for L_0 was 0.41, χ^2 for J_0 was 0.77, χ^2 for D was 3.15; thus substitution of L_0 resulted in the lowest χ^2 value (best fit) for these data. **B,** Po vs. voltage as in panel A, except with solid line showing Po predicted using Fit B. Po with 30 μM NS11021 was generated with Fit B by substituting L_0 = $2.1x10^{-4}$ (solid thin line), J_0 = 0.36 (dashed line), or D = 61 (dotted line). Using parameters from Fit B resulted in a χ^2 value of 1.18. χ^2 values obtained by changing only one parameter to account for Po at 30 μM NS11021: χ^2 for L_0 was 1.91, χ^2 for J_0 was 3.13, χ^2 for D was 4.73. Again substitution of L_0 resulted in the lowest χ^2 value (best fit) for these data. All χ^2 values for A and B based on 5 total data points. **C,** τ (macroscopic time constant of relaxation) vs. voltage from patches with nominally 0 μM Ca^{2+}, with 0 (filled circles) or 30 μM NS11021 (open circles). Lines represent the rate-limiting time constant from fits using Scheme 2 using parameters in Table 3-3: 0 μM NS11021, red; 30 μM NS11021, adjust γ δ, solid blue; adjust α, β, γ, δ, D, dashed blue. D, τ (macroscopic time constant of relaxation) vs. voltage as in panel C, with solid lines representing the rate-limiting time constant showing fits with Scheme 2 using parameters in Table 3-3: 0 μM NS11021, red; 30 μM: α β, green; 30 μM: D, purple. Adjusting α and β or D can only describe the gating kinetics with 30 μM NS11021 at negative voltages (<-100 mV) when in combination

transition) by 5.5-fold (**Table 3-3; Figure 3-7 C**). The ratio of rate constants δ/γ theoretically corresponds to equilibrium constant L_0, both describing the C-O transition of the pore. However, since the rates in Scheme 2 are constrained additionally by kinetic data at very negative voltages (to -240 mV, which are not used in determining the parameters of Scheme 1), direct comparison between estimates of the equilibrium constant L_0 from Scheme 1 (2.8×10^{-6} for Fit A, 1.0×10^{-5} for Fit B) and the ratio of δ/γ (2.7×10^{-4}) with 0 μM NS11021 reveals discrepancies between the two approaches. However, addition of 30 μM NS11021 resulted in a 9-fold increase in the ratio of δ/γ (to 2.4×10^{-3}) in Scheme 2, which is comparable to the increases in L_0 observed with 30 μM NS11021 using Scheme 1 (30-fold and 20-fold for Fits A and B, respectively). Whereas these results reveal potential limitations of these simplified gating models, they are consistent with the observation that the major effects of NS11021 can be described by a change in L_0. In contrast, changing only α and β, which underlie the equilibrium constant J_0 from Scheme 1 to represent voltage sensor-activation, or changing the allosteric constant D (corresponding to f^2 in Scheme 2), could not recapitulate the effects of increasing [NS11021] on time constants at negative voltages (**Figure 3-7 D**).

In summary, using a 10-state model (Scheme 2) to describe voltage-dependent activation of BK channels in the absence of Ca^{2+} suggests that NS11021 is likely altering the C-O equilibrium of the pore (parameters δ and γ). Changing voltage-sensitive parameters (α, β, or D) cannot account for the observed slowing of deactivation caused by 30 μM NS11021 at very negative (≤ -100 mV) voltages (**Figure 3-7 D**, green and purple lines).

Table 3-3: Fitted parameters for Scheme 2 constrained by time constants acquired with nominally 0 Ca^{2+}, and either 0 or 30 μM NS11021

Parameter	0 μM:	30 μM: α β γ δ D	30 μM: γ δ	30 μM: α β	30 μM: D
α (s^{-1})	1,162	**3,387**	1,162	**4,833**	1,162
β (s^{-1})	31,020	**17,200**	31,020	**11,230**	31,020
γ (s^{-1})	5,128	**1,154**	**925.0**	5,128	5,128
δ (s^{-1})	1.40	**1.45**	**2.23**	1.40	1.40
$z_α$ (e_0)	0.31	0.31	0.31	0.31	0.31
$z_β$ (e_0)	-0.31	-0.31	-0.31	-0.31	-0.31
$z_γ$ (e_0)	-0.025	-0.025	-0.025	-0.025	-0.025
$z_δ$ (e_0)	0.29	0.29	0.29	0.29	0.29
D	8.2	**5.2**	8.2	8.2	**25.6**
χ^2	13.4	14.9	82.5	167.4	209.7

Time constants of activation/deactivation over voltages ranging from -240 to 230 mV, with nominally 0 Ca^{2+} and 0 μM NS11021 used to estimate parameters for Scheme 2, as described in Methods. These were used as a base set of parameters to describe the time constants in the presence of 30 μM NS11021, by adjusting parameters corresponding to J_0 (α, β), L_0 (γ, δ), or D. These adjusted values are shown in **bold**. Thus adjusting γ and δ yielded the lowest χ^2 value among these three comparisons, although adjusting these parameters in combination yielded a statistically better fit. Time constants predicted by these parameters superimposed on experimental data are shown in Figure 9. χ^2 based on 41 experimental data points each for 0 and 30 μM NS11021.

Potential NS11021 actions at other gating modules

Although the above results suggest that the effects of NS11021 cannot be explained without a direct effect at the PGD, we could not rule out the possibility that NS11021 might also affect VSD activation in combination with an effect at the PGD. To test this idea, we carried out additional fitting with Scheme 1 over a range of [NS11021] using the parameters illustrated in **Table 3-1** and values for L_0 in **Figures 3-1 F and 3-2 F**, while allowing either J_0 or D to be adjusted as a function of [NS11021]. These results (summarized in **Table 3-2**) show that with the parameters of Fit A, adjusting either J_0 or D did not improve the description of the data beyond the fit determined by adjusting only L_0 (shown in **Figure 3-7**).

However, the adjusting either J_0 or D in combination with L_0 did result in a marginally better description of the data using parameters from Fit B. Specifically, slight increases J_0 from 0.06 to 0.073 and 0.085 at 10 μM and 30 μM NS11021, respectively, yielded better descriptions of the G-V relations at these higher NS11021 concentrations (**Figure 3-8**). Likewise, increasing D from 16 to 19 and 21 yielded better descriptions of the G-V relations at the same higher NS11021 concentrations (**Figure 3-9**). These results seemed to suggest the additional actions of NS11021 may be detectable at these higher drug concentrations. These effects could arise either from the actions of NS11021 at a single site at the interface between the PGD and VSD, or from separate actions of NS11021 through at least two different sites, with one at the PGD and a lower-affinity site at the VSD. The possibility of a mechanism involving the VSD is supported by a fit with Scheme 2 that allows α, β, γ, δ, and D to all be adjusted to account for the gating kinetics at 30 μM NS11021 (illustrated in **Table 3-3** and **Figure 3-7C**, *dashed blue line*). This fit yielded a description of the data that was especially improved at depolarized voltages

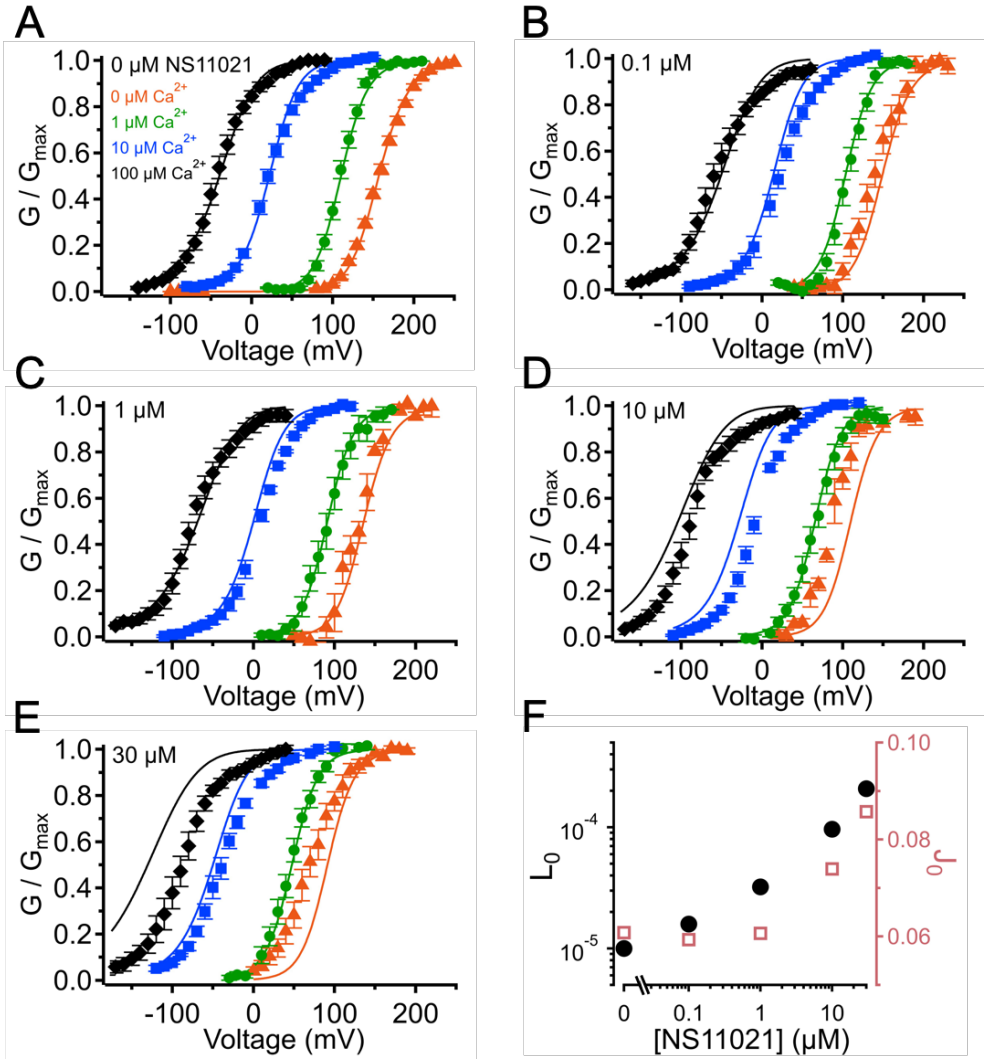


Figure 3-8: G-V-Ca^{2+} relations from Scheme 1 using parameters from Fit B to describe changing parameter J_0 in addition to L_0.

A, G-V-Ca^{2+} relations in the absence of NS11021 bathed in 0 μM (orange triangle), 1 μM (green circle), 10 μM (blue square), and 100 μM Ca^{2+} (black diamond). Parameters are listed in Table 1, Fit B. Lines represent the predictions with L_0 = 1.0 x 10^{-5}, J_0 = 0.06. **B-E,** G-V-Ca^{2+} relations in 0.1, 1, 10, and 30 μM NS11021, respectively. Lines represent predictions with L_0 = 1.6 x 10^{-5}, J_0 = 0.06 for 0.1 μM NS11021: L_0 = 3.2 x 10^{-5}, J_0 = 0.06 for 1 μM NS11021: L_0 = 9.6 x 10^{-5}, J_0 = 0.07 for 10 μM NS11021: L_0 = 2.1 x 10^{-4}, J_0 = 0.09 for 30 μM NS11021: **F,** Fitted values for L_0 (black) and J_0 (red) vs. [NS11021].

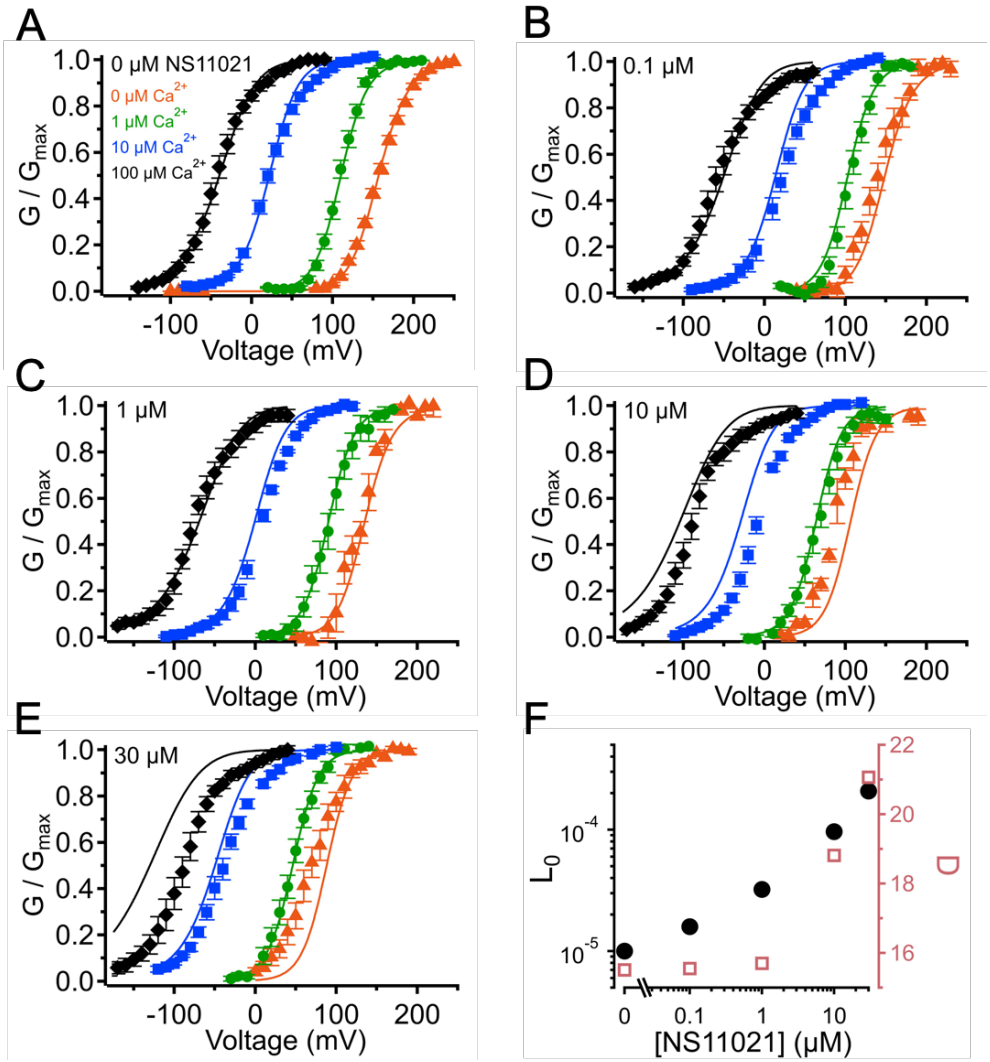


Figure 3-9: G-V-Ca^{2+} relations from Scheme 1 using parameters from Fit B to describe changing parameter D in addition to L_0.

A, G-V-Ca^{2+} relations in the absence of NS11021 bathed in 0 μM (orange triangle), 1 μM (green circle), 10 μM (blue square), and 100 μM Ca^{2+} (black diamond). Parameters are listed in Table 1, Fit B. Lines represent the predictions with $L_0 = 1.0 \times 10^{-5}$, D = 15. **B-E,** G-V-Ca^{2+} relations in 0.1, 1, 10, and 30 μM NS11021, respectively. Lines represent predictions with $L_0 = 1.6 \times 10^{-5}$, D = 16 for 0.1 μM NS11021: $L_0 = 3.2 \times 10^{-5}$, D = 16 for 1 μM NS11021: $L_0 = 9.6 \times 10^{-5}$, D = 19 for 10 μM NS11021: $L_0 = 2.1 \times 10^{-4}$, D = 21 for 30 μM NS11021: **F,** Fitted values for L_0 (black) and D (red) vs. [NS11021].

(where the VSD is activated) compared with the best fit with adjustment of only γ and δ (**Figure 3-7 C**, *solid blue line*).

To further explore the possibility of NS11021 effects on VSD activation, we performed further fitting of Scheme 2 by allowing γ and δ to be adjusted in combination with either α and β or D. Each of these combinations yielded marginally better descriptions of the gating kinetics with 30 μM NS11021 than fitting γ and δ alone (**Table 3-5** and **Figure 3-10**), again supporting the idea that NS11021 may impact VSD activation at higher drug concentrations (>10 μM).

Although the robust activating effects of NS11021 in the nominal absence of Ca^{2+} and on truncated BK channels from which the CSD is deleted suggest that NS11021 actions do not require the CSD, examination of the $V_{1/2}$ vs. [NS11021] relation suggests that the spacing between G-V curves as a function of [Ca^{2+}] may be slightly decreased at high drug concentrations, and this action is not predicted by an increase in L_0 alone with increasing [NS11021] (**Figure 3-11**). This opens that possibility that higher concentrations of NS11021 may also have some impact on Ca^{2+} sensitivity of the channel. To test this idea, we carried out additional fitting with Scheme 1 over a range of [NS11021] using the parameters illustrated in **Table 3-1** and values for L_0 in **Figures 3-1 F and 3-2 F**, while allowing either K_D or C to be adjusted as a function of [NS11021]. These results (summarized in **Table 3-4**) show that with either the parameters of Fit A or Fit B, adjusting C did yield marginal improvement in the description of the data beyond the fit determined by adjusting only L_0 (**Figures 3-12** and **3-13**, respectively). The values determined for C were not well-conserved between these two parameters sets; however, interestingly, the fractional change in C as a function of increasing [NS11021] was remarkably similar, with a 16% decrease (from 2.5 to 2.1) for Fit A and 12% decrease (from 17 to 15) for Fit B over the range of 0 to 30 μM NS11021. Together, these results are consistent with the idea that

Table 3-4: Results of changing L_0 plus a second parameter in Scheme 1 to account for effect of NS11021

Parameter	Fit A		Fit B	
	Value at 30 μM NS11021	χ^2	Value at 30 μM NS11021	χ^2
J_0	0.08	2.16	0.09	3.21
C	2.1	1.83	15	3.20
D	29	2.16	21	3.02

Simultaneous (global) fitting was performed using G-V relations acquired over a range of Ca^{2+} with [NS11021] at 0.1, 1, 10, and 30 μM, with parameters shown above adjusted in addition to L_0, as described in Methods. Adjusted parameter values at 30 μM NS11021 are shown above, along with summed χ^2 values for fitting over the entire range of [NS11021] and [Ca^{2+}] (based on 410 experimental data points). The other parameters for Scheme 1 at 30 μM NS11021 can be found in Table 1, except $L_0 = 8.7 \times 10^{-5}$ for Fit A and 2.1×10^{-4} for Fit B. Changing L_0 alone resulted in a χ^2 value of 2.16 for Fit A and 3.77 for Fit B. Starting values for each parameter can be found in Table 1, Figure 7, and Figure 8.

Table 3-5: Additional sets of fitted parameters for Scheme 2 constrained by time constants acquired with nominally 0 Ca^{2+}, and either 0 or 30 μM NS11021

Parameter	0 μM:	30 μM: α β γ δ D	30 μM: α β γ δ	30 μM: γ δ D	30 μM: γ δ
α (s^{-1})	1,162	**3,387**	**3,202**	1,162	1,162
β (s^{-1})	31,020	**17,200**	**30,860**	31,020	31,020
γ (s^{-1})	5,128	**1,154**	**1,444**	**854.6**	**925.0**
δ (s^{-1})	1.40	**1.45**	**1.36**	**2.41**	**2.23**
$z_α$ (e_0)	0.31	0.31	0.31	0.31	0.31
$z_β$ (e_0)	-0.31	-0.31	-0.31	-0.31	-0.31
$z_γ$ (e_0)	-0.025	-0.025	-0.025	-0.025	-0.025
$z_δ$ (e_0)	0.29	0.29	0.29	0.29	0.29
D	8.2	**5.2**	8.2	**5.6**	8.2
χ^2	13.4	14.9	47.6	72.3	82.5

Time constants of activation/deactivation over voltages ranging from -240 to 230 mV, with nominally 0 Ca^{2+} and 0 μM NS11021 were used to estimate parameters for Scheme 2, as described in Methods. These were used as a base set of parameters to describe the time constants in the presence of 30 μM NS11021, by adjusting parameters corresponding to J_0 (α, β), L_0 (γ, δ), or D (where D = f^2). These adjusted values are shown in **bold**. Adjusting model parameters in addition to γ and δ yielded lower χ^2 values, suggesting that VSD activation, or coupling between VSD activation and PGD opening, may be affected by NS11021. Time constants predicted by these parameters superimposed on experimental data are shown in Figure 3-10. χ^2 based on 41 experimental data points each for 0 and 30 μM NS11021.

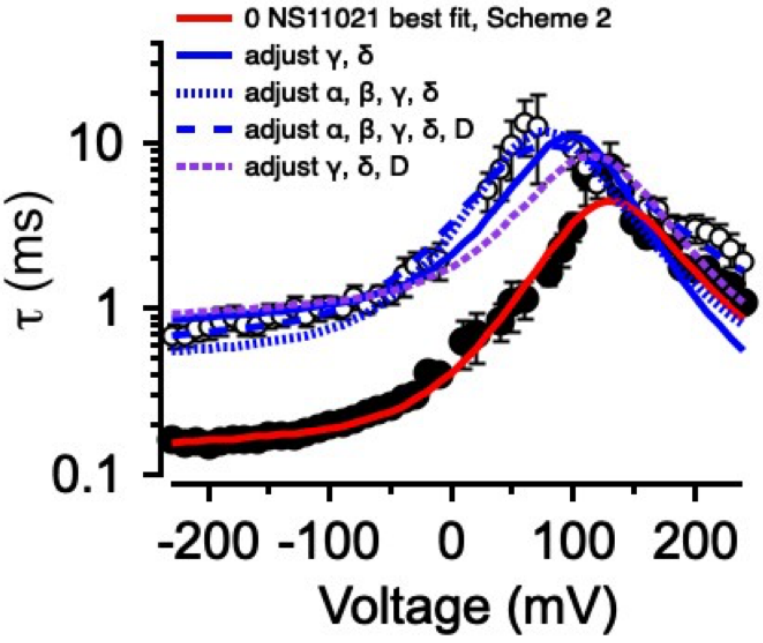


Figure 3-10: Additional fits of τ vs. voltage using scheme 2

Description of BK channel activity at nominally 0 Ca^{2+} with 0 or 30 μM NS11021 using Scheme 2. τ vs. voltage from patches with nominally 0 μM Ca^{2+}, with 0 (filled circles) or 30 μM NS11021 (open circles). Lines represent fits with Scheme 2 using parameters in Table 3 and Table 3-5: 0 μM NS11021, red; 30 μM NS11021, adjust γ δ, solid blue; adjust α, β, γ, δ, dotted blue; adjust α, β, γ, δ, D, dashed blue; adjust γ, δ, D, dashed purple.

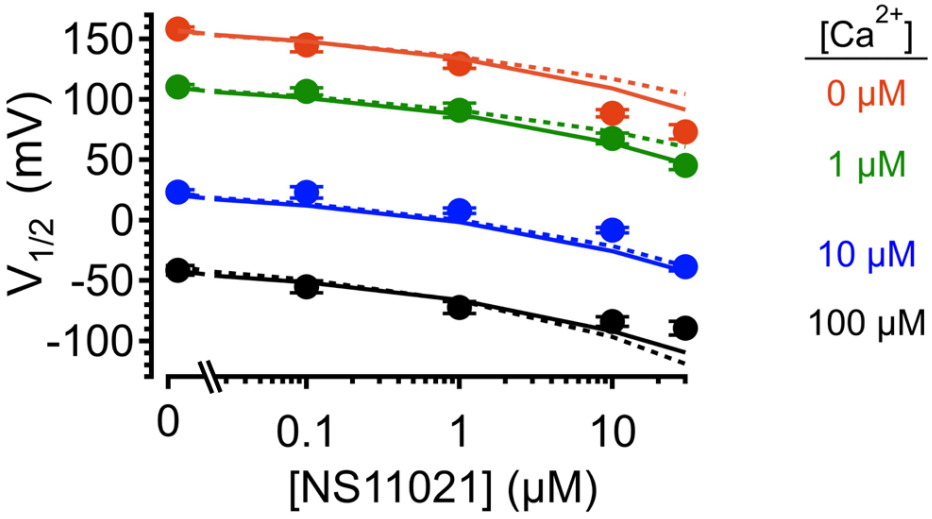


Figure 3-11: Effect of NS11021 on $V_{1/2}$ over a range of $[Ca^{2+}]$.

$V_{1/2}$ plotted as a function of [NS11021] for nominally 0 (orange), 1 (green), 10 (blue), and 100 μM Ca^{2+} (black). Experimental data (mean ± SEM, ●) are plotted along with the predicted $V_{1/2}$ for Fit A (solid line) and Fit B (dotted line), with adjustable parameter L_0.

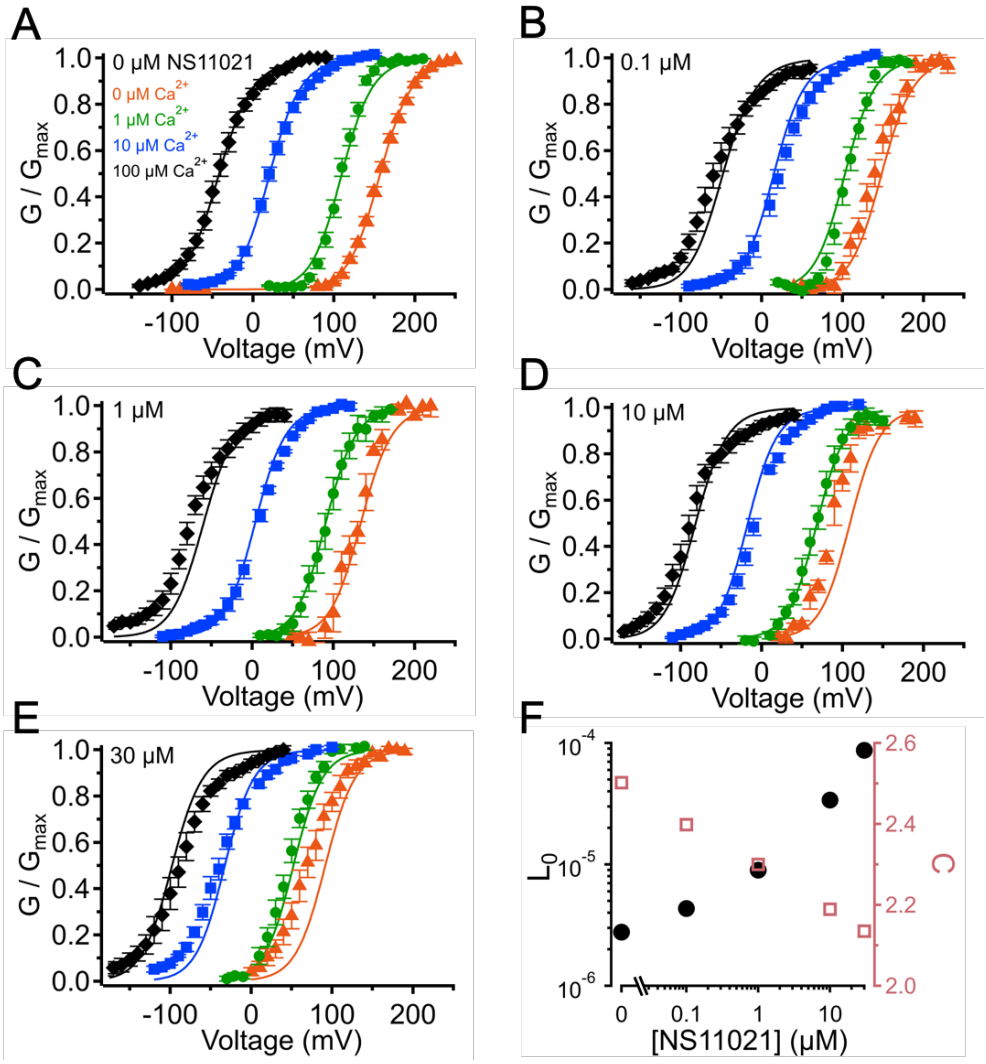


Figure 3-12: G-V-Ca^{2+} relations from Scheme 1 using parameters from Fit A to describe changing parameter C in addition to L_0.

A, G-V-Ca^{2+} relations in the absence of NS11021 bathed in 0 μM (orange triangle), 1 μM (green circle), 10 μM (blue square), and 100 μM Ca^{2+} (black diamond). Parameters are listed in Table 1, Fit A. Lines represent the predictions with $L_0 = 2.8 \times 10^{-6}$, C = 2.5. **B-E,** G-V-Ca^{2+} relations in 0.1, 1, 10, and 30 μM NS11021, respectively. Lines represent predictions with $L_0 = 4.3 \times 10^{-6}$, C = 2.4 for 0.1 μM NS11021: $L_0 = 9.0 \times 10^{-6}$, C = 2.3 for 1 μM NS11021: $L_0 = 3.4 \times 10^{-5}$, C = 2.2 for 10 μM NS11021: $L_0 = 8.7 \times 10^{-5}$, C = 2.1 for 30 μM NS11021: **F,** Fitted values for L_0 (black) and C (red) vs. [NS11021].

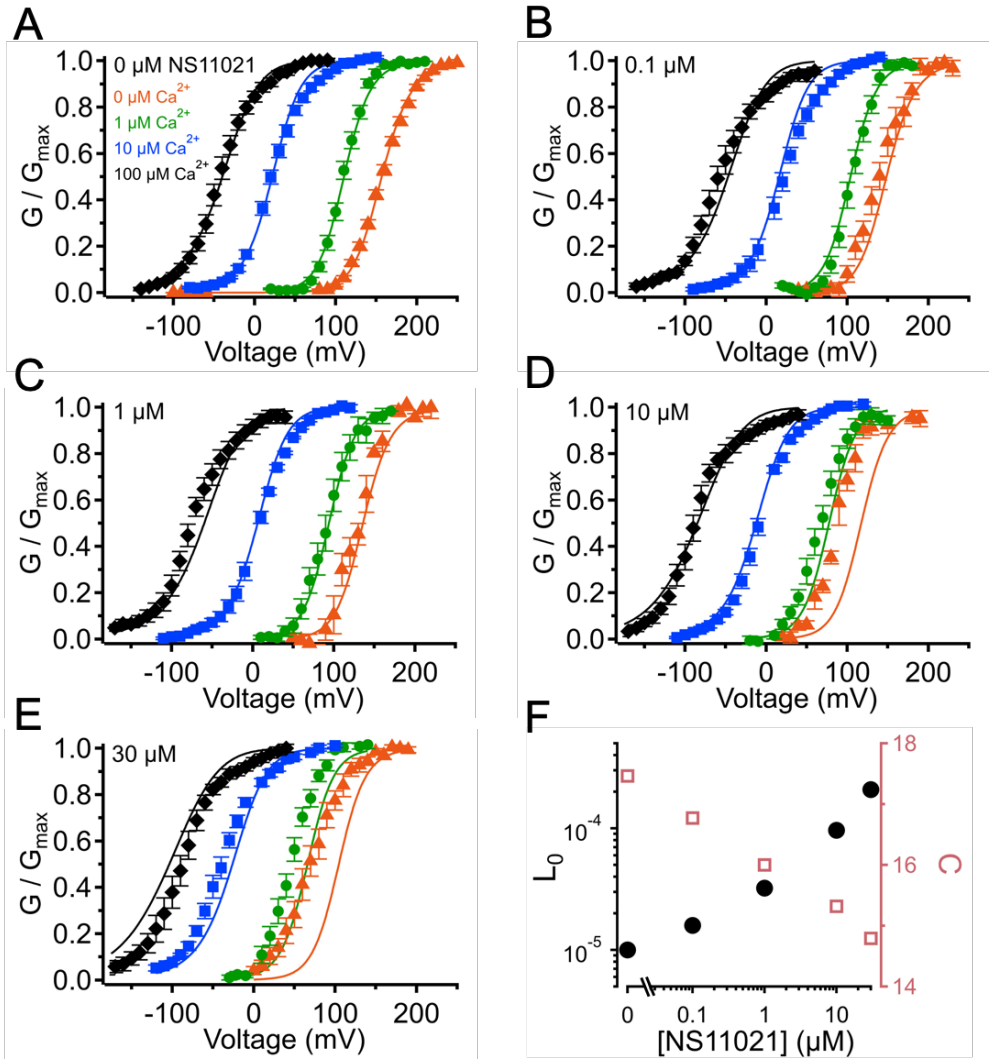


Figure 3-13: G-V-Ca^{2+} relations from Scheme 1 using parameters from Fit B to describe changing parameter C in addition to L_0.

A, G-V-Ca^{2+} relations in the absence of NS11021 bathed in 0 μM (orange triangle), 1 μM (green circle), 10 μM (blue square), and 100 μM Ca^{2+} (black diamond). Parameters are listed in Table 1, Fit B. Lines represent the predictions with $L_0 = 1.0 \times 10^{-5}$, C = 18. **B-E,** G-V-Ca^{2+} relations in 0.1, 1, 10, and 30 μM NS11021, respectively. Lines represent predictions with $L_0 = 1.6 \times 10^{-5}$, C = 17 for 0.1 μM NS11021: $L_0 = 3.2 \times 10^{-5}$, C = 16 for 1 μM NS11021: $L_0 = 9.6 \times 10^{-5}$, C = 15 for 10 μM NS11021: $L_0 = 2.1 \times 10^{-4}$, C = 15 for 30 μM NS11021: F, Fitted values for L_0 (black) and C (red) vs. [NS11021].

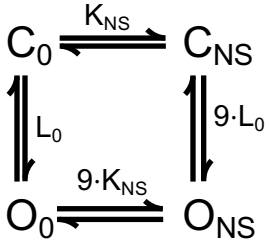
the most consistent and salient effect of NS11021 is at the PGD, whereas we cannot rule out the possibility of actions that may involve activation of the VSD and slightly decreased coupling of the CSD at higher NS11021 concentrations.

Discussion

Mechanism of NS11021 activation

Our results are consistent with the idea that NS11021 facilitates voltage-dependent BK channel opening primarily by slowing the deactivation kinetics of the channel. This is achieved through an action that does not strictly require the presence of the CSD or activation of the VSD, and thus at a minimum involves a direct action at the PGD. This effect can best be observed as an increase in the time constant of channel deactivation at negative voltages and in the nominal absence of Ca^{2+}, conditions where the Ca^{2+}- and voltage-sensors are not substantially contribution to channel activation (**Figure 3-7 C**).

In terms of mechanism, if we assume that a single NS11021 molecule is required to act on the channel to drive the C-O equilibrium toward the open state, then the experimental data suggest that each NS11021 molecule may shift the C-O equilibrium toward the open state by at least 9-fold ($2.4x10^{-3}/2.7x10^{-4}$). In this case, the activating mechanism of NS11021 for channels with both the VSD and CSD at rest can be summarized by a four-state model:



Thus the NS11021 affinity is approximately 9-fold higher in the open state compared to the closed state. Based on this scheme, we can make a minimal estimate of the energetics of NS11021 action using $\Delta\Delta G_{NS11021} = -RTln(9) = -1.3\ kcal/mole$. This estimate is based on rate constants determined from kinetic data obtained at 0 and 30 μM NS11021 using Scheme 2. Lower-limit estimates of this NS11021 coupling factor using Scheme 1, constrained by steady-state kinetic data, are 21 and 31 for Fit A and Fit B, respectively ($\Delta\Delta G_{NS11021}$ = -1.8 and -2.0 kcal/mole). Together these estimates can thus provide a range for the energetic impact of NS11021 on gating of the BK channel pore.

Because of the limited aqueous solubility of NS11021, it was not possible to perform reliable experiments with [NS11021]>30 μM, making it difficult to estimate the EC_{50} for the drug. However, based on dose-response relations obtained over a range of voltage and [Ca^{2+}], it was possible to estimate EC_{50} values for NS11021 that ranged up to 28 μM, as defined in our analysis (**Figure 2-7**). Thus if one assumes that BK channels gate primarily between one open and one closed state in the absence of NS11021, then the four-state model (above) would predict that the channel would gate among at least two open and two closed states with 30 μM NS11021, Consistent with this prediction, we observed an increase in the numbers of open and closed states in these conditions (**Figure 2-6**), which may thus reflect gating among NS11021-bound and unliganded open and closed conformations.

Physical mechanism of NS11021 activation

Permeant ions like Rb^+ are observed to increase the open probability of K^+ channels. Rb^+ can enter the K^+ channel selectivity filter, and is thought to stabilize the open state because of its higher affinity for the pore relative to K^+ (Swenson and Armstrong, 1981; Demo and Yellen, 1992). This mechanism of activation for Rb^+ and other

permeant ions has been described as a 'foot in the door' (Mienville and Clay, 1996; Swenson and Armstrong, 1981; Thompson and Begenisich, 2012; Piskorowski and Aldrich, 2006; Demo and Yellen, 1992). Similar to NS11021, Rb^+ activates BK channels and is observed to slow the closing rate while not substantially impacting the opening rate to increase the single channel mean open time. Recently it was proposed that several K^+ channels (including BK, TREK-1, and hERG) are activated by "negatively-charged activators" (NCA's) like NS11021 and permeant ions like Rb^+ through a similar mechanism (Schewe et al., 2019).

Specifically, it was hypothesized that NS11021 binds in a region lining the cavity of the channel, where the negatively charged tetrazole moiety of NS11021 can interact with permeant ions to promote K^+ binding within the pore cavity and stabilize the open state. This model would suggest that NS11021 might increase the single-channel conductance in BK channels, as observed with the action of the NCA BL-1249 on TREK-2 channels (Schewe et al., 2019). In our experiments, we did not observe a substantial increase in BK single-channel conductance at negative voltages, whereas the mean open times were increased (**Figure 2-5 C**). Structural differences between BK and hERG/TREK-2 channels may explain the different effects of NCAs on conductance. BK channels contain negatively charged residues (E321 and E324) near the "bundle crossing" of the intracellular gate of the channel which are not found in lower-conductance K^+ channels like hERG and TREK-2 (Jiang et al., 2002b; Brelidze et al., 2003; Nimigean et al., 2003). A mechanism where NCAs increase conductance by a similar mechanism as E321 and E324 in BK channels, where negative charge near the inner helix attract K^+ to the vestibule of the channel, can account for NS11021 activating with similar features of other NCAs while not substantially altering BK channel conductance.

Limitations of the proposed mechanism

Although increasing L_0 with increasing [NS11021] in the context of Scheme 1 can describe the major features of NS11021 action on BK channel gating in our experiments, we observe that the model falls short of the best possible description of the data at higher [NS11021]. Specifically, Scheme 1 predicts slightly greater Ca^{2+}-dependent shifts in $V_{1/2}$ values at 10 μM and 30 μM NS11021 than are observed in the experimental data (**Figure 3-1 D/E, Figure 3-2 D/E, Figure 3-11**). We have presented model fits using two different sets of parameters, and we have minimized the possibility that we have not identified the best parameters by re-fitting Scheme 1 many times with different sets of starting parameters. Nonetheless, it is possible that there are additional, unknown sets of parameters that might yield better descriptions of the data.

The observation that changing only L_0 is insufficient to perfectly recapitulate the Ca^{2+}-dependent spacing among G-V relations at higher [NS11021] further leaves open the possibility that at higher drug concentrations, NS11021 may act at alternative sites to effect voltage or Ca^{2+} sensing. The presumed lower-affinity NS11021 sites may be poorly accessible from the aqueous solution, or they may be characterized by weak chemical interactions to yield a rapid off-rate. It is also possible that the slightly decreasing effect of high [NS11021] on $V_{1/2}$ values with high [Ca^{2+}] (illustrated in **Figure 3-11**) could be that activation of the Ca^{2+}-sensor allosterically inhibits the action of NS11021, through an action distinct from the activating effect of the drug.

While direct assessment of allosteric coupling between the VSD and CSD (parameter “E”) has historically been challenging, it was recently hypothesized that VSD-CSD coupling may have a substantial impact on voltage-dependent activation (Lorenzo-Ceballos et al., 2019; Hite et al., 2016). Consistent with this idea, although both Fit A and Fit B describe G-V-Ca^{2+} relations similarly well at 0 μM NS11021 (**Table 3-1**), changing L_0

in Fit A better accounts for the observed effects of NS11021 at higher drug concentrations and contains an 18-fold greater value for parameter E than Fit B (**Table 3-1, Figure 3-7 A, Figure 3-8 A**).

It is important to acknowledge that Scheme 1 itself is a simplification of the underlying gating mechanism, as it is well established that BK channels contain multiple Ca^{2+} binding sites on each channel subunit that are not accounted for or described by Scheme 1. However, fitting with a more complex model with multiple Ca^{2+} binding sites would give rise to additional difficulty in identification of parameters, as the additional parameters would be poorly constrained by our experimental data. Scheme 1 should thus be considered a "working hypothesis". Similarly Scheme 2 should be considered a simplification of gating in the absence of Ca^{2+}, as it cannot account for gating transitions of the CSD that could arise from infrequent Ca^{2+} binding events at very low [Ca^{2+}]. In addition, simplifying assumptions are made in the parameterization of both Schemes 1 and 2; for example, voltage and Ca^{2+} dependent transitions in these schemes are modeled as being independent for each subunit, whereas they may exhibit cooperativity (Niu and Magleby, 2002; Qian et al., 2006; Shelley et al., 2010). Despite these shortcomings, Schemes 1 and 2 enable the quantitative testing of mechanism that support the idea that NS acts relatively selectively at C-O gating transitions.

Conclusions

We show that NS11021 activation of BK channel is largely due to stabilization of an open state at the PGD. Whereas it has been hypothesized that NS11021 binds within the cavity of the channel (Schewe et al., 2019), the actions of the small-molecule activators Cym04 and NS1619 appear to depend on the integrity of the C-linker of the BK channel, and can be disrupted by single amino acid substitutions in that region (Gessner et al.,

2012). The functional effects of Cym04 and NS1619 also appear distinct from those of NS11021; Cym04 and NS1619 primarily act by shifting the voltage-sensor equilibrium toward the activated state (increasing "J_0" in Scheme 1), whereas the Po is altered little under conditions where the VSD is at rest. It will be important to identify molecular mechanisms of additional structural classes of BK channel targeting drugs to improve our understanding of BK channel gating, as well as the development of novel therapies to treat disease.

CHAPTER 4

EFFECTS OF P805L, A LOSS-OF-FUNCTION MUTATION LINKED TO HUMAN DISEASE

Introduction

Diseases linked to specific ion channel defects, known as “channelopathies”, can present with a wide-range of potential phenotypes (Kim, 2014). Due to the pervasive role of ion channels in electrical signaling and modulation of action potentials, channelopathies altering excitability can lead to the development of arrhythmia, seizure, or skeletal muscle disorders (Olson et al., 2006; Bernard et al., 2004). Channelopathies can occur *de novo* or by familial inheritance, and are classified as either gain-of-function (GOF) or loss-of-function (LOF) based on the specific alteration to the ion channel function. In terms of mechanism, channelopathies can alter channel conductance and selectivity, open probability, the rate at which channels open and close, trafficking, and protein folding. Characterizing each mutation based on specific effects to channel function is an important first step into understanding the pathogenesis of disease and will improve our understanding about ion channel gating.

Mutations to the KCNMA1 gene encoding large conductance Ca^{2+}-activated K^+ channels (BK channels) have been associated with seizure, movement disorders, developmental delay, and intellectual disability (Bailey et al., 2019). While at least 21 KCNMA1 mutations have been identified in humans, the specific alterations to channel function underlying each mutation remain largely unresolved for LOF mutations. BK channels are activated by both cytosolic Ca^{2+} and a depolarized membrane potential to participate in important physiological processes such as regulation of smooth muscle contraction and neuronal excitability (Latorre et al., 2016). Structurally, BK channels exist

as tetramers with each subunit containing a voltage-sensing domain (VSD; S0-S6), Ca^{2+}-sensing domain (CSD; RCK1/RCK2), and a pore-forming/gate domain (PGD; P5-S6). Mutations may alter either Ca^{2+} sensitivity, voltage sensor activation, intrinsic opening of the pore, or the allosteric coupling between each of these domains (Horrigan and Aldrich, 2002).

In the present chapter, I characterize a LOF mutation, P805L, that was previously identified in a patient presenting with severe speech delay, developmental delay, and intellectual disability (Liang et al., 2019). P805L has been shown to change the conductance-voltage (G-V) relationship in addition to having less abundant expression (Liang et al., 2019). In terms of alterations to BK channel gating, our data suggest that P805L channels activate less rapidly than WT BK channels. Alterations to gating persist in the nominal absence of Ca^{2+}, indicating that the P805L mutation must alter properties independent of Ca^{2+}-sensitivity. Finally, we show that P805L channels can be activated by the small molecule NS11021, indicating that pharmacological modulation of BK channelopathies may have the ability to ameliorate altered BK currents to more resemble wild-type.

Materials and Methods

Electrophysiology

KCNMA1 α-subunit (Genbank accession no. U11058.2) was subcloned into pcDNA3.1/Zeo(+), with the following modification: the “1D4” epitope tag (amino acid sequence TETSQVAPA) was incorporated in-frame, immediately following the C-terminal sequence (SRDKQKYVQEERL). Addition of the 1D4 tag was performed to aid with potential purification of protein for future biochemical experiments

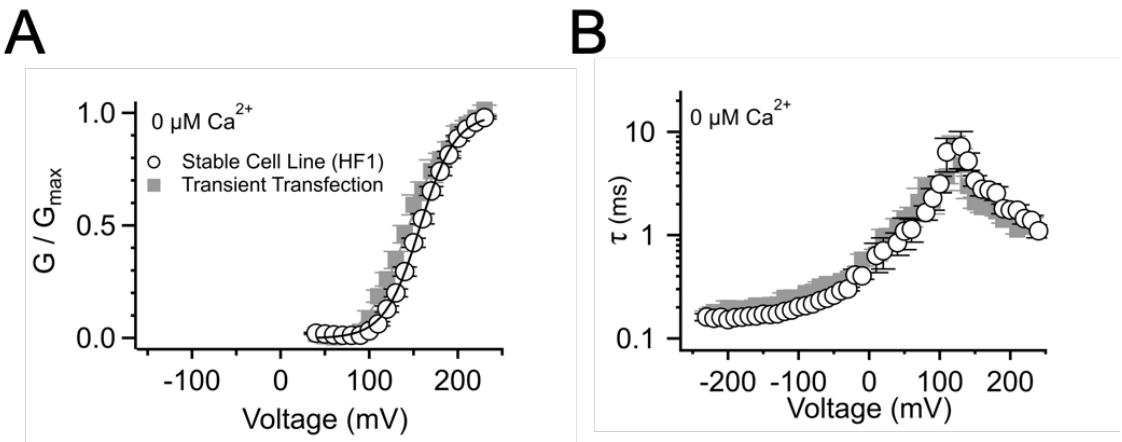


Figure 4-1: BK channel gating was unaffected by the deletion of the N-terminal myc tag and addition of the C-terminal 1D4 tag.

A, G-V relations of a cell line stably expressing myc-tagged WT BK channels (HF1; white circles, $V_{1/2}$ = 158.2 ± 1.7 mV; z = 1.39 ± 0.03 e_0; n=54) and cells which have been transiently transfected with 1D4-tagged BK channels (gray circles, $V_{1/2}$ = 145.3 ± 3.4 mV; z = 1.16 ± 0.08 e_0; n=4) in the presence of nominally 0 Ca^{2+}. **B,** Time constant of relaxation (τ) for activation and deactivation kinetics plotted as a function of voltage. $\tau_{-200\ mV}$ = 0.15 ± 0.01 ms and $\tau_{+200\ mV}$ = 1.75 ± 0.19 ms for the stable cell line and $\tau_{-200\ mV}$ = 0.20 ± 0.03 ms and $\tau_{+200\ mV}$ = 1.64 ± 0.30 ms for cells which have been transiently transfected with BK channels.

(Molday and Molday, 2014). This differed slightly from BK channels overexpressed in HF1 cells, as the HF1 construct contained a myc-epitope tag (MGAEEQKLISEEDL) immediately preceding residue M1, with no C-terminal 1D4-epitope tag. The P805L mutation was incorporated in the 1D4-tagged construct by PCR-based mutagenesis and verified by DNA sequencing (GenScript, Piscataway NJ). Human embryonic kidney cells (HEK-293T; American Type Culture Collection) were transiently transfected with KCNMA1 expression plasmid and GFP (10:2 µg ratio), and electrophysiological experiments were performed 1-3 days after transfection. Transient transfection of wild-type KCNMA1 containing the 1D4-tag (without the N-terminal myc-tag) resulted in channels with nearly identical gating properties as those recorded using stably-expressing HF1 cells (**Figure 4-1**; Meera et al., 1997; Rockman et al., 2020). Experiments were performed using excised inside-out patches as described in Chapter 2. Data for WT channels is the same as those presented in Chapters 2 and 3 (Rockman et al., 2020).

Data Analysis

G/Gmax and macroscopic time constants were analyzed as described in Chapter 2 “Materials and Methods”. Additionally, amplitude histograms for single channel recordings were created, with bin width of 0.2 pA. To facilitate comparison between patches, all distributions were normalized to contain 100 total events. Each distribution was fit with a Gaussian function:

$$H * \exp\{ -(\frac{a-a_0}{2\sigma^2})^2 \}$$

(Eq. 4.1)

where H represents the height of the curve peak, a_0 represents the unitary current amplitude at the center of the peak, and σ represents the standard deviation of the sample, proportional to the width of the distribution. For single channel recordings with few (< 5)

active channels, the number of channels was determined by counting the maximum number of observed open levels (Horn, 1991).

Kinetic Modeling

Global fitting of G-V relations of P805L mutant channels in nominally 0, 1, 10, and 100 μM Ca^{2+} was performed using the dual allosteric model (Scheme 1) as described in Chapter 3 "Materials and Methods." Using parameter values for Fit A and Fit B used to describe activity of WT channels (**Table 3-1**), parameters (J_0, K_D, L_0, etc.) were systemically adjusted to determine which parameters could best describe the changes in BK channel gating caused by the P805L mutation. Parameters were initially adjusted individually, and subsequent adjusted in pairs. Quality of fit was evaluated by the sum of χ^2 for fits at all [Ca^{2+}].

Similarly, the time course of channel activation in the nominal absence of Ca^{2+} was analyzed using a 10-state model (Scheme 2) as described in Chapter 3 "Materials and Methods." To describe the time course of P805L mutant channels, parameters describing activity of WT channels were initially used (**Table 3-3**), then systematically adjusted to determine which parameters could best describe changes caused by the P805L mutation. Parameters were initially adjusted individually with all other parameters constrained to WT values. Subsequent fits were generated with parameters adjusted in pairs and groups, corresponding to alterations to specific gating modules.

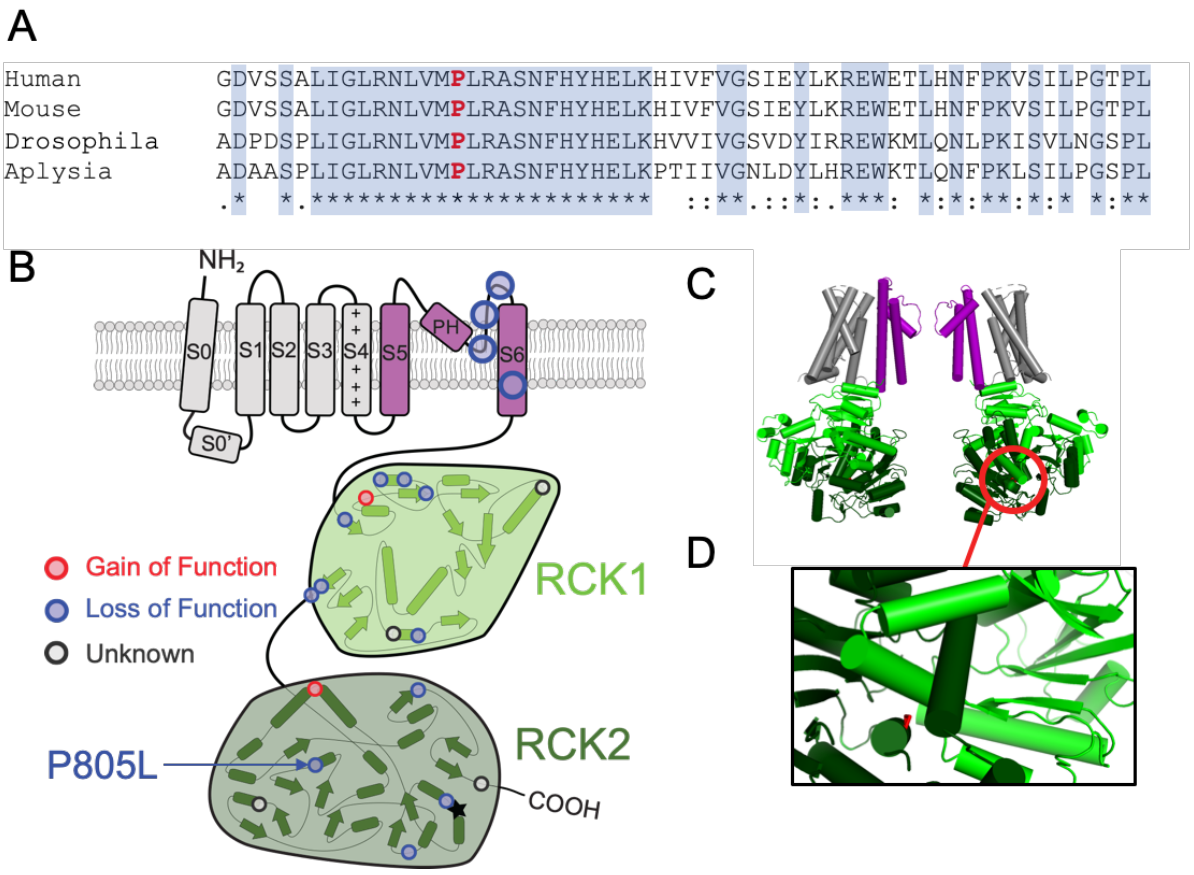


Figure 4-2: P805 is a highly conserved residue in RCK2.

A, Sequence alignment of BK channel encoding genes from human (*homo sapian*, KCNMA1 gene), mouse (*Mus musculus*, KCNMA1 gene), fly (*Drosophila melanogaster,* slo1), and slug (*Aplysia californica ,* slo1)*,* species which are commonly used to study BK channels. Residues conserved in all four species are shaded in blue. P805 (red) is conserved in all four species. **B**, Cartoon of BK channels domain architecture with approximate secondary structure. Voltage sensing domain (S0-S4) is shown in gray, pore-forming/gate domain (S5-PH-S6) is shown in purple, RCK1 is shown in light green, and RCK2 is shown in dark green. α-helices are denoted by a rectangle and β-strands are denoted by a triangle. Approximate locations of 21 known mutations are denoted with a circle. Red represents a gain-of-function mutation, blue represents a loss-of-function mutation, and gray represents mutations that have not yet been functionally characterized. P805 is located in an α-helix in RCK2. Location of the high-affinity Ca^{2+} bowl indicated by ★. **C**, *Aplysia* BK channel structure bound to Ca^{2+} shown with same color scheme. (Adapted from pdb: 5TJ6; Tao *et al.* 2016). Approximate location of P805 shown with red circle, P805 is also shown in red. **D**, View of P805 in proximity with additional secondary structural elements.

Results

P805 is a highly conserved residue in RCK2

Alignment of the human KCNMA1 protein sequence with multiple species shows remarkable homology in the region of P805, which is found near the center of a 22 residue region fully conserved in the human, mouse, *Drosophila*, and *Aplysia* sequences (**Figure 4-2A).** Structurally, P805 is located in RCK2 in an α-helix that associates closely with a helix-loop-helix motif of RCK1 **(Figure 4-2 B-D).** Residues within 4 Å include A495 and Q496 from RCK1, along with I941 and T942 from RCK 2.

P805L alters voltage-dependent gating in the presence of 10 µM Ca^{2+}

To quantify how the P805L mutation alters gating of BK channels, we compared currents from excised inside-out patches in HEK293T cells overexpressing P805L channels with currents from cells overexpressing WT BK channels. As can be seen in the raw macroscopic current recordings, P805L channels bathed in 10 µM Ca^{2+} activate with a slower time course of activation compared to WT **(Figure 4-3 A/B)**. Conductance–voltage (G-V) relations show that P805L channels activate with a 41 mV right-shift in the voltage of half-maximal activation ($V_{1/2}$) from 23.4 ± 2.0 mV (n=20) in WT channels to 64.1 ± 2.3 mV in P805L channels (n=25) (**Figure 4-3 C**). This shift in $V_{1/2}$ is associated with an 18% decrease in gating valence, which was 1.29 ± 0.05 e_0 in WT channels and 1.05 ± 0.06 in P805L channels.

We next quantified the time constant of relaxation (τ) for both activation and deactivation kinetics as described previously (**Chapter 2**; Horrigan et al., 1999; Cui et al., 1997; Rockman et al., 2020). We observed that P805L channels activated more slowly while deactivating at a similar rate (**Figure 4-3 D**). Deactivation $\tau_{-200\ mv}$ remained approximately the same, changing from 0.46 ± 0.04 (n=9) ms for WT channels to 0.44 ±

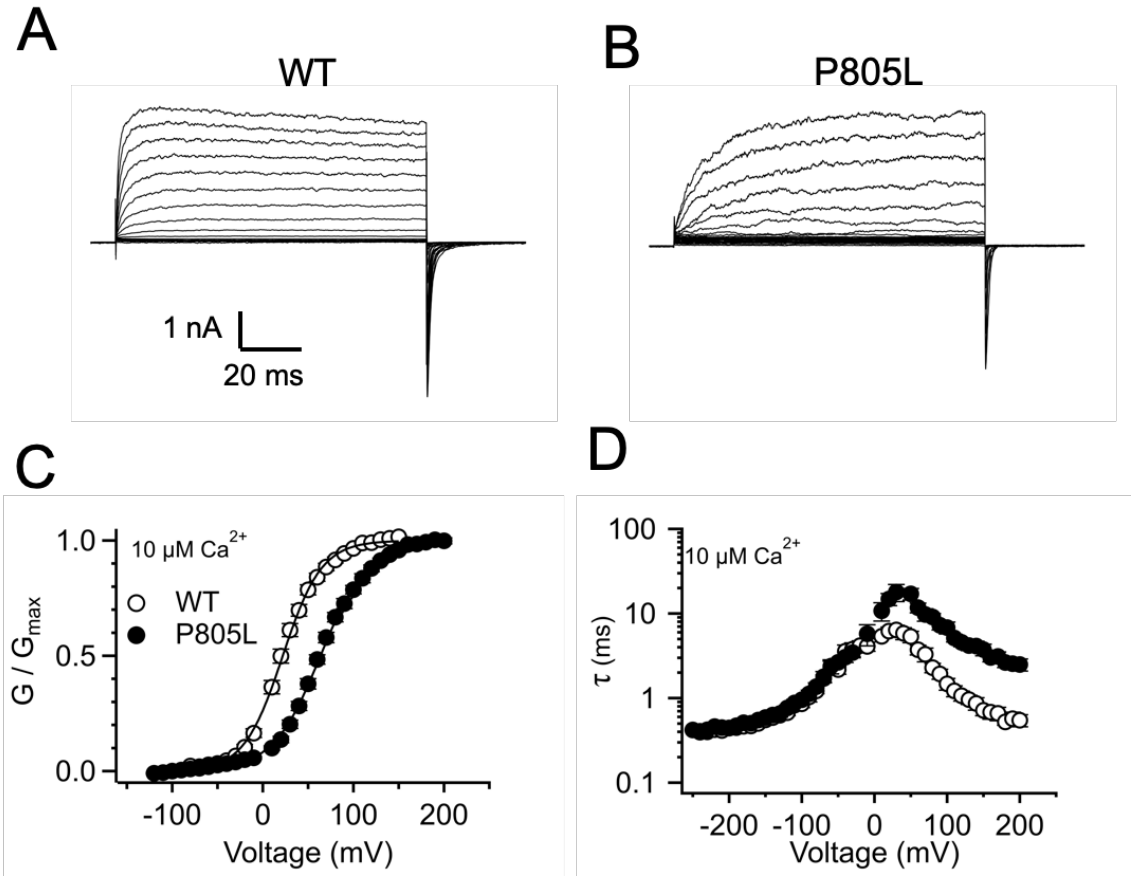


Figure 4-3: P805L mutation slows activation

A, Representative currents of WT BK channels. Channels were initially held at -100 mV and stepped to a series of depolarized voltages shown between -90 mV and +110 mV in 10 mV increments before quantifying tail current amplitude at -90 mV. **B,** Representative currents of P805L channels at the same voltages as panel A. **C,** Conductance vs voltage (G-V) relations and Boltzmann fit for WT ($V_{1/2}$ = 23.4 ± 2.0; z = 1.29 e_0 ± 0.05; n=20) and P805L channels ($V_{1/2}$ = 64.1 ± 2.3; z = 1.05 e_0 ± 0.06; n=25). **D,** Time constant of relaxation (τ) plotted as a function of voltage for WT ($\tau_{+200\ mV}$ = 0.55 ± 0.09) and P805L channels ($\tau_{+200\ mV}$ = 2.5 ± 0.39).

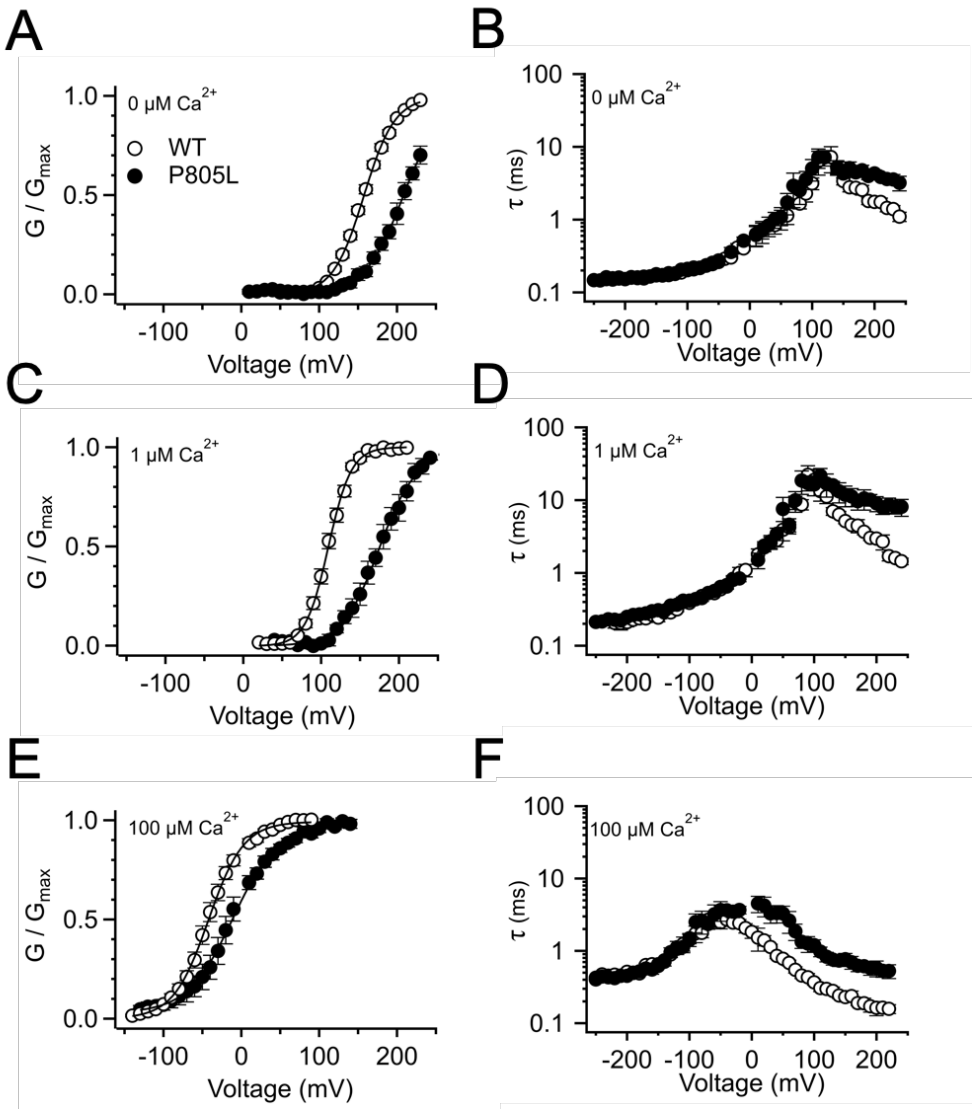


Figure 4-4: P805L mutation alters BK channel activity over a range of [Ca^{2+}].

A, Conductance vs. voltage (G-V) relations and Boltzmann fit from patches with nominally 0 μM Ca^{2+} for WT ($V_{1/2}$ = 158.2 ± 1.7; z = 1.39 e_0 ± 0.03; n=54) and P805L channels ($V_{1/2}$ = 209.5 ± 3.7; z = 1.12 e_0 ± 0.15; n=9). **B,** Time constant (τ) vs. voltage from patches with nominally 0 Ca^{2+} for WT ($\tau_{+200\ mV}$ = 0.89 ± 0.01) and P805L channels ($\tau_{+200\ mV}$ = 4.31 ± 0.51). **C,** (G-V) relations and Boltzmann fit from patches with 1 μM Ca^{2+} for WT ($V_{1/2}$ = 110.3 ± 2.1; z = 1.97 e_0 ± 0.05; n=26) and P805L channels ($V_{1/2}$ = 176.8 ± 5.5; z = 1.13 e_0 ± 0.13; n=6). **D,** τ vs. voltage from patches with 1 μM Ca^{2+} for WT ($\tau_{+200\ mV}$ = 0.99 ± 0.01) and P805L channels ($\tau_{+200\ mV}$ = 9.06 ± 2.11). **E,** (G-V) relations and Boltzmann fit from patches with 100 μM Ca^{2+} for WT ($V_{1/2}$ = -41.5 ± 4.0; z = 1.27 e_0 ± 0.09; n= 15) and P805L channels ($V_{1/2}$ = -13.7 ± 5.5; z = 0.92 e_0 ± 0.12; n=7). **F,** τ vs. voltage from patches with 100 μM Ca^{2+} for WT ($\tau_{+200\ mV}$ = 0.16 ± 0.04) and P805L channels ($\tau_{+200\ mV}$ = 0.58 ± 0.14).

0.04 ms (n=25) for P805L channels, while activation $\tau_{+200\ mv}$ increased 4.6-fold from 0.55 ± 0.09 ms for WT channels to 2.50 ± 0.39 ms for P805L channels.

P805L channels exhibit altered gating over a range of $[Ca^{2+}]$

To determine how the altered gating of P805L channels is dependent on $[Ca^{2+}]$, we next assayed P805L channel activity over a range of $[Ca^{2+}]$. Since P805 is located in RCK2, we hypothesized that the magnitude of alterations to gating would likely change with $[Ca^{2+}]$. Furthermore, if P805L only impacted Ca^{2+}-dependent aspects of gating (K_D, C, or E in the dual-allosteric model), then a minimal change to gating in nominally zero Ca^{2+} could potentially be observed. Instead, we observed a right-shift in the $V_{1/2}$ for P805L channels in nominally zero (chelated with 5 mM EGTA; see methods), 1, 10, and 100 μM Ca^{2+} (**Figure 4-4 A/C/E**). In terms of kinetics, a similar increase in the activation time constant without a corresponding change to the deactivation time constant was also observed at all $[Ca^{2+}]$ tested **(Figure 4-4 B/D/F).** The observed changes in gating of P805L channels at nominally zero Ca^{2+} suggest that Ca^{2+} binding to the Ca^{2+} sensor is not strictly required for the P805L mutation to alter channel gating, and that the mutation must alter aspects of gating independent of Ca^{2+}-sensitivity.

P805L mutation does not substantially alter single-channel conductance

Additional mechanistic information was obtained by analyzing single channel recordings. In **Figure 4-5,** channel activity in a patch containing one active P805L mutant channel is compared to activity in a patch expressing one WT channel. Recordings at +50 mV in in 10 μM Ca^{2+} show that unitary current amplitude is not substantially changed between WT (13.2 ± 1.8; n=3 patches) and P805L channels (14.2 ± 1.4; n=3 patches) (**Figure 4-5 A-D**). Furthermore, the distribution of unitary current amplitudes for both WT

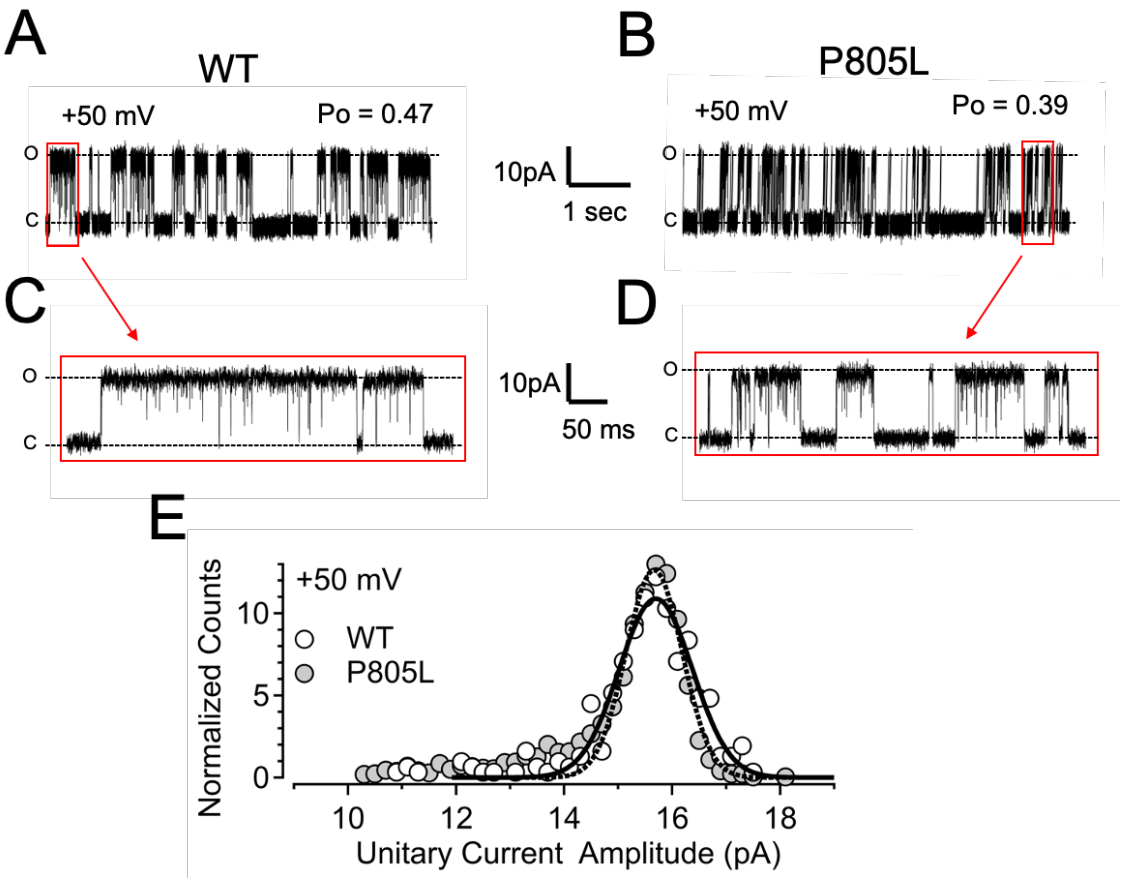


Figure 4-5: P805L mutation does not alter unitary current amplitude.

A. Single channel recording from a patch with one active WT BK channel in 10 μM cytosolic Ca^{2+}. Mean amplitude of each opening was 15.6 ± 1.0 pA (mean ± standard deviation) based on 323 openings. **B,** Single channel recording from a patch with one active P805L channel in 10 μM cytosolic Ca^{2+}. Mean amplitude of each opening was 15.2 ± 1.2 pA (mean ± standard deviation) based on 2,473 openings. **C and D**, Activity from panels A and B shown at an expanded time scale. **E,** Histogram of unitary current amplitudes fit with a Gaussian function for WT (solid line; H = 10.9; center = 15.7; σ = 0.69) and P805L channels (dashed line; H = 12.6; center = 15.7; σ = 0.60). Each condition normalized to have 100 events. Each bin is 0.2 pA in width.

and P805L channels is centered at 15.7 pA in the shown recordings, indicating that the P805L mutation is not altering the overall distribution of current amplitudes (**Figure 4-5 E**). Similar results were observed in two additional patches for both WT and p805L mutant channels. In summary, these results suggest that the P805L mutation does not alter the single channel conductance of BK channels.

Changes in Ca^{2+}-sensitivity alone cannot describe behavior of P805L channels

To gain further insight into specific gating steps altered by the P805L mutation, we used a quantitative approach based on the well-established dual allosteric model (**Scheme 1**). Although BK channels are known to contain a total of eight high-affinity Ca^{2+} binding sites per channel, the model here utilizes a simplified gating scheme where four Ca^{2+} binding sites and four voltage sensors activate independently to contribute to pore opening (Rothberg and Magleby, 1999; Horrigan and Aldrich, 2002). Our approach was to initially start with the two parameter sets identified in **Chapter 3** to describe G-V relations over a rage of [Ca^{2+}] (G-V-Ca^{2+}-relations) for WT BK channels (**Table 3-1**). Our goal was to evaluate whether G-V-Ca^{2+}-relations for P805L channels could be described by only changing a single parameter (J_0, K_D, L_0, etc.) while constraining the remaining parameters. Our results indicate that changing J_0, L_0, or D result in the best fits for Fit A and Fit B, as evaluated by the χ^2 statistic (**Table 4-1**, **Figure 4-6**, **Figure 4-7**). Values obtained for J_0, L_0, or D were decreased by 2.6-, 9.1-, and 1.8-fold respectively for Fit A and 4.2-, 11.9-, and 2.4-fold respectively for Fit B. By contrast, changing parameters involved in Ca^{2+}-sensitivity (either K_D, C, or E) could not account for the observed 51.3 mV right-shift in $V_{1/2}$ caused by the P805L mutation in nominally 0 Ca^{2+} when changed alone (**Figure 4-7**). While this does not eliminate the possibility that Ca^{2+}-sensitivity is altered, these data suggest that the P805L mutation must at a minimum alter either J_0, L_0, D, z_J, or z_L.

Table 4-1: Results of changing individual parameters in Scheme 1 to account for altered gating in P805L channels

Parameter	Fit A			Fit B		
	WT Value	P805L Value	χ^2	WT Value	P805L Value	χ^2
J_0	0.08	0.03	0.68	0.06	0.01	0.31
K_D (μM)	41	156	2.90	17	65	3.05
L_0	2.8×10^{-6}	3.0×10^{-7}	0.69	1.0×10^{-5}	8.4×10^{-7}	0.93
z_J (e_0)	0.30	0.14	2.01	0.41	0.20	1.86
z_L (e_0)	0.57	0.19	2.11	0.58	0.20	2.71
C	2.5	1.4	3.49	17	9.2	3.90
D	29	16	0.68	16	6.5	0.36
E	20	7.6	3.27	1.1	0.20	2.78

Simultaneous (global) fitting was performed using G-V relations acquired for 0, 1, 10, and 10 μM Ca^{2+} with WT parameters adjusted individually, as described in Methods. Adjusted parameter for P805L channels, along with summed χ^2 values for fitting over the entire range of [Ca^{2+}] (based on 105 experimental data points).

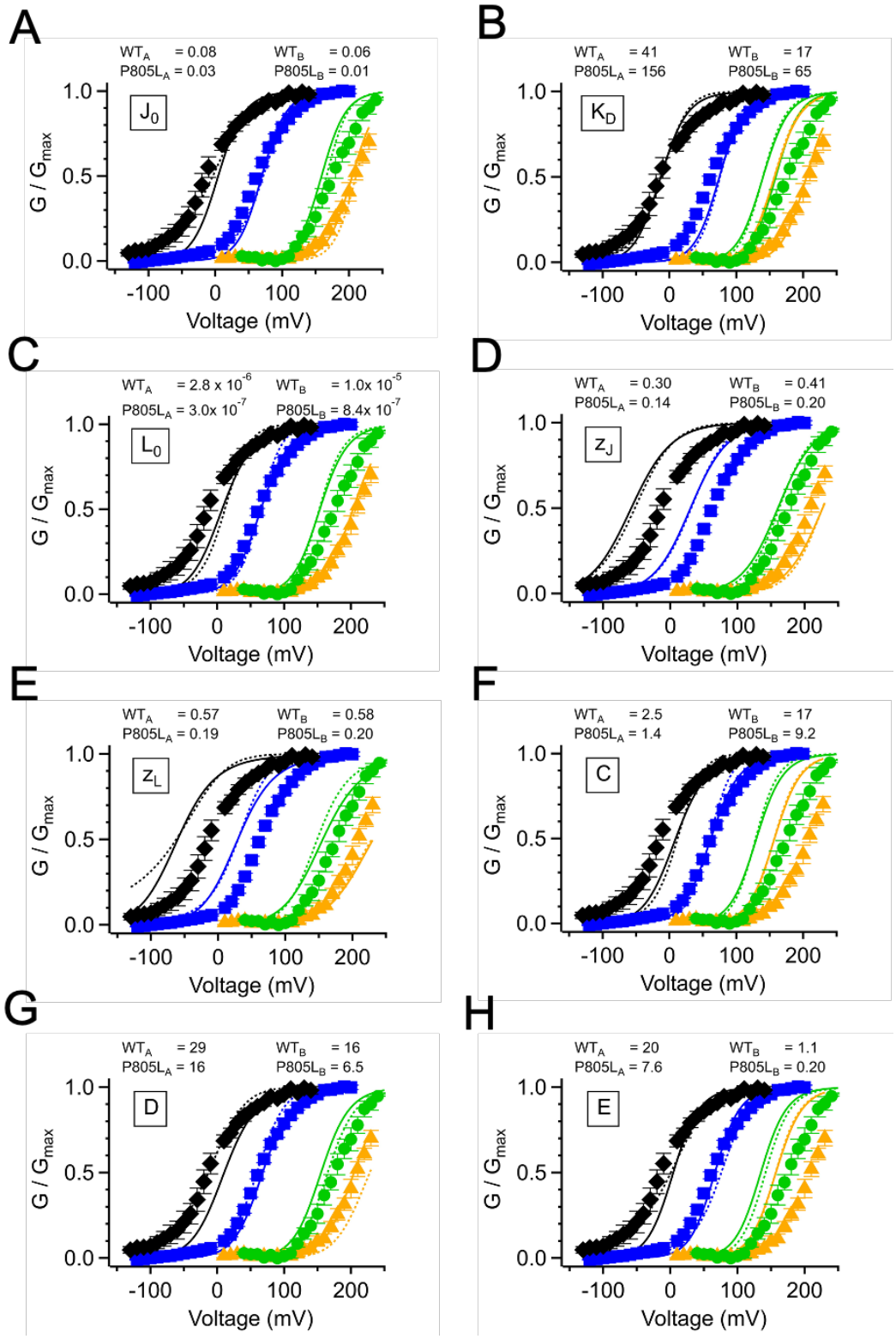


Figure 4-6: G-V-Ca^{2+} relations for P805L channels changing a single parameter from Fit A and Fit B.

G-V-Ca^{2+} relations for P805L channels bathed in 0 μM (orange triangle), 1 μM (green circle), 10 μM (blue square), and 100 μM Ca^{2+} (black diamond). Lines represent fits with parameters for Fit A (solid line) and Fit B (dashed line) while changing only the parameter indicated on the top left corner of each plot. Value of changed parameter can be found above each panel, χ^2 for each fit can be found in **Table 4-1**.

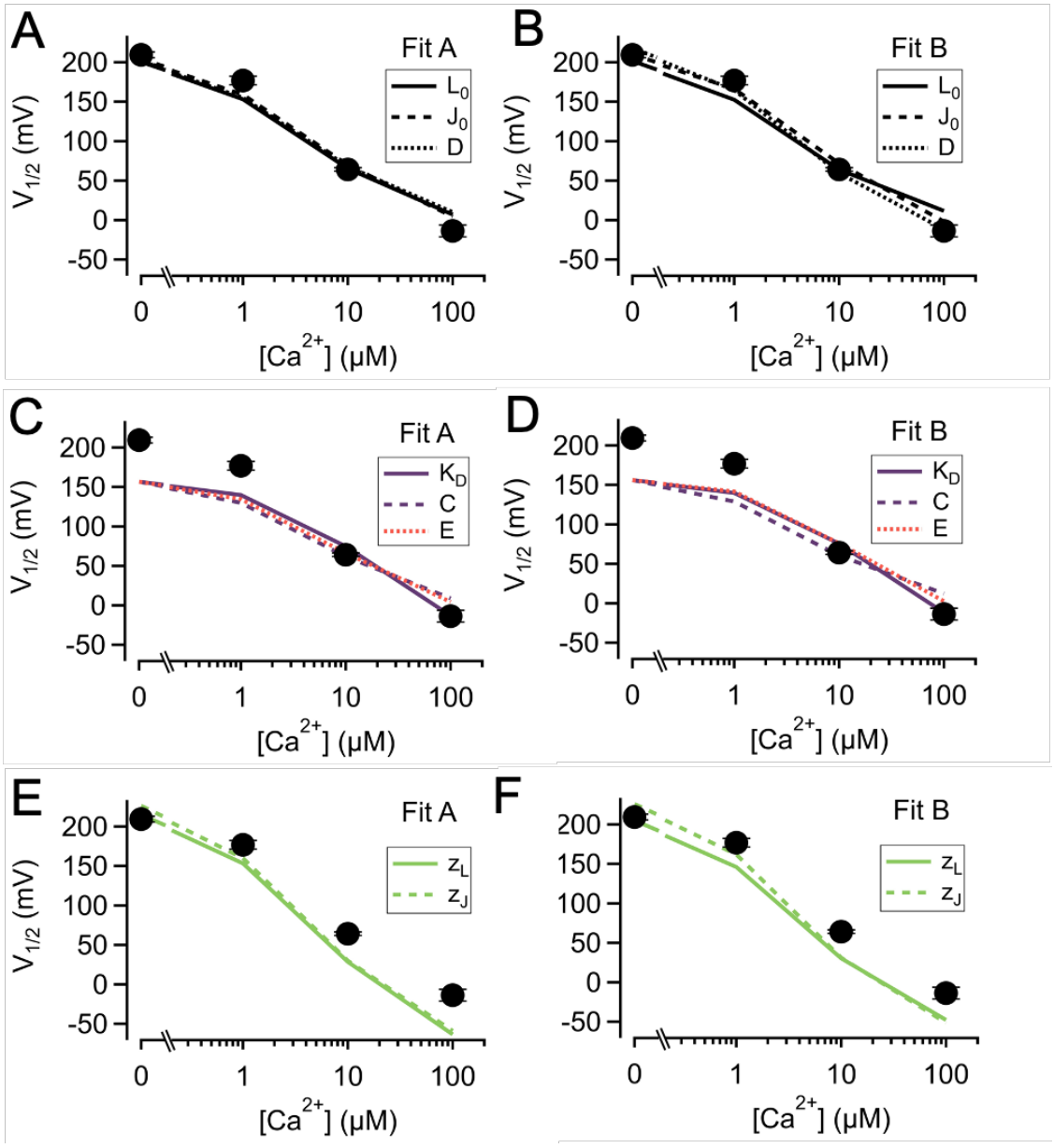


Figure 4-7: $V_{1/2}$ predicted by changing individual parameters in Scheme 1.

Experimentally determined $V_{1/2}$ plotted as a function of $[Ca^{2+}]$ for P805L channels (black circles). Lines represent predictions of $V_{1/2}$ obtained by changing the indicated parameter for Fit A (left) and Fit B (right). **A and B,** Changing only parameter J_0, L_0, or D. **C and D** Changing only a Ca^{2+}-sensative parameter K_D, C, or E. **E and F,** Changing only parameters associated with gating valence z_L, or z_J . Values and χ^2 for each parameter can be found in **Table 3-1** and **Table 4-1**, and the corresponding plot of G-V-Ca^{2+} relations can be found in **Figure 4-6.**

We next wanted to determine whether changing a second parameter could better account for observed changes caused by the P805L mutation. **Table 4-2, Table 4-3,** and **Table 4-4** show parameter values and error obtained from changing an additional parameter from the best fits altering J_0, L_0, and D. Results indicate that while changing most parameters did not improve quality of the fit, changing the gating valence for J and L (z_J and z_L) modestly reduced the χ^2 in most combinations. Physically, z_J and z_L represent the charge associated with voltage-sensor activation and pore opening, respectively (Horn, 1996; Noceti et al., 1996).

To determine the mechanism underlying the slowed activation of P805L mutant channels in terms of a quantitative kinetic scheme, we next fit time constants in the nominal absence of Ca^{2+} using the 10-state model that can describe BK channel gating in the absence of Ca^{2+} (**Scheme 2**). Our first approach was to start with parameter values which best describe activity of WT BK channels (**Table 3-3**), and initially change only one parameter (α, β, etc.) while constraining the remaining parameters to describe the activation and deactivation time constants of channels with the P805L mutation (**Table 4-5**). Evaluation of χ^2 for each fit indicates that decreasing either the C $\rightarrow$ O rate constant (δ) or charge-movement ("gating valence") associated with the C $\rightarrow$ O transition (z_δ), can best account for slowed channel activation (**Figure 4-8 A**).

To facilitate comparison between results obtained modeling macroscopic rate constants (Scheme 2) to results from modeling G-V-Ca^{2+} relations (Scheme 1), rate constants can be changed in pairs. For example, rate constants α and β are associated with parameter J_0 for voltage-sensor activation, rate constants γ and δ are associated with L_0 for intrinsic pore opening, and D in both schemes represents coupling of voltage sensor activation to pore opening. Of these combinations, decreasing γ and δ (χ^2 = 16.6) could best account for macroscopic time constant data in nominally 0 Ca^{2+} when ratio of δ to γ

Table 4-2: Results of changing J_0 (equilibrium of voltage-sensor activation) plus a second parameter in Scheme 1 to account for effects of P805L mutation

Parameter	Fit A			Fit B		
	WT Value	P805L Value	χ^2	WT Value	P805L Value	χ^2
J_0	0.08	0.07	0.67	0.06	0.06	0.92
K_D (μM)	41	48	0.65	17	21	0.85
z_J (e_0)	0.30	0.26	0.47	0.41	0.36	0.72
z_L (e_0)	0.57	0.49	0.47	0.58	0.50	0.75
C	2.5	2.6	0.68	17	18	0.92
D	29	29	0.68	16	14	0.88
E	20	20.1	0.69	1.1	0.9	1.00

Simultaneous (global) fitting was performed using G-V relations for P805L channels acquired over a range of Ca^2, with parameters shown above adjusted in addition to J_0, as described in Methods. Adjusted parameter for P805L channels are shown above, along with summed χ^2 values for fitting [Ca^{2+}] (based on 105 experimental data points). The other parameters for Scheme 1 can be found in Table 3-1, except J_0 = 0.029 for Fit A 0.014 for Fit B. Changing J_0 alone resulted in a χ^2 value of 0.68 for Fit A and 0.31 for Fit B.

Table 4-3: Results of changing L_0 (equilibrium of pore opening) plus a second parameter in Scheme 1 to account for effects of P805L mutation

Parameter	Fit A			Fit B		
	WT Value	P805L Value	χ^2	WT Value	P805L Value	χ^2
J_0	0.08	0.07	0.67	0.06	0.06	0.92
K_D (μM)	41	48	0.65	17	21	0.85
z_J (e_0)	0.30	0.26	0.47	0.41	0.36	0.72
z_L (e_0)	0.57	0.49	0.47	0.58	0.50	0.75
C	2.5	2.6	0.68	17	18	0.92
D	29	29	0.68	16	14	0.88
E	20	20.1	0.69	1.1	0.9	1.00

Simultaneous (global) fitting was performed using G-V relations for P805L channels acquired over a range of Ca^2, with parameters shown above adjusted in addition to L_0, as described in Methods. Adjusted parameter for P805L channels are shown above, along with summed χ^2 values for fitting [Ca^{2+}] (based on 105 experimental data points). The other parameters for Scheme 1 can be found in Table 3-1, except L_0 = 3.0 x 10^{-7} for Fit A and 8.4 x 10^{-7} for Fit B. Changing L_0 alone resulted in a χ^2 value of 0.69 for Fit A and 0.93 for Fit B.

Table 4-4: Results of changing D (coupling between voltage sensors and pore opening) plus a second parameter in Scheme 1 to account for effects of P805L mutation

Parameter	Fit A			Fit B		
	WT Value	P805L Value	χ^2	WT Value	P805L Value	χ^2
J_0	0.08	0.08	0.68	0.06	0.06	0.35
K_D (μM)	41	44	0.68	17	20	0.29
L_0	2.8×10^{-6}	2.8×10^{-6}	0.68	1.0×10^{-5}	9.7×10^{-6}	0.36
z_J (e_0)	0.30	0.26	0.53	0.41	0.41	0.36
z_L (e_0)	0.57	0.51	0.54	0.58	0.59	0.36
C	2.5	2.7	0.63	9.2	16	0.32
E	10	21.2	0.67	1.1	0.9	0.27

Simultaneous (global) fitting was performed using G-V relations for P805L channels acquired over a range of Ca^2, with parameters shown above adjusted in addition to D, as described in Methods. Adjusted parameter for P805L channels are shown above, along with summed χ^2 values for fitting [Ca^{2+}] (based on 105 experimental data points). The other parameters for Scheme 1 can be found in Table 3-1, except D = 16.10 for Fit A and 6.45 for Fit B. Changing D alone resulted in a χ^2 value of 0.68 for Fit A and 0.36 for Fit B.

Table 4-5: Results of changing individual parameters in Scheme 2 to account for altered gating in P805L mutant channels

Parameter	WT	P805L α	P805L β	P805L γ	P805L δ	P805L z_α	P805L z_β	P805L z_γ	P805L z_δ	P805L D
α (s^{-1})	1,162	**771**	1,162	1,162	1,162	1,162	1,162	1,162	1,162	1,162
β (s^{-1})	31,020	31,020	**47,101**	31,020	31,020	31,020	31,020	31,020	31,020	31,020
γ (s^{-1})	5,128	5,128	5,128	**4,966**	5,128	5,128	5,128	5,128	5,128	5,128
δ (s^{-1})	1.40	1.40	1.40	1.40	**0.38**	1.40	1.40	1.40	1.40	1.40
z_α (e_0)	0.31	0.31	0.31	0.31	0.31	**0.20**	0.31	0.31	0.31	0.31
z_β (e_0)	-0.31	-0.31	-0.31	-0.31	-0.31	-0.31	**-0.31**	-0.31	-0.31	-0.31
z_γ (e_0)	-0.025	-0.025	-0.025	-0.025	-0.025	-0.025	-0.025	**-0.060**	-0.025	-0.025
z_δ (e_0)	0.29	0.29	0.29	0.29	0.29	0.29	0.29	0.29	**0.12**	0.29
D	8.2	8.2	8.2	8.2	8.2	8.2	8.2	8.2	8.2	**6.1**
χ^2	13.4	100.4	100.8	121.7	19.2	62.5	121.9	118.8	11.7	98.2

Time constants of activation/deactivation over voltages ranging from -240 to 230 mV in nominally 0 Ca^{2+}, as described by Scheme 2 (see methods). Using WT parameters as a framework, time constants for P805L channels were described by altering adjusting a single parameter, indicated in bold. Adjusting δ or z_δ, associated with pore C → O rate constant and respective valence, resulted in the lowest χ^2. The rate-limiting time constants from these fits from -240 to 230 mV are shown in **Figure 4-9 A**

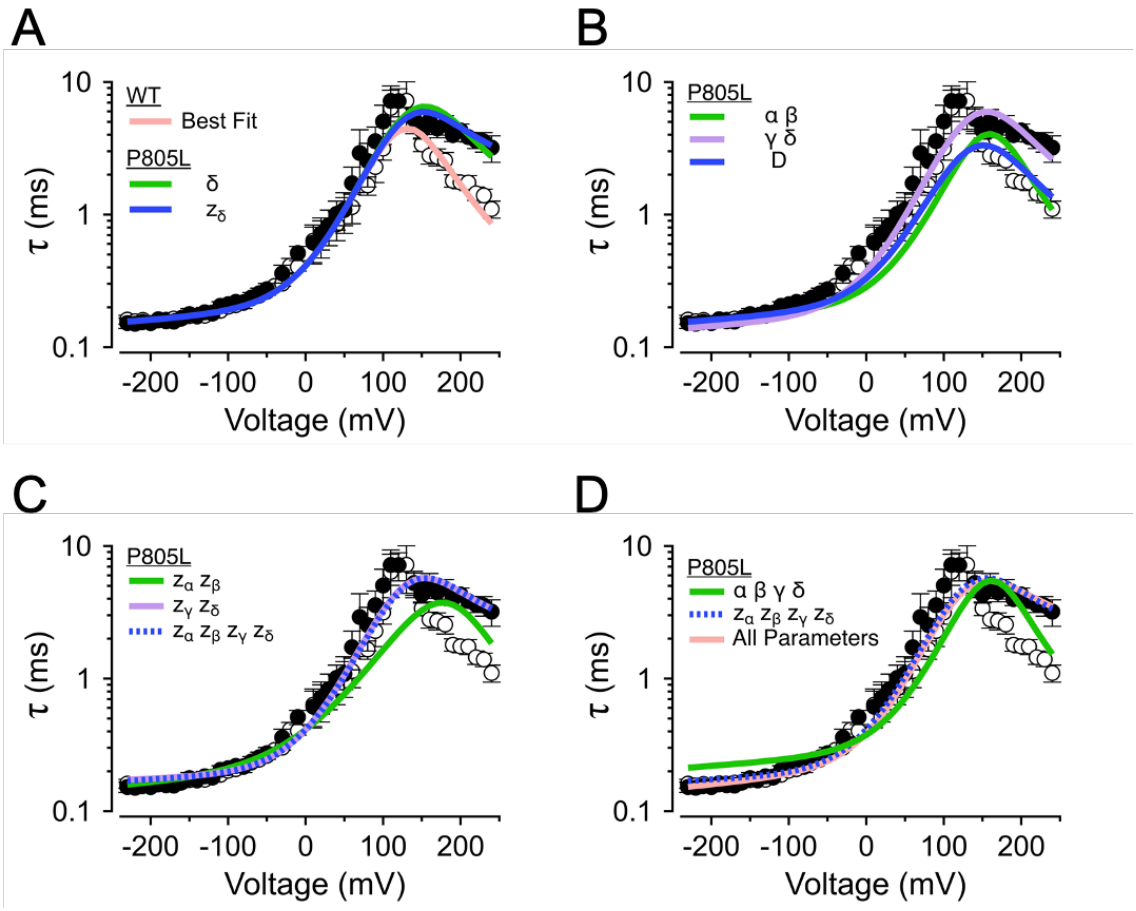


Figure 4-8: Description of macroscopic time constants in the absence of Ca^{2+} using Scheme 2

τ vs voltage plotted for WT (white) and P805L (black) channels in all panels. **A,** Effect of changing one parameter (indicated in legend) to account for changes in activation of P805L channels. Pink line represent best fit with Scheme 2 for WT channels using parameters in **Table 4-5**. Plots for activation of P805L channels changing only γ (green) or z_γ (blue) are shown. **B,** Effect of changing parameters associated with J_0 (α and β; green), L_0 (γ and δ; purple), and D (green) to account for activation of P805L channels. **C,** Effect of changing gating valence for voltage-sensor activation (z_α z_β; green), pore opening (z_γ z_δ; purple), or all valences (z_α z_β z_γ z_δ; blue) to account for activation of P805L channels. **D,** Effect of allowing α β γ δ (green), z_α z_β z_γ z_δ (blue) or all parameters (pink) to account for activation of P805L channels. Parameter values and associated χ^2 for panels B-D can be found in **Table 4-5** and **Table 4-6**.

was decreased by approximately 4.0-fold for P805L channels, suggesting potential changes to the intrinsic opening of the pore (**Table 4-6; Figure 4-8 B**). A similar approach for pairs of gating valences indicates that changing the valences of pore closing and opening, z_γ and z_δ, could best account for observed time constant data in P805L channels, with the ratio of z_δ to z_γ decreased by 1.4-fold (**Table 4-6; Figure 4-8 C**). Interestingly, changing z_γ and z_δ (χ^2 = 13.0) resulted in a better fit than changing γ and δ (χ^2 = 16.6).

In contrasting the results obtained from Schemes 1 & 2, it is notable that changes in valence for pore opening and closing (z_γ and z_δ) resulted in the best fit for Scheme 2, while changes to the corresponding parameter z_L alone in Scheme 1 do not properly describe G-V-Ca^{2+} relations at larger [Ca^{2+}] (**Figure 4-6 D/E** and **Figure 4-6 E/F**). Since Scheme 2 only describes activity in the absence of Ca^{2+}, we hypothesized that changing z_L in combination with a Ca^{2+}-sensitive parameter (K_D, C, or E) in Scheme 1 could potentially provide a better description of G-V-Ca^{2+} relations. We found that decreasing C in combination with z_L resulted in the best fit of any two-parameter combination tested, with a with a χ^2 of 0.26 for Fit A and 0.20 for Fit B (**Table 4-7: Figure 4-9**). Changes to E with z_L by comparison resulted in a χ^2 of 0.27 for Fit A and 0.55 for Fit B, whereas changes to K_D with z_L did not adequately describe G-V-Ca^{2+} relations and resulted in a χ^2 of 0.68 for Fit A and 0.70 for Fit B (**Figure 4-9**).

In summary, the results obtained by modeling Schemes 1 & 2 are not conclusive but demonstrate that changes in gating caused by the P805L mutation cannot be explained by altered Ca^{2+}-sensitivity alone. Describing time constants of activation and deactivation in the absence of Ca^{2+} with Scheme 2 suggests that the P805L mutation may be changing the equilibrium constant for pore opening (L) by altering L_0 or z_L. A mechanism by which the P805L mutation causes channels to activate with altered gating valence of pore opening (z_L) in combination with decreased coupling between the Ca^{2+}-sensing domain and pore opening (C) can account for changes in G-V-Ca^{2+} relations and

Table 4-6: Results of changing multiple parameters in Scheme 2 to account for altered gating in P805 channels

Parameter	WT	P805L $\alpha\ \beta$	P805L $\gamma\ \delta$	P805L D	P805L $z_\alpha\ z_\beta$	P805L $z_\gamma\ z_\delta$	P805L $z_\alpha z_\beta z_\gamma z_\delta$	P805L $\alpha\ \beta\ \gamma\ \delta$	P805L All
α (s^{-1})	1,162	**626**	1,162	1,162	1,162	1,162	1,162	**1,208**	**3,116**
β (s^{-1})	31,020	**38,248**	31,020	31,020	31,020	31,020	31,020	**34,259**	**75,349**
γ (s^{-1})	5,128	5,128	**5,746**	5,128	5,128	5,128	5,128	**5,684**	**4,842**
δ (s^{-1})	1.40	1.40	**0.39**	1.40	1.40	1.40	1.40	**1.38**	**1.25**
z_α (e_0)	0.31	0.31	0.31	0.31	**0.20**	0.31	**0.30**	0.31	**0.31**
z_β (e_0)	-0.31	-0.31	-0.31	-0.31	**-0.20**	-0.31	**-0.30**	-0.31	**-0.31**
z_γ (e_0)	-0.025	-0.025	-0.025	-0.025	-0.025	**-0.014**	**-0.016**	-0.025	**-0.034**
z_δ (e_0)	0.29	0.29	0.29	0.29	0.29	**0.12**	**0.12**	0.29	**0.15**
D	8.2	8.2	8.2	**6.1**	8.2	8.2	8.2	8.2	**7.0**
χ^2	13.4	99.5	16.6	98.2	63.2	13.0	11.2	15.8	9.9

Time constants of activation/deactivation over voltages ranging from -240 to 230 mV in nominally 0 Ca^{2+}, as described by Scheme 2 (see methods). Using WT parameters as a framework, time constants for P805L channels were described by altering adjusting multiple parameter, indicated in bold. time constants from these fits are shown in **Figure 4-9 B-D**

Table 4-7: Results of changing z_L plus a second parameter in Scheme 1 to account for effects of P805L mutation

Parameter	Fit A			Fit B		
	WT Value	P805L Value	χ^2	WT Value	P805L Value	χ^2
J_0	0.08	0.03	0.68	0.06	0.08	0.24
K_D (μM)	41	84	0.68	17	39	0.70
L_0	2.8×10^{-6}	1.2×10^{-6}	0.95	1.0×10^{-5}	3.2×10^{-6}	0.91
z_J (e_0)	0.30	0.30	2.11	0.41	0.38	2.59
C	2.5	1.8	0.26	17	11.4	0.20
D	29	24	0.96	16	11.3	1.74
E	20	11	0.27	1.1	0.5	0.55

Simultaneous (global) fitting was performed using G-V relations for P805L channels acquired over a range of Ca^2, with parameters shown above adjusted in addition to z_L, as described in Methods. Adjusted parameter for P805L channels are shown above, along with summed χ^2 values for fitting [Ca^{2+}] (based on 105 experimental data points). The other parameters for Scheme 1 can be found in Table 3-1, except z_L = 0.188 for Fit A and 0.202 for Fit B. Changing z_L alone resulted in a χ^2 value of 2.11 for Fit A and 2.71 for Fit B.

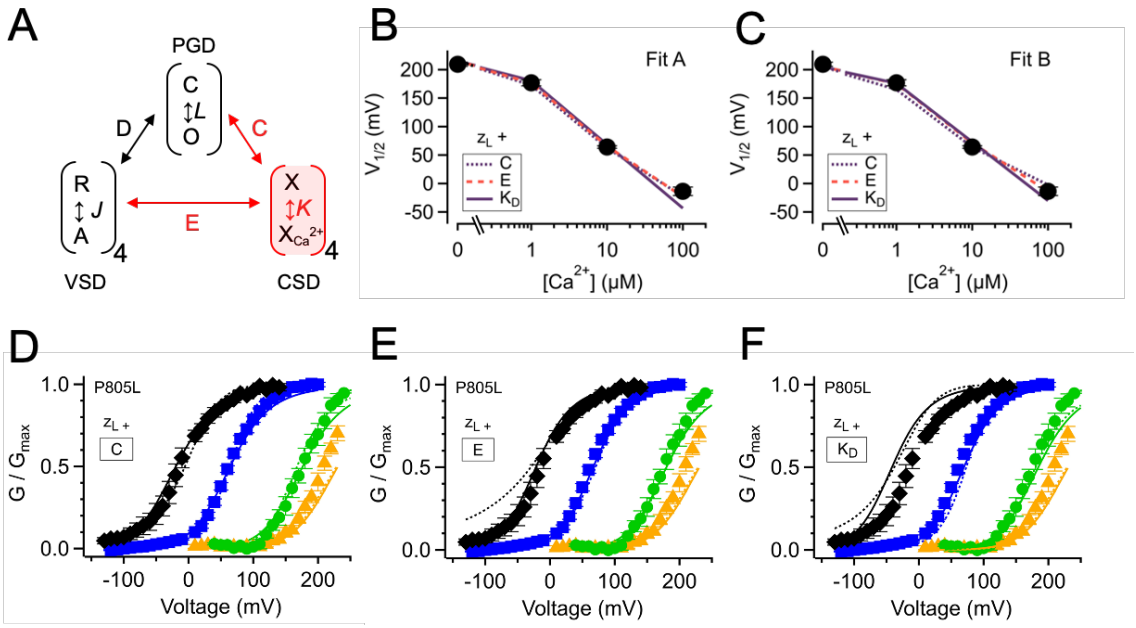


Figure 4-9: Altering Ca^{2+}-sensitive parameters in addition to z_L in Scheme 1 to account for changes in gating of P805L channels

A, Dual-allosteric model (Scheme 10 describing BK channel activation. Ca^{2+}-sensitive parameters C, E, and K_D are denoted in red. **B and C,** Experimentally determined $V_{1/2}$ plotted as a function of $[Ca^{2+}]$ for P805L channels (black circles). Lines represent predictions by changing z_L and the indicated parameter for Fit A (Panel B) and Fit B (Panel C). **D – F,** G-V-Ca^{2+} relations for P805L channels bathed in 0 μM (orange triangle), 1 μM (green circle), 10 μM (blue square), and 100 μM Ca^{2+} (black diamond). Lines represent fits with parameters for Fit A (solid line) and Fit B (dashed line) while changing the indicated parameter and Z_L from WT value. Values and χ^2 for each parameter can be found in **Table 4-7**.

macroscopic time constant data in P805L channels (**Figure 4-9 D**). However, we cannot eliminate the possibility that the P805L mutation alters L_0.

NS11021 activates P805L mutant channels

To determine whether known BK channel activators can ameliorate currents in channels with the P805L mutation, we applied NS11021 to the cytosolic face of patches in 10 μM Ca^{2+}. NS11021 is a BK channel specific activator which stabilizes an open state of the pore by slowing channel deactivation (Rockman et al., 2020). Our results indicate that addition of NS11021 to P805L channels causes a left-shift in the $V_{1/2}$ in a dose-dependent manner (**Figure 4-9 A**). The 58.1 ± 11.2 mV (n=4) shift caused by 30 μM NS11021 in P805L channels was not significantly different than the 61.9 ± 3.4 mV (n=14) shift observed in WT BK channels (p=0.67, two-tailed t-test). However, τ – V relations indicate that adding NS11021 to P805L channels does not restore basal kinetics, and instead results in a channel with slowed activation and deactivation kinetics (**Figure 4-9 B**).

Addition of 10 μM NS11021 results in a nearly equivalent shift in $V_{1/2}$ as is caused by the P805L mutation in 10 μM Ca^{2+} (p=0.58, two-tailed t-test). However, the associated slope of the Boltzmann fit remained 1.5-fold lower, the deactivation time constant remained 4.2-fold slower at -200 mV, and the activation time constant remained 4.2-fold slower at +200 mV compared to values obtained for WT channels. To our knowledge, this is the first demonstrated use of selective small molecule-activator to successfully ameliorate shifts in $V_{1/2}$ caused by a LOF BK channelopathy, however, the physiological implications of these findings are not known.

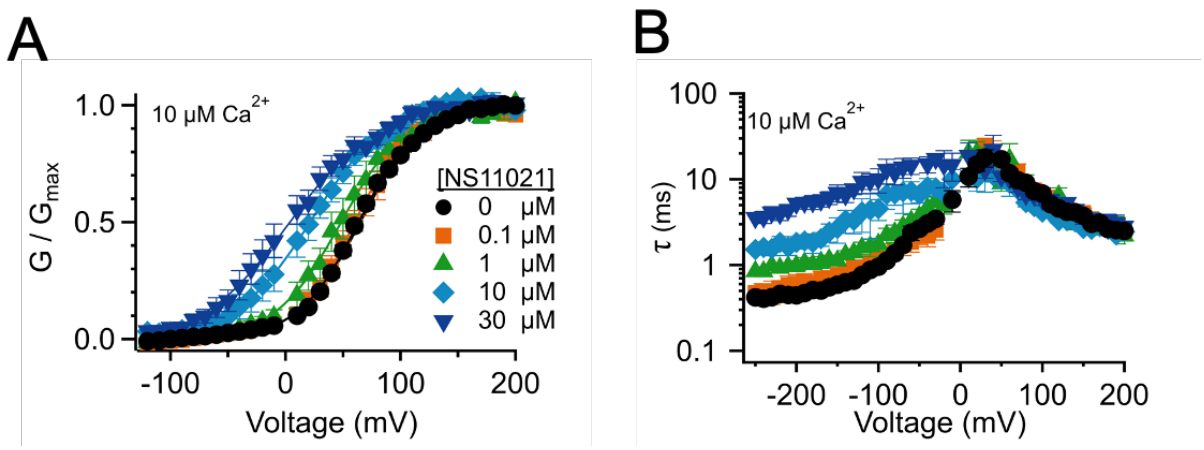


Figure 4-10: NS11021 activates BK channels with the P805L mutation

A, G-V relations of P805L channels with NS11021 added to the cytosolic face. $V_{1/2}$ for each [NS11021]: 0 μM = 64.1 ± 2.3 mV (n=25); 0.1 μM = 60.7 ± 4.5 mV (n=4); 1 μM = 48.9 ± 8.2 mV (n=5); 10 μM = 19.9 ± 9.4 mV (n=5); 30 μM = 6.0 ± 11.2 mv (n=5). **B,** τ vs V relations of P805L channels with NS11021 added to the cytosolic face. $\tau_{-200\ mV}$ for each [NS11021]: 0μM = 0.46 ± 0.59 mV (n=16); 0.1 μM = 0.59 ± 0.18 (n=3); 1 μM = 0.98 ± 1.83 (n=3); 10 μM = 1.83 ± 0.31 (n=3); 30 μM = 5.23 ± 0.72 (n=4).

Discussion

Structural changes caused by the P805L mutation

P805 is a highly conserved residue in RCK2 located in an α-helix. Proline is the only amino acid to contain a secondary amine group, called an imine, which cannot act as a hydrogen bond donor in α-helices. Steric clashing between the proline ring and backbone carbonyl further destabilizes proline-containing helices (Chang et al., 1999). Because of reduced stabilization, proline residues have been known to introduce kinks and swivels to the secondary structure of α-helices (Cordes et al., 2002; Ballesteros and Weinstein, 1992; Chang et al., 1999). Loss of these unique structural properties could potentially explain the large changes in gating caused by the P805L mutation. Alternatively, steric clashing between the bulky tertbutyl sidechain of leucine and Q496 or I941 could also contribute to altered gating. Additional mutagenesis experiments changing P805 to other residues could further elucidate the structural basis for altered gating in channels with the P805L mutation.

Gating mechanism of BK channels with the P805L mutation

BK channels with the P805L mutation activate slower and with a $V_{1/2}$ shifted towards depolarized voltages. These alterations persist in the nominal absence of Ca^{2+}, indicating that changes in gating caused by the P805L mutation cannot be described by changing Ca^{2+} sensitivity alone, and must alter either voltage-sensing or intrinsic opening of the pore.

In terms of mechanism, the slowed activation rate observed in channels with the P805L mutation in the nominal absence of Ca^{2+} could best be accounted for by adjusting either the closed-to-open rate constant (δ) or corresponding gating valence (z_δ) in Scheme 2 (**Figure 4-8 A**). However, adjusting z_δ better described the apparent voltage-

dependence of activation time constants at depolarized voltages (>150 mV) in P805L mutant channels, resulting in a lower χ^2 (**Figure 4-8 A**). Adjusting parameters in pairs similarly indicated that changing z_δ and z_γ, corresponding to parameter z_L in Scheme 1, could best account for the observed changes caused by the P805L mutation in nominally zero Ca^{2+} (**Figure 4-8 C**). Using these results as a framework to describe gating at higher [Ca^{2+}], we found that a mechanism where the P805L mutation is altering z_L in addition to C in Scheme 1 could best describe alterations in G-V-Ca^{2+} relations, even though changing either z_L or C alone remained insufficient (**Figure 4-9 C; Figure 4-6**). These results suggest that the P805L mutation may be reducing coupling between the Ca^{2+}-sensor and the pore in addition to the reducing the valence of pore opening.

RCK domains are well-established to facilitate BK channel activation in response to Ca^{2+} binding (Xia et al., 2002; Zeng et al., 2005; Yang et al., 2008; Miranda et al., 2016; Schreiber and Salkoff, 1997). Structural data suggests that Ca^{2+}-binding to RCK domains causes large conformational changes including a tilt in RCK1 away from the pore (Tao et al., 2016; Hite et al., 2016). The Ca^{2+}-bound state brings RCK1 in close proximity to transmembrane domains of the voltage-sensor in addition to membrane lipids, providing a structural basis for a mechanism where RCK domains alter voltage-dependent aspects of gating (Hite et al., 2016; Fan et al., 2020). Electrophysiological data further suggest a strong coupling between Ca^{2+}- and voltage-sensors may be preset in BK channels, since measurements of gating currents show strong Ca^{2+}-dependencies of voltage-sensor activation. (Lorenzo-Ceballos et al., 2019; Sweet and Cox, 2009; Miranda et al., 2018; Zhang et al., 2017; Geng et al., 2020). Activity of a truncation construct of BK channels containing removal of both RCK domains (slo1c-Kv-minT) shows that z_L may be altered in these truncated channels, in addition to more prominent changes to parameters C, D, E, and K_D (Zhang et al., 2017). In summary, our data indicate that the P805L mutation may alter coupling between Ca^{2+}-sensors and the pore (C) and the gating valence associated

pore opening (z_L). However, we cannot eliminate the possibility that the P805L mutation instead alters L_0, J_0, or D.

Other loss-of-function BK channelopathies

While the mechanisms of activation for GOF BK channel mutations like D369G have been known for the past decade, mechanisms of how LOF mutations alter channel gating are only now beginning to be uncovered (Wang et al., 2009; Moldenhauer et al., 2020; Du et al., 2020). Multiple disease-associated mutations in RCK2 have been identified to adjust different aspects of Ca^{2+}- and voltage-dependent aspects of gating. The D965V mutation causes a +40 mV shift in the $V_{1/2}$ and slows the time course of activation 6.5-fold in 10 μM Ca^{2+} (Moldenhauer et al., 2020). Meanwhile, R1079H was observed to shift the $V_{1/2}$ only in the presence of low (1 μM) Ca^{2+}, an effect which was not observed at higher (10 μM) Ca^{2+} (Moldenhauer et al., 2020). Occurring in RCK1, the H444Q mutation shifts the $V_{1/2}$ +30 mV, slows channel activation, and causes deactivation to occur more rapidly (Moldenhauer et al., 2020). Finally, a mutation in the selectivity filter, G354S, decreased both K^+ selectivity and conductance (Du et al., 2020). The results of LOF mutations P805L, R1079H, H444Q, D965V, and G354S, indicate that a wide spectrum of observed phenotypes may be caused by LOF mutations. Additionally, hundreds of non-pathological single nucleotide polymorphisms have been identified which further add the diversity of “wild-type” BK channels (Plante et al., 2019a).

Pharmacological activation of a loss-of-function BK channelopathy

Addition of NS11021 to channels with the P805L mutation causes a left-shift in the $V_{1/2}$ such that addition of 10 μM NS11021 returns the $V_{1/2}$ to the value of WT channels in 10 μM Ca^{2+} ($p=0.67$). To my knowledge, this is the first example of the $V_{1/2}$ in a LOF BK channelopathy being ameliorated by a BK channel-specific activator. A recent study

characterizing the selectivity filter mutation G354S attempted to activate channels with the activator NS1619 (Du et al., 2020). Interestingly, addition of 10 μM NS1619 appeared to have the opposite effect, blocking the channels and decreasing the overall current (Du et al., 2020). Although the $V_{1/2}$ shifts to near WT levels for P805L channels in the presence of 10 μM NS11021, the rates of both activation and deactivation remain slowed by approximately 4-fold (**Figure 4-10**). The physiological implications of slowed kinetics in a potential treatment option for patients with this mutation remain unclear, and would likely require in-vivo data to determine whether restoring physiological $V_{1/2}$ alone is sufficient to improve symptoms for patients with this mutation.

Conclusions

In this chapter, alterations to gating in the loss-of-function BK channelopathy P805L are characterized using electrophysiology. P805L channels activate with $V_{1/2}$ shifted towards depolarized voltages and with a slower rate of activation. These changes persist in the presence and nominal absence of Ca^{2+}, indicating that alterations to gating cannot be explained by changes to Ca^{2+}-sensitivity alone. The single-channel current amplitude is unchanged in P805L channels, indicating that the P805L mutation is unlikely to substantially alter single-channel conductance. Kinetic modeling suggests that a mechanism altering either voltage-sensitivity (J_0 or D) or the closed-to-open equilibrium constant (L_0) alone can account for changes to G-V relations over a range of $[Ca^{2+}]$ caused by the P805L mutation. However, a mechanism including changes to both gating valence of the pore opening (z_L) and coupling between Ca^{2+}-sensors and the pore (C) provides the most complete description of G-V relations over a range of Ca^{2+} and activation time constants in the nominal absence of Ca^{2+}. Addition of a specific BK channel activator NS11021 (10μM) to channels with the P805L mutation restores $V_{1/2}$ to near WT levels.

These findings increase our understanding about a BK channel mutation associated with human disease and contribute to our knowledge about BK channel gating.

CHAPTER 5

DISCUSSION

Mechanisms of activation similar to NS11021

Ion channels are a large superfamily of membrane proteins that currently comprise the second most common target of all FDA approved drugs (Santos et al., 2016). Drugs currently used in clinical practice that target ion channels include the anticonvulsant gabapentin (a calcium channel inhibitor), the anxiolytic alprazolam (GABA receptor agonist), and the antiarrhythmic amiodarone (potassium and calcium channel inhibitor) (Fuentes et al., 2018). Structurally distinct classes of drugs can activate the same ion channel with different mechanisms of action, leading to different clinical uses. For example, the barbiturate and benzodiazepine classes of drugs both activate $GABA_A$ channels, yet the benzodiazepine family of drugs is often preferred clinically due to a lower risk of adverse events (MacDonald et al., 1989; Rogers et al., 1994; Hendey et al., 2011). It is therefore important to determine the mechanism of action for structurally distinct activators, like NS11021, in order to potentially identify novel mechanisms of channel activation.

Drugs acting on other classes of ion channels may activate using a similar mechanism as NS11021 activation of BK channels. Roscovitine, first discovered as a cyclin-dependent kinase inhibitor (Meijer et al., 1997; Benson et al., 2007), can activate N-type and P-type Ca^{2+} channels by slowing channel deactivation and increasing mean open time (Yan et al., 2002; Buraei et al., 2005; DeStefino et al., 2010). Similarly, pyrantel activates nicotinic acetylcholine receptors (AChR) at low doses by increasing the mean open time and increasing the number of openings per burst (Aubry et al., 1970; Robertson et al., 1994). In summary, BK channels may contain an activation gate at the selectivity

filter similar to TREK and hERG channels (Schewe et al., 2016, 2019). Activators of other ion channels alter dwell time distributions and the deactivation rate, similar to NS11021.

Loss-of-function BK channelopathies

At least 15 BK channelopathies have been characterized as loss-of-function (Bailey et al., 2019; Du et al., 2020). Mechanisms by which function is decreased in LOF mutations include complete inhibition of current, decreased maximum current, altered $V_{1/2}$ or kinetics of gating, and changes to single-channel conduction and selectivity. LOF mutations have been identified in the pore domain, S6-RCK linker, and RCK domains: mutations to the voltage-sensing domain (S0-S4) have not been identified.

Certain mutations to the pore, including S351Y and G356R, have been shown to completely abolish BK channel current. (Liang et al., 2019). The S351Y mutation was identified in a 10-year-old female patient with ataxia, tremor, apraxia, and multiple limb abnormalities, while the G356R mutation was identified in a 12-year-old male affected by mild cognitive delay, axial hypotonia, and ataxia (Liang et al., 2019). Interestingly, patients with either mutation did not have epilepsy, a symptom identified in half of patients with a LOF mutation and nearly all patients with a GOF mutation (Liang et al., 2019; Bailey et al., 2019). Meanwhile, the S6 mutation G375R also resulted in complete inhibition of current and was identified in three unrelated patients (Liang et al., 2019). This mutation presented with a syndrome consisting of facial and visceral malformations, developmental delay, intellectual disability, and axial hypotonia (Liang et al., 2019). Two of the patients presented with episodic seizures (Liang et al., 2019).

One patient was identified to contain two inherited mutant alleles from asymptomatic parents (Liang et al., 2019). Containing both the frameshift mutation N449fs and missense mutation C413Y, the patient presented with multiple congenital

abnormalities, developmental delay, and intellectual disability (Liang et al., 2019). Electrophysiological characterization of N449fs channels shows complete inhibition of BK channel current, while channels with the C413Y mutation activate with reduced current and a $V_{1/2}$ shifted towards additional depolarized voltages, features that are similarly seen with the P805L mutation characterized in **Chapter 4** (Liang et al., 2019). Finally, the D984N mutation in RCK2 was observed to decreased the maximum macroscopic current amplitude without a shift in $V_{1/2}$. It was identified in a patient with moderate intellectual disability, severe language impairment, and multifocal epilepsy (Liang et al., 2019).

The pharmacology of mutant BK channels can exhibit significant differences compared to the pharmacology of "wild-type" BK channels. The G354S mutation was identified in a child with progressive cerebellar ataxia and cognitive impairment (Du et al., 2020). Located in the selectivity filter, BK channels with the G354S mutation activate with significantly reduced single-channel conductance and K^+ selectivity (Du et al., 2020). G354S channels traffic normally to the plasma membrane, although cells with G354S mutant channels show reduced cell viability and neurite outgrowth, a process by which neurons produce new neuronal attachments as they grow (Du et al., 2020; Bovolenta et al., 1992; Lipton and Kater, 1989). Surprisingly, addition of NS1619 paradoxically inhibits G354S channels, rather than activating (Du et al., 2020). Still, NS1619 addition to cells containing G354S mutant channels improved cell viability and neuronal development (Du et al., 2020). The authors suggest that NS1619 addition may be advantageous to cells with the G354S channels in two ways: by blocking aberrant current from the G354S mutant channels, and by activating the remaining WT BK channels (Du et al., 2020). These results suggest that the link between channel mutations and disease phenotypes need to be further investigated, as symptoms might arise due to the reduced contribution of WT BK channels, or due to the addition of mutant channels with aberrant activation.

The wide spectrum of potential phenotypes associated with LOF BK channelopathies begs the question: are symptoms caused by haploinsufficiency or the presence of channels which activate in anomalous conditions? Knockout of the BK channel α-subunit in mice results in ataxia and muscle weakness in addition to bladder contractions and altered circadian rhythms (Meredith et al., 2004, 2006; Sausbier et al., 2004; Rüttiger et al., 2004; Wang et al., 2020). These results suggest that some symptoms of a LOF channelopathy such as ataxia and hypotonia can be explained by loss of basal BK channel function. The link between seizures and decreased BK channel function is less established, although knockout of the β4 subunit leads to an epileptic phenotype (Brenner et al., 2005; Pacheco Otalora et al., 2008; Shruti et al., 2012; Whitmire et al., 2017; Wang et al., 2006). In summary, the causal link between alterations to BK channel gating and specific disease phenotypes remain insufficiently defined. Based on the spectrum of symptoms in patients with LOF BK channel mutations, a combination of haploinsufficiency and aberrant BK channel gating may underlie disease phenotypes.

Whereas addition of 10 μM NS11021 could ameliorate the shift in $V_{1/2}$ caused by the P805L mutation, it is unknown whether this alone could improve symptoms in affected patients since the kinetics remained dissimilar to WT channels (**Figure 4-12**). Based on the observation that a LOF BK channelopathy impeded the growth of neurons in neuronal precursor cells, certain symptoms such as intellectual disability and developmental delay may be less likely be improved with NS11021 administration to a patient with these symptoms (Du et al., 2020). Pharmacological treatment for these symptoms, if possible at all, would potentially require identification and intervention early in development. On the other hand, symptoms associated with excitability such as seizures and movement disorders may be more likely to improve upon addition of an activating drug. Certain patients with an epilepsy-associated Na^+ channelopathy can show dramatic improvement

when given a Na^+ channel modulating drug (Imbrici et al., 2016; Kratschmer et al., 2020). Additional in-vivo experiments with knock-in of P805L mutant channels would be the next step to determine the translational potential of NS11021 to improve symptoms in patients with this mutation.

Based on the mechanism of NS11021 activating BK channels by stabilizing an open state of the pore, NS11021 could potentially act as a “master key” activator for many LOF mutations. Quantifying the activity of NS11021 acting on channels with the selectivity filter mutation G354S would be informative about mechanism, since NS11021 is proposed to bind near the selectivity filter (Schewe et al., 2019; Du et al., 2020). If NS11021 causes a left-shift in $V_{1/2}$ without restoring current to WT levels, this would suggest that NS11021 activation is not dependent on G354S. On the other hand, if NS11021 causes a left-shift in $V_{1/2}$ and restores current to WT levels, this would suggest that NS11021 stabilizes the open state in a mechanism that can ameliorate altered gating in G354S channels. Meanwhile if NS11021 does not change $V_{1/2}$ or current, this would imply that G354S causes a change in gating which cannot be restored by NS11021 or that G354 mediates NS11021 activation of BK channels. Finally, if NS11021 inhibits S351Y channels similar to the effect of NS1619, this would suggest a shared mechanism by which NS11021 and NS1619 interact with G354S mutant channels.

Other BK channel specific activators with different mechanisms of action may act on P805L channels differently. It would be worth determining whether the inhibitory effects of NS1619 on G354S channels could be recapitulated on other BK channelopathies, since use of NS1619 in cells homozygous for the G354S mutation provides dual benefit by both reducing mutant BK currents and increasing WT BK current (Du et al., 2020). NS1619 and Cym04 are both proposed to alter voltage-sensing (J_0 or D), likely by interacting with a different site than NS11021 (Gessner et al., 2012). Mallotoxin administration to P805L

mutant channels may result in a similar phenotype as NS11021 on P805L channels, since both drugs primarily act by prolonging channel deactivation (Zakharov et al., 2005; Gessner et al., 2012; Rockman et al., 2020). Moreover, GoSlo-SR-5-6 could also potentially activate similarly to NS11021 since both drugs are proposed to alter intrinsic gating of the pore (Webb et al., 2015). Since drug activity on mutant channels for each compound would likely depend primarily on the integrity of the binding site, it is difficult to speculate the effect of drug on mutant channels without experimental data.

In summary, at least 15 loss-of-function BK channelopathies have been identified with different mechanisms of inhibition (Bailey et al., 2019; Liang et al., 2019; Du et al., 2020). Mutations including N449fs, S351Y, and G356R completely inhibit BK channel current, while others like D984N decrease the maximum current. Some mutations including P805L and C413Y decrease maximum current ***and*** shift the $V_{1/2}$ to more depolarized voltages, while the selectivity-filter mutation G354S alters single-channel conductance as well as K^+-selectivity (Liang et al., 2019; Du et al., 2020). Similarly, diversity exists between the mechanisms of activators NS11021 and mallotoxin, NS1619 and Cym04, and GoSlo-SR-5-6. The ability of BK channel activators to ameliorate currents in mutant channels remains less defied, since NS11021 activates P805L channels and NS1619 inhibits G354S channels (**Figure 4-12**; Du et al., 2020). These results support the potential for certain BK channel activators to restore current in LOF BK channelopathies and should invite additional investigations into the potential of these drugs to treat disease.

Challenges to treating BK channelopathies

The development of specific treatment options for BK channelopathies has been limited by four major factors: 1) lack of FDA-approved selective BK channel modulators, 2) lack of a classical presentation of core symptoms, 3) lack of established causality

between specific alterations in channel activity and symptoms, and 4) inadequate medical assessment identifying the loci for symptoms in patients (Bailey et al., 2019). For these reasons, treatment of BK channelopathies has largely been limited to anti-seizure medications like valproic acid and lamotrigine that have provided varying success in controlling seizure (Du et al., 2005; Bailey et al., 2019).

In the context of identifying novel BK channel modulators, multiple types of ***selectivity*** exist. BK channel modulators should be selective for BK channels over other K^+ channels to reduce the likelihood of unintended effects. Splice variants represent another form of selectivity, since BK channels in different tissues exhibit different Ca^{2+}- and voltage-sensitivities, as well as different responses to post-translational modifications (Kyle and Braun, 2014). For example, the stress axis hormone-regulated exon, STREX, is a 59 amino acid insert identified in some neurons and vascular smooth muscle cells to cause increased phosphorylation by cAMP/PKA to decrease BK channel activity (Shipston, 2001; Xie and McCobb, 1998; Tian et al., 2001). Another brain-specific splice variant defined by insertion of four residues, SRKR, causes a right-shift in the $V_{1/2}$ and alters activation and deactivation kinetics (Shelley et al., 2013). Expression of the BK_{SRKR} variant increases the during daylight hours, and BK_{SRKR} is thought to contribute to observed circadian properties of BK channels (Shelley et al., 2013; Montgomery et al., 2013; Meredith et al., 2006). Certain BK channel specific drugs, like Cym04, show different activity acting on BK channel variants, highlighting the potential role of post-translational modifications on modulating selectivity of BK channel modulators (Cui et al., 2009; Gessner et al., 2012).

Assembly with auxiliary β or γ subunits can add an additional form of selectivity to pharmacological modulation of BK channels (Gonzalez-Perez and Lingle, 2019). For example, the β1 subunit is found in smooth muscle and enhances the apparent Ca^{2+}-

sensitivity, causing a leftward shift in $V_{1/2}$ at high [Ca^{2+}], while no significant shift in the absence of Ca^{2+} (Brenner et al., 2000). Knock-out of the β1 subunit in mice results in channels with diminished activation in response to Ca^{2+} leading to hypertension, while knock-in mice contain lowered blood pressure (Nelson and Bonev, 2004; Fernández-Fernández et al., 2004). In terms of pharmacology, charybdotoxin has 50x higher binding affinity and a longer block time for BK cannels assembled with the β1 subunit (Hanner et al., 1997; Giangiacomo et al., 2000). Similarly, the estrogen receptor partial agonist tamoxifen has an activatory effect on BK channels coassembled with the β1 subunit, but not with BK α-subunit alone (Dick et al., 2001). Taken together, these results indicate that auxiliary subunits generate channels with significantly altered function and pharmacology, adding an additional form of selectivity for BK channel modulating drugs.

Outlook for treating BK channelopathies

In the 15 years since the first BK channelopathy was identified, only a few dozen cases have thus far been diagnosed (Bailey et al., 2019; Du et al., 2005, 2020). The awareness of BK channelopathies has recently increased due to a detailed patient account appearing in the *New York Times Magazine* as well as in a *Netflix* documentary (Sanders, 2018a; b, 2019). The "KCNMA1 channelopathy international advocacy foundation" (KCIAF; https://www.kciaf.org/) was subsequently created in 2019 to "combin[e] and focus our knowledge and experiences [to] help speed the development of safe treatments and make lives better". Establishing a community to bring patients, physicians, and scientists together will hopefully aid the identification of treatment options.

Since the median cost to bring a new drug to market is $985 million, it is unlikely that a pharmaceutical company would invest in development of a novel compound with the sole intention of treating a BK channelopathy (Wouters et al., 2020). However, two

pharmacological approaches would likely increase the probability of obtaining a compound that could benefit patients with a BK channelopathy: 1) develop novel BK channel-specific drugs which could aid treatment of a common disease such as hypertension or epilepsy, or 2) identify existing FDA-approved drugs that can modulate BK channels. Since 68 million Americans are estimated to have hypertension and 3 million Americans are estimated to have a form of epilepsy, developing an anti-convulsant or anti-hypertensive that acts at a new target (BK channels) may be seen as useful for treating patients whose symptoms may not respond to existing drugs (CDC, 2011; Zack and Kobau, 2017). The therapeutic potential of BK channel modulators has been well established and these efforts are likely underway (Bentzen et al., 2014).

Assaying FDA-approved drugs for activity modulating BK channels would likely provide benefit to BK channelopathy patients on a shorter timescale. Multiple drugs have been shown to inhibit BK channels including verapamil (Harper et al., 2001), ketamine (Denson and Eaton, 1994; Denson et al., 1994), clotrimazole (Wu et al., 1999), and loperamide (Vouga et al., 2019). Meanwhile chlorzoxazone (Liu et al., 2003) and natural products dehydrosoyasaponin-I and kamala have shown activatory effects on BK channels (Giangiacomo et al., 1998; Zakharov et al., 2005). Probing these drugs for activity against mutant BK channels and subsequently testing whether pathological changes can be improved using *in vivo* models would be an experimental approach to recognize potential to treat BK channelopathies.

In conclusion, the identification of known BK channelopathies has increased significantly over the past few years. While there are currently no FDA-approved drugs to treat BK channelopathies, focusing on repurposing currently approved drugs could provide relief for these patients. In the long term, gene editing may hopefully provide a treatment for BK channelopathies.

REFERENCES CITED

Adelman, J.P. 2016. SK channels and calmodulin. *Channels*. 10:1–6. doi:10.1080/19336950.2015.1029688.

Adelman, W.J., and D.L. Gilbert. 1990. Electrophysiology and Biophysics of the Squid Giant Axon BT - Squid as Experimental Animals. D.L. Gilbert, W.J. Adelman, and J.M. Arnold, editors. Springer US, Boston, MA. 93–132.

Andersen, O.S. 1983. Ion movement through gramicidin A channels. Studies on the diffusion-controlled association step. *Biophysical Journal*. 41:147–165. doi:https://doi.org/10.1016/S0006-3495(83)84416-6.

Anderson, C.S., R. MacKinnon, C. Smith, and C. Miller. 1988. Charybdotoxin block of single Ca2+-activated K+ channels. Effects of channel gating, voltage, and ionic strength. *The Journal of General Physiology*. 91:317–333.

Armstrong, C.M. 1971. Interaction of tetraethylammonium ion derivatives with the potassium channels of giant axons. *The Journal of general physiology*. 58:413–437.

Armstrong, C.M. 2007. Life Among the Axons. *Annual Review of Physiology*. 69:1–18. doi:10.1146/annurev.physiol.69.120205.124448.

Armstrong, C.M., and F. Bezanilla. 1973. Currents related to movement of the gating particles of the sodium channels. *Nature*. 242:459.

Armstrong, C.M., and F. Bezanilla. 1974. Charge movement associated with the opening and closing of the activation gates of the Na channels. *The Journal of general physiology*. 63:533–552.

Armstrong, C.M., and F. Bezanilla. 1977. Inactivation of the sodium channel. II. Gating current experiments. *The Journal of general physiology*. 70:567–590.

Armstrong, C.M., and L. Binstock. 1965. Anomalous Rectification in the Squid Giant Axon Injected with Tetraethylammonium Chloride . *Journal of General Physiology*. 48:859–872. doi:10.1085/jgp.48.5.859.

Armstrong, C.M., and B. Hille. 1998. Voltage-gated ion channels and electrical excitability. *Neuron*. 20:371–380.

Atkinson, N.S., G.A. Robertson, and B. Ganetzky. 1991. A component of calcium-activated potassium channels encoded by the Drosophila slo locus. *Science*. 253:551 LP – 555. doi:10.1126/science.1857984.

Aubry, M.L., P. Cowell, M.J. Davey, and S. Shevde. 1970. Aspects of the pharmacology of a new anthelmintic: pyrantel. *British Journal of Pharmacology*. 38:332–344.

Bai, X.-C. chen, G. McMullan, and S.H.W. Scheres. 2015. How cryo-EM is revolutionizing structural biology. *Trends in biochemical sciences*. 40:49–57. doi:10.1016/j.tibs.2014.10.005.

Bailey, C.S., H.J. Moldenhauer, S.M. Park, S. Keros, and A.L. Meredith. 2019. KCNMA1-linked channelopathy. *The Journal of General Physiology*. jgp.201912457. doi:10.1085/jgp.201912457.

Ballesteros, J.A., and H. Weinstein. 1992. The role of Pro/Hyp-kinks in determining the transmembrane helix length and gating mechanism of a [Leu] zervamicin channel.

Biophysical journal. 62:110–111.

Bao, L., and D.H. Cox. 2005. Gating and Ionic Currents Reveal How the BK $_{Ca}$ Channel's Ca $^{2+}$ Sensitivity Is Enhanced by its β1 Subunit. *The Journal of General Physiology*. 126:393–412. doi:10.1085/jgp.200509346.

Bao, L., A.M. Rapin, E.C. Holmstrand, and D.H. Cox. 2002. Elimination of the BK $_{Ca}$ Channel's High-Affinity Ca $^{2+}$ Sensitivity. *The Journal of General Physiology*. 120:173–189. doi:10.1085/jgp.20028627.

Benson, C., J. White, J. De Bono, A. O'Donnell, F. Raynaud, C. Cruickshank, H. McGrath, M. Walton, P. Workman, and S. Kaye. 2007. A phase I trial of the selective oral cyclin-dependent kinase inhibitor seliciclib (CYC202; R-Roscovitine), administered twice daily for 7 days every 21 days. *British journal of cancer*. 96:29–37.

Bentzen, B.H., A. Nardi, K. Calloe, L.S. Madsen, S.S.-P. Olesen, and M. Grunnet. 2007. The Small Molecule NS11021 Is a Potent and Specific Activator of Ca2+-Activated Big-Conductance K+ Channels. *Molecular Pharmacology*. 72:1033–1044. doi:10.1124/mol.107.038331.

Bentzen, B.H., S.P. Olesen, L.C.B. Ronn, and M. Grunnet. 2014. BK channel activators and their therapeutic perspectives. *Frontiers in Physiology*. 5:1–12. doi:10.3389/fphys.2014.00389.

Bentzen, B.H., O. Osadchii, T. Jespersen, R.S. Hansen, S.-P.P. Olesen, and M. Grunnet. 2009. Activation of big conductance Ca2+-activated K+ channels (BK) protects the heart against ischemia–reperfusion injury. *Pflügers Archiv - European Journal of Physiology*. 457:979–988. doi:10.1007/s00424-008-0583-5.

Bernard, C., A. Anderson, A. Becker, N.P. Poolos, H. Beck, and D. Johnston. 2004. Acquired Dendritic Channelopathy in Temporal Lobe Epilepsy. *Science*. 305:532 LP – 535. doi:10.1126/science.1097065.

Berridge, M.J. 1993. A tale of two messengers. *Nature*. 365:388–389. doi:10.1038/365388a0.

Bers, D.M., C.W. Patton, and R. Nuccitelli. 2010. A practical guide to the preparation of Ca2+ buffers.

Bett, G.C.L., A. Lis, H. Guo, M. Liu, Q. Zhou, and R.L. Rasmusson. 2012. Interaction of the S6 Proline Hinge with N-Type and C-Type Inactivation in Kv1.4 Channels. *Biophysical Journal*. 103:1440–1450. doi:https://doi.org/10.1016/j.bpj.2012.08.036.

Bezanilla, F. 2018. Influences: The Cell Physiology Laboratory in Montemar, Chile. *The Journal of General Physiology*. 150:1464 LP – 1468. doi:10.1085/jgp.201812157.

Blatt, M.R., and G. Thiel. 1993. Hormonal control of ion channel gating. *Annual review of plant biology*. 44:543–567.

Blatz, A.L., and K.L. Magleby. 1984. Ion conductance and selectivity of single calcium-activated potassium channels in cultured rat muscle. *Journal of General Physiology*. 84:1–23. doi:10.1085/jgp.84.1.1.

Bond, C.T., P.S. Herson, T. Strassmaier, R. Hammond, R. Stackman, J. Maylie, and J.P. Adelman. 2004. Small Conductance Ca2+-Activated K+ Channel Knock-Out Mice Reveal the Identity of Calcium-Dependent Afterhyperpolarization Currents. *The Journal of Neuroscience*. 24:5301 LP – 5306. doi:10.1523/JNEUROSCI.0182-

04.2004.

Bond, C.T., J. Maylie, and J.P. Adelman. 2005. SK channels in excitability, pacemaking and synaptic integration. *Current Opinion in Neurobiology*. 15:305–311. doi:https://doi.org/10.1016/j.conb.2005.05.001.

Bovolenta, P., F. Wandosell, and M. Nieto-Sampedro. 1992. CNS glial scar tissue: a source of molecules which inhibit central neurite outgrowth. *In* Progress in brain research. Elsevier. 367–379.

Brelidze, T.I., and K.L. Magleby. 2005. Probing the Geometry of the Inner Vestibule of BK Channels with Sugars. *The Journal of General Physiology*. 126:105 LP – 121. doi:10.1085/jgp.200509286.

Brelidze, T.I., X. Niu, and K.L. Magleby. 2003. A ring of eight conserved negatively charged amino acids doubles the conductance of BK channels and prevents inward rectification. *Proceedings of the National Academy of Sciences*. 100:9017 LP – 9022. doi:10.1073/pnas.1532257100.

Brenner, R., Q.H. Chen, A. Vilaythong, G.M. Toney, J.L. Noebels, and R.W. Aldrich. 2005. BK channel β4 subunit reduces dentate gyrus excitability and protects against temporal lobe seizures. *Nature neuroscience*. 8:1752–1759. doi:10.1038/nn1573.

Brenner, R., G.J. Per,z, A.D. Bonev, D.M. Eckman, J.C. Kosek, S.W. Wiler, A.J. Patterson, M.T. Nelson, and R.W. Aldrich. 2000. Vasoregulation by the β1 subunit of the calcium-activated potassium channel. *Nature*. 407:870–876.

Budelli, G., Y. Geng, A. Butler, K.L. Magleby, and L. Salkoff. 2013. Properties of Slo1 K+ channels with and without the gating ring. *Proceedings of the National Academy of Sciences*. 110:16657–16662. doi:10.1073/pnas.1313433110.

Buraei, Z., M. Anghelescu, and K.S. Elmslie. 2005. Slowed N-type calcium channel (CaV2.2) deactivation by the cyclin-dependent kinase inhibitor roscovitine. *Biophysical Journal*. 89:1681–1691. doi:10.1529/biophysj.104.052837.

Butler, A., S. Tsunoda, D.P. McCobb, A. Wei, and L. Salkoff. 1993. mSlo, a complex mouse gene encoding "maxi" calcium-activated potassium channels. *Science*. 261:221 LP – 224. doi:10.1126/science.7687074.

Calderone, V., R. Spogli, A. Martelli, G. Manfroni, L. Testai, S. Sabatini, O. Tabarrini, and V. Cecchetti. 2008. Novel 1,4-Benzothiazine Derivatives as Large Conductance Ca2+-Activated Potassium Channel Openers. *Journal of Medicinal Chemistry*. 51:5085–5092. doi:10.1021/jm701605f.

Campos, J.A., O.S. Silva, M. Lopez, and L. Freire-Maia. 1980. Signs, symptoms and treatment of severe scorpion poisoning in children. *In* Natural toxins. Elsevier. 61–68.

Candia, S., M.L. Garcia, and R. Latorre. 1992. Mode of action of iberiotoxin, a potent blocker of the large conductance Ca(2+)-activated K+ channel. *Biophysical Journal*. 63:583–590. doi:https://doi.org/10.1016/S0006-3495(92)81630-2.

Carbone, E., E. Wanke, G. Prestipino, L.D. Possani, and A. Maelicke. 1982. Selective blockage of voltage-dependent K+ channels by a novel scorpion toxin. *Nature*. 296:90–91. doi:10.1038/296090a0.

Carrisoza-Gaytan, R., E.C. Ray, D. Flores, A.L. Marciszyn, P. Wu, L. Liu, A.R.

Subramanya, W. Wang, S. Sheng, L.J. Nkashama, J. Chen, E.K. Jackson, S.M. Mutchler, S. Heja, D.E. Kohan, L.M. Satlin, and T.R. Kleyman. 2020. Intercalated cell BKα subunit is required for flow-induced K+ secretion. *JCI Insight*. doi:10.1172/jci.insight.130553.

Catterall, W.A. 1977. Membrane potential-dependent binding of scorpion toxin to the action potential Na+ ionophore. Studies with a toxin derivative prepared by lactoperoxidase-catalyzed iodination. *Journal of Biological Chemistry*. 252:8660–8668.

CDC. 2011. Vital signs: prevalence, treatment, and control of hypertension--United States, 1999-2002 and 2005-2008. *MMWR. Morbidity and mortality weekly report*. 60:103.

Chang, D.-K., S.-F. Cheng, V.D. Trivedi, and K.-L. Lin. 1999. Proline affects oligomerization of a coiled coil by inducing a kink in a long helix. *Journal of structural biology*. 128:270–279.

Chen, R.-S., Y. Geng, and K.L. Magleby. 2011. Mg(2+) binding to open and closed states can activate BK channels provided that the voltage sensors are elevated. *The Journal of general physiology*. 138:593–607. doi:10.1085/jgp.201110707.

Cheney, J.A., J.D. Weisser, F.M. Bareyre, H.L. Laurer, K.E. Saatman, R. Raghupathi, V. Gribkoff, J.E. Starrett, and T.K. McIntosh. 2001. The maxi-K channel opener BMS-204352 attenuates regional cerebral edema and neurologic motor impairment after experimental brain injury. *Journal of cerebral blood flow and metabolism : official journal of the International Society of Cerebral Blood Flow and Metabolism*. 21:396–403. doi:10.1097/00004647-200104000-00008.

Clapham, D.E. 2007. Calcium Signaling. *Cell*. 131:1047–1058. doi:https://doi.org/10.1016/j.cell.2007.11.028.

Colquhoun, D. 1971. Lectures on biostatistics; an introduction to statistics with applications in biology and medicine [by] D. Colquhoun. Clarendon Press, Oxford.

Colquhoun, D., and A.G. Hawkes. 1981. On the Stochastic Properties of Single Ion Channels. *Proceedings of the Royal Society B: Biological Sciences*. 211:205–235. doi:10.1098/rspb.1981.0003.

Colquhoun, D., and A.G. Hawkes. 1982. On the stochastic Properties of burst in single ion channel openings and of clusters of bursts. 300:1–59.

Colquhoun, D., and A.G. Hawkes. 1987. A note on correlations in single ion channel records. *Proceedings of the Royal society of London. Series B. Biological sciences*. 230:15–52.

Colquhoun, D., and A.G. Hawkes. 1994. The interpretation of single channel recordings. *In* Microelectrode Techniques. The Plymouth Workshop Handbook, 2nd Ed. D. Ogden, editor. The Company of Biologists Ltd., Cambridge, UK. 141–188.

Colquhoun, D., and A.G. Hawkes. 1995. A Q-Matrix Cookbook. *In* Single-Channel Recording.

Colquhoun, D., and B. Sakmann. 1981. Fluctuations in the microsecond time range of the current through single acetylcholine receptor ion channels. *Nature*. 294:464–466. doi:10.1038/294464a0.

Contet, C., S.P. Goulding, D.A. Kuljis, and A.L. Barth. 2016. BK Channels in the Central

Nervous System. 128. 1st ed. Elsevier Inc. 281–342 pp.

Cordes, F.S., J.N. Bright, and M.S.P. Sansom. 2002. Proline-induced Distortions of Transmembrane Helices. *Journal of Molecular Biology*. 323:951–960. doi:https://doi.org/10.1016/S0022-2836(02)01006-9.

Coskun, C., H.S. Buyuknacar, F. Cicek, and I. Gunay. 2020. BK channel openers NS1619 and NS11021 reverse hydrogen peroxide-induced membrane potential changes in skeletal muscle. *Journal of Receptors and Signal Transduction*. 1–7. doi:10.1080/10799893.2020.1756324.

Cox, D.H., J. Cui, and R.W. Aldrich. 1997. Allosteric gating of a large conductance Ca-activated K+ channel. *The Journal of general physiology*. 110:257–281. doi:10.1085/jgp.110.3.257.

Cui, J. 2020. The action of a BK channel opener. *Journal of General Physiology*. 152. doi:10.1085/jgp.202012571.

Cui, J., and R.W. Aldrich. 2000. Allosteric linkage between voltage and Ca2+-dependent activation of BK-type mslol K+ channels. *Biochemistry*. 39:15612–15619. doi:10.1021/bi001509+.

Cui, J., D.H. Cox, and R.W. Aldrich. 1997. Intrinsic voltage dependence and Ca2+ regulation of mslo large conductance Ca-activated K+ channels. *Journal of General Physiology*. doi:10.1085/jgp.109.5.647.

Cui, J., H. Yang, and U.S. Lee. 2009. Molecular mechanisms of BK channel activation. *Cellular and Molecular Life Sciences*. 66:852–875. doi:10.1007/s00018-008-8609-x.

Curtis, H.J., and K.S. Cole. 1940. Membrane action potentials from the squid giant axon. *Journal of Cellular and Comparative Physiology*. 15:147–157.

Delmas, P., and B. Coste. 2013. Mechano-gated ion channels in sensory systems. *Cell*. 155:278–284.

Demo, S.D.D., and G. Yellen. 1992. Ion effects on gating of the Ca(2+)-activated K+ channel correlate with occupancy of the pore. *Biophysical Journal*. 61:639–648. doi:10.1016/S0006-3495(92)81869-6.

Denson, D.D., P. Duchatelle, and D.C. Eaton. 1994. The effect of racemic ketamine on the large conductance Ca+2-activated potassium (BK) channels in GH3 cells. *Brain Research*. 638:61–68. doi:https://doi.org/10.1016/0006-8993(94)90633-5.

Denson, D.D., and D.C. Eaton. 1994. Ketamine inhibition of large conductance Ca(2+)-activated K+ channels is modulated by intracellular Ca2+. *American Journal of Physiology-Cell Physiology*. 267:C1452–C1458. doi:10.1152/ajpcell.1994.267.5.C1452.

DeStefino, N.R., A.A. Pilato, M. Dittrich, S. V. Cherry, S. Cho, J.R. Stiles, and S.D. Meriney. 2010. (R)-roscovitine prolongs the mean open time of unitary N-type calcium channel currents. *Neuroscience*. 167:838–849. doi:10.1016/j.neuroscience.2010.02.041.

Dick, G.M., C.F. Rossow, S. Smirnov, B. Horowitz, and K.M. Sanders. 2001. Tamoxifen Activates Smooth Muscle BK Channels through the Regulatory β1 Subunit. *Journal of Biological Chemistry* . 276:34594–34599. doi:10.1074/jbc.M104689200.

Díez-Sampedro, A., W.R. Silverman, J.F. Bautista, and G.B. Richerson. 2006. Mechanism of Increased Open Probability by a Mutation of the BK Channel. *Journal of Neurophysiology*. 96:1507–1516. doi:10.1152/jn.00461.2006.

Donner, B.C., M. Schullenberg, N. Geduldig, A. Hüning, J. Mersmann, K. Zacharowski, A. Kovacevic, U. Decking, M.I. Aller, and K.G. Schmidt. 2011. Functional role of TASK-1 in the heart: Studies in TASK-1-deficient mice show prolonged cardiac repolarization and reduced heart rate variability. *Basic Research in Cardiology*. 106:75–87. doi:10.1007/s00395-010-0128-x.

Doyle, D.A. 1998. The Structure of the Potassium Channel: Molecular Basis of K+ Conduction and Selectivity. *Science*. 280:69–77. doi:10.1126/science.280.5360.69.

Du, W., J.F. Bautista, H. Yang, A. Diez-Sampedro, S.-A. You, L. Wang, P. Kotagal, H.O. Lüders, J. Shi, J. Cui, G.B. Richerson, and Q.K. Wang. 2005. Calcium-sensitive potassium channelopathy in human epilepsy and paroxysmal movement disorder. *Nature Genetics*. 37:733–738. doi:10.1038/ng1585.

Du, X., J.L. Carvalho-de-Souza, C. Wei, W. Carrasquel-Ursulaez, Y. Lorenzo, N. Gonzalez, T. Kubota, J. Staisch, T. Hain, N. Petrossian, M. Xu, R. Latorre, F. Bezanilla, and C.M. Gomez. 2020. Loss-of-function BK channel mutation causes impaired mitochondria and progressive cerebellar ataxia. *Proceedings of the National Academy of Sciences*. 117:6023 LP – 6034. doi:10.1073/pnas.1920008117.

Dworetzky, S.I., J.T. Trojnacki, and V.K. Gribkoff. 1994. Cloning and expression of a human large-conductance calcium-activated potassium channel. *Molecular brain research*. 27:189–193.

Evanson, K.W., J.P. Bannister, M.D. Leo, and J.H. Jaggar. 2014. LRRC26 is a functional BK channel auxiliary gamma subunit in arterial smooth muscle cells. *Circulation Research*. 115:423–431. doi:10.1161/CIRCRESAHA.115.303407.

Faber, E.S.L., and P. Sah. 2002. Physiological Role of Calcium-Activated Potassium Currents in the Rat Lateral Amygdala. *The Journal of Neuroscience*. 22:1618 LP – 1628. doi:10.1523/JNEUROSCI.22-05-01618.2002.

Fan, C., N. Sukomon, E. Flood, J. Rheinberger, T.W. Allen, and C.M. Nimigean. 2020. Ball-and-chain inactivation in a calcium-gated potassium channel. *Nature*. doi:10.1038/s41586-020-2116-0.

Fernández-Fernández, J.M., M. Tomás, E. Vázquez, P. Orio, R. Latorre, M. Sentí, J. Marrugat, and M.A. Valverde. 2004. Gain-of-function mutation in the KCNMB1 potassium channel subunit is associated with low prevalence of diastolic hypertension. *The Journal of Clinical Investigation*. 113:1032–1039. doi:10.1172/JCI20347.

French, R.J., and J.J. Shoukimas. 1981. Blockage of squid axon potassium conductance by internal tetra-N-alkylammonium ions of various sizes. *Biophysical journal*. 34:271–291.

Fuentes, A. V, M.D. Pineda, and K.C.N. Venkata. 2018. Comprehension of Top 200 Prescribed Drugs in the US as a Resource for Pharmacy Teaching, Training and Practice. *Pharmacy (Basel, Switzerland)*. 6:43. doi:10.3390/pharmacy6020043.

Furuya, K., K. Enomoto, S. Furuya, S. Yamagishi, C. Edwards, and T. Oka. 1989. Single calcium-activated potassium channel in cultured mammary epithelial cells. *Pflügers*

Archiv. 414:118–124. doi:10.1007/BF00580952.

Galvez, A., G. Gimenez-Gallego, J.P. Reuben, L. Roy-Contancin, P. Feigenbaum, G.J. Kaczorowski, and M.L. Garcia. 1990. Purification and characterization of a unique, potent, peptidyl probe for the high conductance calcium-activated potassium channel from venom of the scorpion Buthus tamulus. *Journal of Biological Chemistry* . 265:11083–11090.

Geng, Y., Z. Deng, G. Zhang, G. Budelli, A. Butler, P. Yuan, J. Cui, L. Salkoff, and K.L. Magleby. 2020. Coupling of Ca2+ and voltage activation in BK channels through the αB helix/voltage sensor interface. *bioRxiv*. 2020.01.29.925545. doi:10.1101/2020.01.29.925545.

Gessner, G., Y.Y.-M. Cui, Y. Otani, T. Ohwada, M. Soom, T. Hoshi, and S.H. Heinemann. 2012. Molecular mechanism of pharmacological activation of BK channels. *Proceedings of the National Academy of Sciences*. 109:3552–3557. doi:10.1073/pnas.1114321109.

Gessner, G., K. Schoonherr, M. Soom, A. Hansel, M. Asim, A. Baniahmad, C. Derst, T. Hoshi, and S.H. Heinemann. 2005. BKCa channels activating at resting potential without calcium in LNCaP prostate cancer cells. *The Journal of membrane biology*. 208:229–240. doi:10.1007/s00232-005-0830-z.

Giangiacomo, K.M., J. Becker, C. Garsky, W. Schmalhofer, M.L. Garcia, and T.J. Mullmann. 2008. Novel α-KTx Sites in the BK Channel and Comparative Sequence Analysis Reveal Distinguishing Features of the BK and KV Channel Outer Pore. *Cell Biochemistry and Biophysics*. 52:47. doi:10.1007/s12013-008-9026-3.

Giangiacomo, K.M., V. Fremont, T.J. Mullmann, M. Hanner, R.H. Cox, and M.L. Garcia. 2000. Interaction of charybdotoxin S10A with single maxi-K channels: Kinetics of blockade depend on the presence of the beta1 subunit. *Biochemistry*. 39:6115–6122. doi:10.1021/bi992865z.

Giangiacomo, K.M., M.L. Garcia, and O.B. McManus. 1992. Mechanism of iberiotoxin block of the large-conductance calcium-activated potassium channel from bovine aortic smooth muscle. *Biochemistry*. 31:6719–6727.

Giangiacomo, K.M., A. Kamassah, G. Harris, and O.B. McManus. 1998. Mechanism of Maxi-K Channel Activation by Dehydrosoyasaponin-I . *The Journal of General Physiology*. 112:485–501. doi:10.1085/jgp.112.4.485.

Giraldez, T., and B.S. Rothberg. 2017. Understanding the conformational motions of RCK gating rings. *The Journal of General Physiology*. 149:431 LP – 441. doi:10.1085/jgp.201611726.

Gonzalez-Perez, V., M. Ben Johny, X.-M. Xia, and C.J. Lingle. 2018. Regulatory γ1 subunits defy symmetry in functional modulation of BK channels. *Proceedings of the National Academy of Sciences*. 201804560. doi:10.1073/pnas.1804560115.

Gonzalez-Perez, V., and C.J. Lingle. 2019. Regulation of BK Channels by Beta and Gamma Subunits. *Annual Review of Physiology*. 81:113–137. doi:10.1146/annurev-physiol-022516-034038.

Gonzalez-Perez, V., X.-M. Xia, and C.J. Lingle. 2014. Functional regulation of BK potassium channels by γ1 auxiliary subunits. *Proceedings of the National Academy of Sciences of the United States of America*. 111:4868–4873.

doi:10.1073/pnas.1322123111.

Van Goor, F., Y.-X. Li, and S.S. Stojilkovic. 2001. Paradoxical Role of Large-Conductance Calcium-Activated K⁺ (BK) Channels in Controlling Action Potential-Driven Ca²⁺ Entry in Anterior Pituitary Cells. *The Journal of Neuroscience*. 21:5902 LP – 5915. doi:10.1523/JNEUROSCI.21-16-05902.2001.

Grimm, P.R., D.L. Irsik, D.C. Settles, J.D. Holtzclaw, and S.C. Sansom. 2009. Hypertension of Kcnmb1−/− is linked to deficient K secretion and aldosteronism. *Proceedings of the National Academy of Sciences*. 106:11800 LP – 11805. doi:10.1073/pnas.0904635106.

Grimm, P.R., and S.C. Sansom. 2007. BK channels in the kidney. *Current Opinion in Nephrology and Hypertension*. 16.

Grimm, P.R., and S.C. Sansom. 2010. BK channels and a new form of hypertension. *Kidney International*. 78:956–962. doi:https://doi.org/10.1038/ki.2010.272.

Gu, N., K. Vervaeke, and J.F. Storm. 2007. BK potassium channels facilitate high-frequency firing and cause early spike frequency adaptation in rat CA1 hippocampal pyramidal cells. *The Journal of Physiology*. 580:859–882. doi:10.1113/jphysiol.2006.126367.

Guan, X., Q. Li, and J. Yan. 2017. Relationship between auxiliary gamma subunits and mallotoxin on BK channel modulation. *Scientific Reports*. 7:42240. doi:10.1038/srep42240.

Gurrola, G.B., and L.D. Possani. 1995. Structural and functional features of noxiustoxin: a K+ channel blocker. *Biochemistry and molecular biology international*. 37:527–535.

Guy, H.R., S.R. Durell, J. Warmke, R. Drysdale, and B. Ganetzky. 1991. Similarities in amino acid sequences of Drosophila eag and cyclic nucleotide-gated channels. *Science*. 254:730–731.

Ha, T.S., S.Y. Jeong, S.-W. Cho, H. Jeon, G.S. Roh, W.S. Choi, and C.-S. Park. 2000. Functional characteristics of two BKCa channel variants differentially expressed in rat brain tissues. *European Journal of Biochemistry*. 267:910–918. doi:10.1046/j.1432-1327.2000.01076.x.

Hamill, O.P., A. Marty, E. Neher, B. Sakmann, and F.J. Sigworth. 1981. Improved patch clamp techniques for high resolution current recording from cells and cell-free membranes. *Pflugers Archiv : European journal of physiology*. 391:85–100. doi:10.1007/BF00656997.

Hanner, M., W.A. Schmalhofer, P. Munujos, H.-G. Knaus, G.J. Kaczorowski, and M.L. Garcia. 1997. The β subunit of the high-conductance calcium-activated potassium channel contributes to the high-affinity receptor for charybdotoxin. *Proceedings of the National Academy of Sciences*. 94:2853 LP – 2858. doi:10.1073/pnas.94.7.2853.

Harper, A.A., L. Catacuzzeno, C. Trequattrini, A. Petris, and F. Franciolini. 2001. Verapamil Block of Large-Conductance Ca-Activated K Channels in Rat Aortic Myocytes. *The Journal of Membrane Biology*. 179:103–111. doi:10.1007/s002320010041.

Hartmann, H.A., G.E. Kirsch, J.A. Drewe, M. Taglialatela, R.H. Joho, and A.M. Brown. 1991. Exchange of conduction pathways between two related K+ channels. *Science*.

251:942–944.

Heginbotham, L., T. Abramson, and R. MacKinnon. 1992. A functional connection between the pores of distantly related ion channels as revealed by mutant K+ channels. *Science*. 258:1152 LP – 1155. doi:10.1126/science.1279807.

Heginbotham, L., and R. MacKinnon. 1992. The aromatic binding site for tetraethylammonium ion on potassium channels. *Neuron*. 8:483–491.

Hendey, G.W., R.A. Dery, R.L. Barnes, B. Snowden, and P. Mentler. 2011. A prospective, randomized, trial of phenobarbital versus benzodiazepines for acute alcohol withdrawal. *The American Journal of Emergency Medicine*. 29:382–385. doi:10.1016/j.ajem.2009.10.010.

Hille, B. 1967. The selective inhibition of delayed potassium currents in nerve by tetraethylammonium ion. *The Journal of general physiology*. 50:1287–1302.

Hille, B. 1977. Local anesthetics: hydrophilic and hydrophobic pathways for the drug-receptor reaction. *The Journal of general physiology*. 69:497–515.

Hille, B. 1978. Ionic channels in excitable membranes. Current problems and biophysical approaches. *Biophysical Journal*. 22:283–294.

Hille, B. 2001. Ion Channel Excitable Membranes. *Sunderland Massachusetts USA*. 1–37. doi:10.1007/3-540-29623-9_5640.

Hille, B., C.M. Armstrong, and R. MacKinnon. 1999. Ion channels: From idea to reality. *Nature Medicine*. 5:1105–1109. doi:10.1038/13415.

Hille, B., and W. Schwarz. 1978. Potassium channels as multi-ion single-file pores. *The Journal of General Physiology*. 72:409–442.

Hite, R.K., X. Tao, R. Mackinnon, R.K. Hite, R. Mackinnon, and K. Ca. 2016. Structural basis for gating the high-conductance Ca2+-activated K+ channel. *Nature*. 1–15. doi:10.1038/nature20775.

Ho, K., C.G. Nichols, W.J. Lederer, J. Lytton, P.M. Vassilev, M. V Kanazirska, and S.C. Hebert. 1993. Cloning and expression of an inwardly rectifying ATP-regulated potassium channel. *Nature*. 362:31–38. doi:10.1038/362031a0.

Hodgkin, A.L., and A.F. Huxley. 1939. Action potentials recorded from inside a nerve fibre. *Nature*. 144:710.

Hodgkin, A.L., and A.F. Huxley. 1952. A Quantitative Description of Membrane Current and its Application to Conduction and Excitation in Nerves. *J. Physiol.* 117:500–544. doi:10.1016/S0092-8240(05)80004-7.

Hodgkin, A.L., and R.D. Keynes. 1955. The potassium permeability of a giant nerve fibre. *The Journal of physiology*. 128:61–88. doi:10.1113/jphysiol.1955.sp005291.

Horn, R. 1991. Estimating the number of channels in patch recordings. *Biophysical Journal*. 60:433–439. doi:https://doi.org/10.1016/S0006-3495(91)82069-0.

Horn, R. 1996. Counting charges. *The Journal of general physiology*. 108:129–132.

Horn, R., and K. Lange. 1983. Estimating kinetic constants from single channel data. *Biophysical Journal*. 43:207–223. doi:10.1016/S0006-3495(83)84341-0.

Horrigan, F.T., and R.W. Aldrich. 1999. Allosteric Voltage Gating of Potassium Channels

II. *The Journal of General Physiology*. 114:305–336. doi:10.1085/jgp.114.2.305.

Horrigan, F.T., and R.W. Aldrich. 2002. Coupling between Voltage Sensor Activation, Ca 2+ Binding and Channel Opening in Large Conductance (BK) Potassium Channels. *The Journal of General Physiology*. 120:267–305. doi:10.1085/jgp.20028605.

Horrigan, F.T., J. Cui, and R.W. Aldrich. 1999. Allosteric voltage gating of potassium channels I. Mslo ionic currents in the absence of Ca(2+). *The Journal of general physiology*. 114:277–304. doi:10.1085/jgp.114.2.305.

Horrigan, F.T., and Z. Ma. 2008. Mg 2 + Enhances Voltage Sensor / Gate Coupling in BK Channels. *J. Gen. Physiol.* 131:13–32. doi:10.1085/jgp.200709877.

Horrocks, L.A., and Y.K. Yeo. 1999. HEALTH BENEFITS OF DOCOSAHEXAENOIC ACID (DHA). *Pharmacological Research*. 40:211–225. doi:https://doi.org/10.1006/phrs.1999.0495.

Hoshi, T., and S.H. Heinemann. 2016. Chapter Six - Modulation of BK Channels by Small Endogenous Molecules and Pharmaceutical Channel Openers. *In* Big on Bk. C.B.T.-I.R. of N. Contet, editor. Academic Press. 193–237.

Hoshi, T., B. Wissuwa, Y. Tian, N. Tajima, R. Xu, M. Bauer, S.H. Heinemann, and S. Hou. 2013. Omega-3 fatty acids lower blood pressure by directly activating large-conductance Ca²⁺-dependent K⁺ channels. *Proceedings of the National Academy of Sciences*. 110:4816 LP – 4821. doi:10.1073/pnas.1221997110.

Hu, H., L.-R. Shao, S. Chavoshy, N. Gu, M. Trieb, R. Behrens, P. Laake, O. Pongs, H.G. Knaus, O.P. Ottersen, and J.F. Storm. 2001. Presynaptic Ca2+-Activated K+ Channels in Glutamatergic Hippocampal Terminals and Their Role in Spike Repolarization and Regulation of Transmitter Release. *The Journal of Neuroscience*. 21:9585 LP – 9597. doi:10.1523/JNEUROSCI.21-24-09585.2001.

Hu, L., H. Yang, J. Shi, and J. Cui. 2005. Effects of Multiple Metal Binding Sites on Calcium and Magnesium-dependent Activation of BK Channels . *Journal of General Physiology*. 127:35–50. doi:10.1085/jgp.200509317.

Hu, S., M.Z. Labuda, M. Pandolfo, G.G. Goss, H.E. McDermid, and D.W. Ali. 2003. Variants of the KCNMB3 regulatory subunit of maxi BK channels affect channel inactivation. *Physiological Genomics*. 15:191–198. doi:10.1152/physiolgenomics.00110.2003.

Huang, X., and L.Y. Jan. 2014. Targeting potassium channels in cancer. *Journal of Cell Biology*. 206:151–162. doi:10.1083/jcb.201404136.

Hunter, M., A.G. Lopes, E.L. Boulpaep, and G.H. Giebisch. 1984. Single channel recordings of calcium-activated potassium channels in the apical membrane of rabbit cortical collecting tubules. *Proceedings of the National Academy of Sciences*. 81:4237 LP – 4239. doi:10.1073/pnas.81.13.4237.

Imbrici, P., A. Liantonio, G.M. Camerino, M. De Bellis, C. Camerino, A. Mele, A. Giustino, S. Pierno, A. De Luca, D. Tricarico, J.-F. Desaphy, and D. Conte. 2016. Therapeutic Approaches to Genetic Ion Channelopathies and Perspectives in Drug Discovery. *Frontiers in pharmacology*. 7:121. doi:10.3389/fphar.2016.00121.

Imlach, W.L., S.C. Finch, Y. Zhang, J. Dunlop, and J.E. Dalziel. 2011. Mechanism of action of lolitrem B, a fungal endophyte derived toxin that inhibits BK large conductance

Ca2+-activated K+ channels. *Toxicon*. 57:686–694. doi:https://doi.org/10.1016/j.toxicon.2011.01.013.

Ishii, T.M., C. Silvia, B. Hirschberg, C.T. Bond, J.P. Adelman, and J. Maylie. 1997. A human intermediate conductance calcium-activated potassium channel. *Proceedings of the National Academy of Sciences*. 94:11651 LP – 11656. doi:10.1073/pnas.94.21.11651.

Jaggar, J.H., V.A. Porter, W.J. Lederer, and M.T. Nelson. 2000. Calcium sparks in smooth muscle. *American Journal of Physiology-Cell Physiology*. 278:C235–C256. doi:10.1152/ajpcell.2000.278.2.C235.

Jaggar, J.H., A.S. Stevenson, and M.T. Nelson. 1998. Voltage dependence of Ca2+ sparks in intact cerebral arteries. *American Journal of Physiology-Cell Physiology*. 274:C1755–C1761.

Jensen, B.S. 2002. BMS-204352: A Potassium Channel Opener Developed for the Treatment of Stroke. *CNS Drug Reviews*. 8:353–360. doi:10.1111/j.1527-3458.2002.tb00233.x.

Jiang, Y., A. Lee, J. Chen, M. Cadene, B.T. Chait, and R. MacKinnon. 2002a. Crystal structure and mechanism of a calcium-gated potassium channel. *Nature*. 417:515–22. doi:10.1038/417515a.

Jiang, Y., A. Lee, J. Chen, M. Cadene, B.T. Chait, and R. MacKinnon. 2002b. The open pore conformation of potassium channels. *Nature*. 417:523–526. doi:10.1038/417523a.

John, S.A., L.-H. Xie, and J.N. Weiss. 2004. Mechanism of inward rectification in Kir channels. *The Journal of general physiology*. 123:623–625.

Joiner, W.J., L.-Y. Wang, M.D. Tang, and L.K. Kaczmarek. 1997. hSK4, a member of a novel subfamily of calcium-activated potassium channels. *Proceedings of the National Academy of Sciences*. 94:11013–11018.

Jones, E.M.C., M. Gray-Keller, and R. Fettiplace. 1999. The role of Ca2+-activated K+ channel spliced variants in the tonotopic organization of the turtle cochlea. *The Journal of Physiology*. 518:653–665. doi:10.1111/j.1469-7793.1999.0653p.x.

Kaupp, U.B., and R. Seifert. 2002. Cyclic nucleotide-gated ion channels. *Physiological reviews*. 82:769–824.

Kim, J.-B. 2014. Channelopathies. *Korean journal of pediatrics*. 57:1–18. doi:10.3345/kjp.2014.57.1.1.

Knaus, H.-G., O.B. McManus, S.H. Lee, W.A. Schmalhofer, M. Garcia-Calvo, L.M.H. Helms, M. Sanchez, K. Giangiacomo, and J.P. Reuben. 1994. Tremorgenic indole alkaloids potently inhibit smooth muscle high-conductance calcium-activated potassium channels. *Biochemistry*. 33:5819–5828.

Köhler, M., B. Hirschberg, C.T. Bond, J.M. Kinzie, N. V Marrion, J. Maylie, and J.P. Adelman. 1996. Small-conductance, calcium-activated potassium channels from mammalian brain. *Science*. 273:1709–1714.

Koval, O.M., Y. Fan, and B.S. Rothberg. 2007. A Role for the S0 Transmembrane Segment in Voltage-dependent Gating of BK Channels. *The Journal of General Physiology*. 129:209–220. doi:10.1085/jgp.200609662.

Kratschmer, P., E. Buhl, K.-F. Chen, S. Lowe, D.M. Kullmann, J.J.L. Hodge, and J.E.C. Jepson. 2020. A knock-in Drosophila model supports a conserved link between potassium channelopathy and involuntary movement. *bioRxiv*.

Kshatri, A., A. Cerrada, R. Gimeno, D. Bartolomé-Martín, P. Rojas, and T. Giraldez. 2020. Differential regulation of BK channels by fragile X mental retardation protein. *Journal of General Physiology*. 152. doi:10.1085/jgp.201912502.

Kshatri, A.S., A. Gonzalez-Hernandez, and T. Giraldez. 2018. Physiological Roles and Therapeutic Potential of Ca2+ Activated Potassium Channels in the Nervous System. *Frontiers in Molecular Neuroscience*. 11:258. doi:10.3389/fnmol.2018.00258.

Kühlbrandt, W. 2014. Microscopy: cryo-EM enters a new era. *Elife*. 3:e03678.

Kun, A., V. V. Matchkov, E. Stankevicius, A. Nardi, A.D. Hughes, H.J. Kirkeby, J. Demnitz, and U. Simonsen. 2009. NS11021, a novel opener of large-conductance Ca2+-activated K+ channels, enhances erectile responses in rats. *British Journal of Pharmacology*. 158:1465–1476. doi:10.1111/j.1476-5381.2009.00404.x.

Kyle, B.D., and A.P. Braun. 2014. The regulation of BK channel activity by pre- and post-translational modifications . *Frontiers in Physiology* . 5:316. doi:10.3389/fphys.2014.00316.

Lagrutta, A.A., K.-Z. Shen, A. Rivard, R.A. North, and J.P. Adelman. 1998. Aromatic residues affecting permeation and gating in dSlo BK channels. *Pflügers Archiv*. 435:731–739. doi:10.1007/s004240050575.

Lang, D.G., and A.K. Ritchie. 1990. Tetraethylammonium blockade of apamin-sensitive and insensitive Ca2 (+)-activated K+ channels in a pituitary cell line. *The Journal of Physiology*. 425:117–132.

Large, R.J., A. Kshatri, T.I. Webb, S. Roy, A. Akande, E. Bradley, G.P. Sergeant, K.D. Thornbury, N.G. McHale, and M.A. Hollywood. 2015. Effects of the novel BK channel opener GoSlo-SR-5-130 are dependent on the presence of BK β subunits. *British Journal of Pharmacology*. 2:n/a-n/a. doi:10.1111/bph.13085.

Latorre, R., K. Castillo, W. Carrasquel-Ursulaez, R. V Sepulveda, F. Gonzalez-Nilo, C. Gonzalez, and O. Alvarez. 2016. Molecular Determinants of BK Channel Functional Diversity and Functioning. *Physiological Reviews*. 97:39–87. doi:10.1152/physrev.00001.2016.

Latorre, R., and C. Miller. 1983. Conduction and selectivity in potassium channels. *The Journal of Membrane Biology*. 71:11–30. doi:10.1007/BF01870671.

Läuger, P. 1976. Diffusion-limited ion flow through pores. *Biochimica et Biophysica Acta (BBA) - Biomembranes*. 455:493–509. doi:https://doi.org/10.1016/0005-2736(76)90320-5.

Layne, J.J., B. Nausch, S.-P. Olesen, and M.T. Nelson. 2010. BK channel activation by NS11021 decreases excitability and contractility of urinary bladder smooth muscle. *American journal of physiology. Regulatory, integrative and comparative physiology*. 298:R378–R384. doi:10.1152/ajpregu.00458.2009.

Lee, C.-H., and R. Mackinnon. 2018. Activation mechanism of a human SK-calmodulin channel complex elucidated by cryo-EM structures. *Science*. 360:508–513. doi:10.1126/SCIENCE.AAS9466.

Lenaeus, M.J., M. Vamvouka, P.J. Focia, and A. Gross. 2005. Structural basis of TEA blockade in a model potassium channel. *Nature Structural & Molecular Biology*. 12:454–459. doi:10.1038/nsmb929.

Lesage, F., E. Guillemare, M. Fink, F. Duprat, M. Lazdunski, G. Romey, and J. Barhanin. 1996. TWIK-1, a ubiquitous human weakly inward rectifying K+ channel with a novel structure. *The EMBO Journal*. 15:1004–1011. doi:10.1002/j.1460-2075.1996.tb00437.x.

Li, Q., and J. Yan. 2016. Modulation of BK Channel Function by Auxiliary Beta and Gamma Subunits. *International Review of Neurobiology*. 128:51–90. doi:10.1016/bs.irn.2016.03.015.

Li, W., D.B. Halling, A.W. Hall, and R.W. Aldrich. 2009. EF hands at the N-lobe of calmodulin are required for both SK channel gating and stable SK–calmodulin interaction. *Journal of General Physiology*. 134:281–293. doi:10.1085/jgp.200910295.

Liang, L., X. Li, S. Moutton, S.A. Schrier Vergano, B. Cogné, A. Saint-Martin, A.C.E. Hurst, Y. Hu, O. Bodamer, J. Thevenon, C.Y. Hung, B. Isidor, B. Gerard, A. Rega, S. Nambot, D. Lehalle, Y. Duffourd, C. Thauvin-Robinet, L. Faivre, S. Bézieau, L.S. Dure, D.C. Helbling, D. Bick, C. Xu, Q. Chen, G.M.S. Mancini, A. Vitobello, and Q.K. Wang. 2019. De novo loss-of-function KCNMA1 variants are associated with a new multiple malformation syndrome and a broad spectrum of developmental and neurological phenotypes. *Human Molecular Genetics*. doi:10.1093/hmg/ddz117.

Liao, M., E. Cao, D. Julius, and Y. Cheng. 2013. Structure of the TRPV1 ion channel determined by electron cryo-microscopy. *Nature*. 504:107.

Lingle, C.J. 2007. Gating Rings Formed by RCK Domains: Keys to Gate Opening . *The Journal of General Physiology*. 129:101–107. doi:10.1085/jgp.200709739.

Lingle, C.J., X.-H. Zeng, J.-P. Ding, and X.-M. Xia. 2001. Inactivation of Bk Channels Mediated by the Nh2 Terminus of the β3b Auxiliary Subunit Involves a Two-Step Mechanism : Possible Separation of Binding and Blockade . *The Journal of General Physiology*. 117:583–606. doi:10.1085/jgp.117.6.583.

Lipton, S.A., and S.B. Kater. 1989. Neurotransmitter regulation of neuronal outgrowth, plasticity and survival. *Trends in Neurosciences*. 12:265–270. doi:10.1016/0166-2236(89)90026-X.

Liu, F., Z. Zhang, L. Csanády, D.C. Gadsby, and J. Chen. 2017. Molecular structure of the human CFTR ion channel. *Cell*. 169:85–95.

Liu, G., S.I. Zakharov, L. Yang, S.-X. Deng, D.W. Landry, A. Karlin, and S.O. Marx. 2008. Position and Role of the BK Channel α Subunit S0 Helix Inferred from Disulfide Crosslinking . *The Journal of General Physiology*. 131:537–548. doi:10.1085/jgp.200809968.

Liu, Y.C., Y.K. Lo, and S.N. Wu. 2003. Stimulatory effects of chlorzoxazone, a centrally acting muscle relaxant, on large conductance calcium-activated potassium channels in pituitary GH3 cells. *Brain Research*. 959:86–97. doi:10.1016/S0006-8993(02)03730-7.

Loeb, J. 1906. The Dynamics of Living Matter. VIII. The MacMillan Company, Norwood, Mass., USA.

Logsdon, N.J., J. Kang, J.A. Togo, E.P. Christian, and J. Aiyar. 1997. A Novel Gene, hKCa4, Encodes the Calcium-activated Potassium Channel in Human T Lymphocytes . *Journal of Biological Chemistry* . 272:32723–32726. doi:10.1074/jbc.272.52.32723.

Long, S.B. 2005. Crystal Structure of a Mammalian Voltage-Dependent Shaker Family K+ Channel. *Science*. 309:897–903. doi:10.1126/science.1116269.

Lopatin, A.N., E.N. Makhina, and C.G. Nichols. 1994. Potassium channel block by cytoplasmic polyamines as the mechanism of intrinsic rectification. *Nature*. 372:366–369.

Lorenzo-Ceballos, Y., W. Carrasquel-Ursulaez, K. Castillo, O. Alvarez, and R. Latorre. 2019. Calcium-driven regulation of voltage-sensing domains in BK channels. *eLife*. 8:e44934. doi:10.7554/eLife.44934.

Lu, Z., A.M. Klem, and Y. Ramu. 2001. Ion conduction pore is conserved among potassium channels. *Nature*. 413:809–813. doi:10.1038/35101535.

Ma, Z., X.J. Lou, and F.T. Horrigan. 2006. Role of Charged Residues in the S1–S4 Voltage Sensor of BK Channels. *The Journal of General Physiology*. 127:309 LP – 328. doi:10.1085/jgp.200509421.

MacDonald, R.L., C.J. Rogers, and R.E. Twyman. 1989. Barbiturate regulation of kinetic properties of the GABAA receptor channel of mouse spinal neurones in culture. *The Journal of Physiology*. 417:483–500. doi:10.1113/jphysiol.1989.sp017814.

MacKinnon, R., and C. Miller. 1988. Mechanism of charybdotoxin block of the high-conductance, Ca2+-activated K+ channel. *The Journal of General Physiology*. 91:335–349. doi:10.1085/jgp.91.3.335.

MacKinnon, R., and C. Miller. 1989. Mutant potassium channels with altered binding of charybdotoxin, a pore-blocking peptide inhibitor. *Science*. 245:1382 LP – 1385. doi:10.1126/science.2476850.

Madden, D.R. 2002. Ion channel structure: the structure and function of glutamate receptor ion channels. *Nature Reviews Neuroscience*. 3:91.

Maioli, E., and G. Valacchi. 2010. Rottlerin: bases for a possible usage in psoriasis. *Current drug metabolism*. 11:425–430.

Malerba, M., M. D`Amato, A. Radaeli, G. Giacovelli, L. Rovati, S. Arshad, and S. Holgate. 2015. Efficacy of Andolast in Mild to Moderate Asthma: A Randomized, Controlled, Double-Blind Multicenter Study (The Andast Trial). *Current Pharmaceutical Design*. 21:3835–3843. doi:10.2174/1381612821666150407101614.

Maljevic, S., and H. Lerche. 2013. Potassium channels: a review of broadening therapeutic possibilities for neurological diseases. *Journal of Neurology*. 260:2201–2211. doi:10.1007/s00415-012-6727-8.

Marks, T.N., and S.W. Jones. 1992. Calcium currents in the A7r5 smooth muscle-derived cell line. An allosteric model for calcium channel activation and dihydropyridine agonist action. *The Journal of General Physiology*. 99:367 LP – 390. doi:10.1085/jgp.99.3.367.

Martin, G.M., C. Yoshioka, E.A. Rex, J.F. Fay, Q. Xie, M.R. Whorton, J.Z. Chen, and S.-L. Shyng. 2017. Cryo-EM structure of the ATP-sensitive potassium channel

illuminates mechanisms of assembly and gating. *Elife*. 6:e24149.

Marty, A. 1981. Ca-dependent K channels with large unitary conductance in chromaffin cell membranes. *Nature*. 291:497–500.

Marty, A. 1983. Ca2+-dependent K+ channels with large unitary conductance. *Trends in Neurosciences*. 6:262–265. doi:https://doi.org/10.1016/0166-2236(83)90115-7.

Marty, A., and E. Neher. 1985. Potassium channels in cultured bovine adrenal chromaffin cells. *The Journal of Physiology*. 367:117–141.

Matsuda, H., A. Saigusa, and H. Irisawa. 1987. Ohmic conductance through the inwardly rectifying K channel and blocking by internal Mg 2+. *Nature*. 325:156–159.

McAfee, D.A., and P.J. Yarowsky. 1979. Calcium-dependent potentials in the mammalian sympathetic neurone. *The Journal of Physiology*. 290:507–523. doi:10.1113/jphysiol.1979.sp012787.

McCobb, D.P., N.L. Fowler, T. Featherstone, C.J. Lingle, M. Saito, J.E. Krause, and L. Salkoff. 1995. A human calcium-activated potassium channel gene expressed in vascular smooth muscle. *American Journal of Physiology-Heart and Circulatory Physiology*. 269:H767–H777. doi:10.1152/ajpheart.1995.269.3.H767.

McManus, O.B., A.L. Blatz, and K.L. Magleby. 1987. Sampling, log binning, fitting, and plotting durations of open and shut intervals from single channels and the effects of noise. *Pflügers Archiv - European Journal of Physiology*. 410:530–553. doi:10.1007/BF00586537.

McManus, O.B., G.H. Harris, K.M. Giangiacomo, P. Feigenbaum, J.P. Reuben, M.E. Addy, J.F. Burka, G.J. Kaczorowski, and M.L. Garcia. 1993. An activator of calcium-dependent potassium channels isolated from a medicinal herb. *Biochemistry*. 32:6128–6133. doi:10.1021/bi00075a002.

Meech, R.W. 1972. Intracellular calcium injection causes increased potassium conductance in Aplysia nerve cells. *Comparative Biochemistry and Physiology Part A: Physiology*. 42:493–499. doi:https://doi.org/10.1016/0300-9629(72)90128-4.

Meech, R.W. 1978. Calcium-dependent potassium activation in nervous tissues. *Annual review of biophysics and bioengineering*. 7:1–18.

Meech, R.W., and N.B. Standen. 1975. Potassium activation in Helix aspersa neurones under voltage clamp: a component mediated by calcium influx. *The Journal of physiology*. 249:211–259.

Meera, P., M. Wallner, M. Song, and L. Toro. 1997. Large conductance voltage- and calcium-dependent K+ channel, a distinct member of voltage-dependent ion channels with seven N-terminal transmembrane segments (S0-S6), an extracellular N terminus, and an intracellular (S9-S10) C terminus. *Proceedings of the National Academy of Sciences*. 94:14066–14071. doi:10.1073/pnas.94.25.14066.

Meera, P., M. Wallner, and L. Toro. 2000. A neuronal β subunit (KCNMB4) makes the large conductance, voltage-and Ca2+-activated K+ channel resistant to charybdotoxin and iberiotoxin. *Proceedings of the National Academy of Sciences*. 97:5562–5567.

Meijer, L., A. Borgne, O. Mulner, J.P.J. Chong, J.J. Blow, N. Inagaki, M. Inagaki, J.-G. Delcros, and J.-P. Moulinoux. 1997. Biochemical and Cellular Effects of Roscovitine,

a Potent and Selective Inhibitor of the Cyclin-Dependent Kinases cdc2, cdk2 and cdk5. *European Journal of Biochemistry*. 243:527–536. doi:10.1111/j.1432-1033.1997.t01-2-00527.x.

Meredith, A.L., K.S. Thorneloe, M.E. Werner, M.T. Nelson, and R.W. Aldrich. 2004. Overactive bladder and incontinence in the absence of the BK large conductance Ca2+-activated K+ channel. *Journal of Biological Chemistry*. 279:36746–36752. doi:10.1074/jbc.M405621200.

Meredith, A.L., S.W. Wiler, B.H. Miller, J.S. Takahashi, A.A. Fodor, N.F. Ruby, and R.W. Aldrich. 2006. BK calcium-activated potassium channels regulate circadian behavioral rhythms and pacemaker output. *Nature Neuroscience*. 9:1041–1049. doi:10.1038/nn1740.

Merino, N., H.S. Aronson, D.P. Bojanova, J. Feyhl-Buska, M.L. Wong, S. Zhang, and D. Giovannelli. 2019. Living at the Extremes: Extremophiles and the Limits of Life in a Planetary Context. *Frontiers in Microbiology*. 10:780.

Mienville, J.M., and J.R. Clay. 1996. Effects of intracellular K+ and Rb+ on gating of embryonic rat telencephalon Ca2+-activated K+ channels. *Biophysical Journal*. 70:778–785. doi:10.1016/S0006-3495(96)79617-0.

Miller, C. 1987. Trapping single ions inside single ion channels. *Biophysical journal*. 52:123–126. doi:10.1016/S0006-3495(87)83196-X.

Miller, C. 1995. The charybdotoxin family of K+ channel-blocking peptides. *Neuron*. 15:5–10.

Miller, C., E. Moczydlowski, R. Latorre, and M. Phillips. 1985. Charybdotoxin, a protein inhibitor of single Ca2+-activated K+ channels from mammalian skeletal muscle. *Nature*. 313:316–318. doi:10.1038/313316a0.

Miranda, P., T. Giraldez, and M. Holmgren. 2016. Interactions of divalent cations with calcium binding sites of BK channels reveal independent motions within the gating ring. *Proceedings of the National Academy of Sciences*. 201611415. doi:10.1073/pnas.1611415113.

Miranda, P., M. Holmgren, and T. Giraldez. 2018. Voltage-dependent dynamics of the BK channel cytosolic gating ring are coupled to the membrane-embedded voltage sensor. *eLife*. 7:e40664. doi:10.7554/eLife.40664.

Miyagawa, Y., Y. Matsushita, H. Suzuki, M. Komatsu, T. Yoshimaru, R. Kimura, A. Yanai, J. Honda, A. Tangoku, and M. Sasa. 2018. Frequent downregulation of LRRC26 by epigenetic alterations is involved in the malignant progression of triple-negative breast cancer. *International journal of oncology*. 52:1539–1558.

Molday, L.L., and R.S. Molday. 2014. 1D4: a versatile epitope tag for the purification and characterization of expressed membrane and soluble proteins. *In* Protein Affinity Tags. Springer. 1–15.

Moldenhauer, H.J., K.K. Matychak, and A.L. Meredith. 2019. Comparative Gain-of-Function Effects of the KCNMA1-N999S Mutation on Human BK Channel Properties. *Journal of Neurophysiology*. doi:10.1152/jn.00626.2019.

Moldenhauer, H.J., S.M. Park, and A.L. Meredith. 2020. Characterization of New Human KCNMA1 Loss-of-function Mutations. *Biophysical Journal*. 118:114a.

Montgomery, J.R., and A.L. Meredith. 2012. Genetic activation of BK currents in vivo generates bidirectional effects on neuronal excitability. *Proceedings of the National Academy of Sciences*. 109:18997 LP – 19002. doi:10.1073/pnas.1205573109.

Montgomery, J.R., J.P. Whitt, B.N. Wright, M.H. Lai, and A.L. Meredith. 2013. Mis-expression of the BK K+ channel disrupts suprachiasmatic nucleus circuit rhythmicity and alters clock-controlled behavior. *AJP: Cell Physiology*. 304:C299–C311. doi:10.1152/ajpcell.00302.2012.

Morrow, J.P., S.I. Zakharov, G. Liu, L. Yang, A.J. Sok, and S.O. Marx. 2006. Defining the BK channel domains required for β1-subunit modulation. *Proceedings of the National Academy of Sciences of the United States of America*. 103:5096 LP – 5101. doi:10.1073/pnas.0600907103.

Morton, M.J., K. Hutchinson, P.W. Mathieson, I.R. Witherden, M.A. Saleem, and M. Hunter. 2004. Human Podocytes Possess a Stretch-Sensitive, Ca²⁺-Activated K⁺ Channel: Potential Implications for the Control of Glomerular Filtration. *Journal of the American Society of Nephrology*. 15:2981 LP – 2987. doi:10.1097/01.ASN.0000145046.24268.0D.

Nagel, G., T. Szellas, W. Huhn, S. Kateriya, N. Adeishvili, P. Berthold, D. Ollig, P. Hegemann, and E. Bamberg. 2003. Channelrhodopsin-2, a directly light-gated cation-selective membrane channel. *Proceedings of the National Academy of Sciences*. 100:13940–13945.

Naranjo, D., H. Moldenhauer, M. Pincuntureo, and I. Díaz-Franulic. 2016. Pore size matters for potassium channel conductance. *The Journal of general physiology*. 148:277–291. doi:10.1085/jgp.201611625.

Nardi, A., and S.-P. Olesen. 2008. BK channel modulators: a comprehensive overview. *Current medicinal chemistry*. 15:1126–1146.

Nausch, B., F. Rode, S. Jørgensen, A. Nardi, M.P. G Korsgaard, C. Hougaard, A.D. Bonev, W.D. Brown, T. Dyhring, D. Strøbaek, S.-P. Olesen, P. Christophersen, M. Grunnet, M.T. Nelson, L.C. B Rønn, M.P.G. Korsgaard, C. Hougaard, A.D. Bonev, W.D. Brown, T. Dyhring, D. Strøbæk, S.-P. Olesen, P. Christophersen, M. Grunnet, M.T. Nelson, and L.C.B. Rønn. 2014. NS19504: A Novel BK Channel Activator with Relaxing Effect on Bladder Smooth Muscle Spontaneous Phasic Contractions. *Journal of Pharmacology and Experimental Therapeutics*. 350:520 LP – 530. doi:10.1124/jpet.113.212662.

Neher, E., and H.D. Lux. 1969. Voltage clamp on helix pomatia neuronal membrane; Current measurement over a limited area of the soma surface. *Pflügers Archiv*. 311:272–277. doi:10.1007/BF00590532.

Neher, E., and B. Sakmann. 1976. Single-channel currents recorded from membrane of denervated frog muscle fibres. *Nature*. 260:799–802. doi:10.1038/260799a0.

Nelson, A.M.T., H. Cheng, M. Rubart, L.F. Santana, A.D. Bonev, H.J. Knot, and W.J. Lederer. 1995. Relaxation of Arterial Smooth Muscle by Calcium Sparks Published by : American Association for the Advancement of Science Stable URL : http://www.jstor.org/stable/2888335 JSTOR is a not-for-profit service that helps scholars , researchers , and students d. 270:633–637.

Nelson, M.T., and A.D. Bonev. 2004. The β1 subunit of the Ca2+-sensitive K+ channel protects against hypertension. *The Journal of Clinical Investigation*. 113:955–957.

doi:10.1172/JCI21388.

Neyton, J. 1996. A Ba2+ chelator suppresses long shut events in fully activated high-conductance Ca(2+)-dependent K+ channels. *Biophysical journal*. 71:220–226. doi:10.1016/S0006-3495(96)79218-4.

Neyton, J., and C. Miller. 1988a. Potassium blocks barium permeation through a calcium-activated potassium channel. *The Journal of General Physiology*. 92:549–567. doi:10.1085/jgp.92.5.549.

Neyton, J., and C. Miller. 1988b. Discrete Ba2+ block as a probe of ion occupancy and pore structure in the high-conductance Ca2+ -activated K+ channel. *The Journal of General Physiology*. 92:569–586. doi:10.1085/jgp.92.5.569.

Neyton, J., and M. Pelleschi. 1991. Multi-ion occupancy alters gating in high-conductance, Ca(2+)-activated K+ channels. *The Journal of General Physiology*. 97:641–665. doi:10.1085/jgp.97.4.641.

Nimigean, C.M., J.S. Chappie, and C. Miller. 2003. Electrostatic Tuning of Ion Conductance in Potassium Channels. *Biochemistry*. 42:9263–9268. doi:10.1021/bi0348720.

Nimigean, C.M., and K.L. Magleby. 1999. The beta subunit increases the Ca2+ sensitivity of large conductance Ca2+-activated potassium channels by retaining the gating in the bursting states. *The Journal of general physiology*. 113:425–40. doi:10.1085/jgp.113.3.425.

Nimigean, C.M., and K.L. Magleby. 2000. Functional Coupling of the β1 Subunit to the Large Conductance Ca2+-Activated K+ Channel in the Absence of Ca2+ : Increased Ca2+ Sensitivity from a Ca2+-Independent Mechanism . *The Journal of General Physiology*. 115:719–736. doi:10.1085/jgp.115.6.719.

Niu, X., and K.L. Magleby. 2002. Stepwise contribution of each subunit to the cooperative activation of BK channels by Ca²⁺ *Proceedings of the National Academy of Sciences*. 99:11441 LP – 11446. doi:10.1073/pnas.172254699.

Noceti, F., P. Baldelli, X. Wei, N. Qin, L. Toro, L. Birnbaumer, and E. Stefani. 1996. Effective gating charges per channel in voltage-dependent K+ and Ca2+ channels. *The Journal of general physiology*. 108:143–155.

Noda, M., S. Shimizu, T. Tanabe, T. Takai, T. Kayano, T. Ikeda, H. Takahashi, H. Nakayama, Y. Kanaoka, and N. Minamino. 1984. Primary structure of Electrophorus electricus sodium channel deduced from cDNA sequence. *Nature*. 312:121.

Noda, S., Y. Suzuki, H. Yamamura, W.R. Giles, and Y. Imaizumi. 2019. Roles of LRRC26 as an auxiliary γ1-subunit of large-conductance Ca2+-activated K+ channels in bronchial smooth muscle cells. *American Journal of Physiology-Lung Cellular and Molecular Physiology*. 318:L366–L375. doi:10.1152/ajplung.00331.2019.

Ohwada, T., T. Nonomura, K. Maki, K. Sakamoto, S. Ohya, K. Muraki, and Y. Imaizumi. 2003. Dehydroabietic acid derivatives as a novel scaffold for large-conductance calcium-activated K+ channel openers. *Bioorganic & medicinal chemistry letters*. 13:3971–3974.

Olesen, S.-P., E. Munch, P. Moldt, and J. Drejer. 1994a. Selective activation of Ca2+ -dependent K+ channels by novel benzimidazolone. *European Journal of Pharmacology*. 251:53–59. doi:10.1016/0014-2999(94)90442-1.

Olesen, S.-P., E. Munch, F. Wätjen, and J. Drejer. 1994b. NS 004--an activator of Ca (2+)-dependent K+ channels in cerebellar granule cells. *Neuroreport*. 5:1001–1004.

Olson, T.M., A.E. Alekseev, X.K. Liu, S. Park, L. V. Zingman, M. Bienengraeber, S. Sattiraju, J.D. Ballew, A. Jahangir, and A. Terzic. 2006. Kv1.5 channelopathy due to KCNA5 loss-of-function mutation causes human atrial fibrillation . *Human Molecular Genetics*. 15:2185–2191. doi:10.1093/hmg/ddl143.

Pacheco Otalora, L.F., E.F. Hernandez, M.F. Arshadmansab, S. Francisco, M. Willis, B. Ermolinsky, M. Zarei, H.-G. Knaus, E.R. Garrido-Sanabria, L.F.P. Otalora, E.F. Hernandez, M.F. Arshadmansab, S. Francisco, M. Willis, B. Ermolinsky, M. Zarei, H.-G. Knaus, and E.R. Garrido-Sanabria. 2008. Down-regulation of BK channel expression in the pilocarpine model of temporal lobe epilepsy. *Brain research*. 1200:116–131. doi:https://doi.org/10.1016/j.brainres.2008.01.017.

Pallotta, B.S., K.L. Magleby, and J.N. Barrett. 1981. Single channel recordings of Ca2+-activated K+ currents in rat muscle cell culture. *Nature*. 293:471–474. doi:10.1017/CBO9781107415324.004.

Pantazis, A., A.P. Kohanteb, and R. Olcese. 2010. Relative motion of transmembrane segments S0 and S4 during voltage sensor activation in the human BKCa channel. *The Journal of General Physiology*. 136:645–657. doi:10.1085/jgp.201010503.

Park, C.-S., and C. Miller. 1992. Interaction of charybdotoxin with permeant ions inside the pore of a K+ channel. *Neuron*. 9:307–313. doi:https://doi.org/10.1016/0896-6273(92)90169-E.

Park, E., E.B. Campbell, and R. MacKinnon. 2017. Structure of a CLC chloride ion channel by cryo-electron microscopy. *Nature*. 541:500.

Pattillo, J.M., B. Yazejian, D.A. DiGregorio, J.L. Vergara, A.D. Grinnell, and S.D. Meriney. 2001. Contribution of presynaptic calcium-activated potassium currents to transmitter release regulation in cultured Xenopus nerve–muscle synapses. *Neuroscience*. 102:229–240. doi:https://doi.org/10.1016/S0306-4522(00)00453-X.

Paulsen, C.E., J.-P. Armache, Y. Gao, Y. Cheng, and D. Julius. 2015. Structure of the TRPA1 ion channel suggests regulatory mechanisms. *Nature*. 520:511.

Pietrzykowski, A.Z., R.M. Friesen, G.E. Martin, S.I. Puig, C.L. Nowak, P.M. Wynne, H.T. Siegelmann, and S.N. Treistman. 2008. Posttranscriptional Regulation of BK Channel Splice Variant Stability by miR-9 Underlies Neuroadaptation to Alcohol. *Neuron*. 59:274–287. doi:https://doi.org/10.1016/j.neuron.2008.05.032.

Piskorowski, R.A., and R.W. Aldrich. 2006. Relationship between Pore Occupancy and Gating in BK Potassium Channels. *The Journal of General Physiology*. 127:557 LP – 576. doi:10.1085/jgp.200509482.

Plante, A.E., M.H. Lai, J. Lu, and A.L. Meredith. 2019a. Effects of Single Nucleotide Polymorphisms in Human KCNMA1 on BK Current Properties . *Frontiers in Molecular Neuroscience* . 12:285.

Plante, A.E., H.J. Moldenhauer, J.R. Harvey, and A.L. Meredith. 2019b. Gain-of-Function Effects of KCNMA1-N999S Mutation on Human BK Channel Properties. *Biophysical Society*.

Pluznick, J.L., and S.C. Sansom. 2006. BK channels in the kidney: role in K+ secretion and localization of molecular components. *American Journal of Physiology-Renal*

Physiology. 291:F517–F529. doi:10.1152/ajprenal.00118.2006.

Qian, X., and K.L. Magleby. 2003. β1 subunits facilitate gating of BK channels by acting through the Ca2+, but not the Mg2+, activating mechanisms. *Proceedings of the National Academy of Sciences*. 100:10061 LP – 10066. doi:10.1073/pnas.1731650100.

Qian, X., C.M. Nimigean, X. Niu, B.L. Moss, and K.L. Magleby. 2002. Slo1 Tail Domains, but Not the Ca2+ Bowl, Are Required for the β1 Subunit to Increase the Apparent Ca2+ Sensitivity of BK Channels . *Journal of General Physiology*. 120:829–843. doi:10.1085/jgp.20028692.

Qian, X., X. Niu, and K.L. Magleby. 2006. Intra- and Intersubunit Cooperativity in Activation of BK Channels by Ca²⁺ *The Journal of General Physiology*. 128:389 LP – 404. doi:10.1085/jgp.200609486.

Quintero-Hernández, V., J.M. Jiménez-Vargas, G.B. Gurrola, H.H. Valdivia, and L.D. Possani. 2013. Scorpion venom components that affect ion-channels function. *Toxicon : official journal of the International Society on Toxinology*. 76:328–342. doi:10.1016/j.toxicon.2013.07.012.

Reimann, F., and F.M. Ashcroft. 1999. Inwardly rectifying potassium channels. *Current opinion in cell biology*. 11:503–508.

Rieg, T., V. Vallon, M. Sausbier, U. Sausbier, B. Kaissling, P. Ruth, and H. Osswald. 2007. The role of the BK channel in potassium homeostasis and flow-induced renal potassium excretion. *Kidney International*. 72:566–573. doi:https://doi.org/10.1038/sj.ki.5002369.

Robertson, S.J., A.J. Pennington, A. Mark Evans, and R.J. Martin. 1994. The action of pyrantel as an agonist and an open channel blocker at acetylcholine receptors in isolated Ascaris suum muscle vesicles. *European Journal of Pharmacology*. 271:273–282. doi:10.1016/0014-2999(94)90784-6.

Rockman, M.E., A.G. Vouga, and B.S. Rothberg. 2020. Molecular mechanism of BK channel activation by the smooth muscle relaxant NS11021. *Journal of General Physiology*. 152. doi:10.1085/jgp.201912506.

Rogers, C.J., R.E. Twyman, and R.L. Macdonald. 1994. Benzodiazepine and beta-carboline regulation of single GABAA receptor channels of mouse spinal neurones in culture. *The Journal of Physiology*. 475:69–82. doi:10.1113/jphysiol.1994.sp020050.

Rogers, J.C., Y. Qu, T.N. Tanada, T. Scheuer, and W.A. Catterall. 1996. Molecular determinants of high affinity binding of α-scorpion toxin and sea anemone toxin in the S3-S4 extracellular loop in domain IV of the Na+ channel α subunit. *Journal of Biological Chemistry*. 271:15950–15962.

Rothberg, B.S., R.A. Bello, and K.L. Magleby. 1997. Two-dimensional components and hidden dependencies provide insight into ion channel gating mechanisms. *Biophysical Journal*. 72:2524–2544. doi:https://doi.org/10.1016/S0006-3495(97)78897-0.

Rothberg, B.S., R.A. Bello, L. Song, and K.L. Magleby. 1996. High Ca2+ concentrations induce a low activity mode and reveal Ca2(+)-independent long shut intervals in BK channels from rat muscle. *The Journal of Physiology*. 493:673–689.

doi:10.1113/jphysiol.1996.sp021414.

Rothberg, B.S., and K.L. Magleby. 1999. Gating kinetics of single large-conductance Ca2+-activated K+ channels in high Ca2+ suggest a two-tiered allosteric gating mechanism. *The Journal of general physiology*. 114:93–124. doi:10.1085/jgp.114.1.93.

Rothberg, B.S., and K.L. Magleby. 2000. Voltage and Ca2+ activation of single large-conductance Ca2+-activated K+ channels described by a two-tiered allosteric gating mechanism. *The Journal of general physiology*. 116:75–99. doi:10.1085/jgp.116.1.75.

Roy, S., R.J. Large, A.M. Akande, A. Kshatri, T.I. Webb, C. Domene, G.P. Sergeant, N.G. McHale, K.D. Thornbury, and M.A. Hollywood. 2014. Development of GoSlo-SR-5-69, a potent activator of large conductance Ca2+-activated K+ (BK) channels. *European Journal of Medicinal Chemistry*. 75:426–437. doi:10.1016/j.ejmech.2014.01.035.

Rüttiger, L., M. Sausbier, U. Zimmermann, H. Winter, C. Braig, J. Engel, M. Knirsch, C. Arntz, P. Langer, B. Hirt, M. Müller, I. Köpschall, M. Pfister, S. Münkner, K. Rohbock, I. Pfaff, A. Rüsch, P. Ruth, and M. Knipper. 2004. Deletion of the Ca2+-activated potassium (BK) α-subunit but not the BKβ1-subunit leads to progressive hearing loss. *Proceedings of the National Academy of Sciences of the United States of America*. 101:12922 LP – 12927. doi:10.1073/pnas.0402660101.

Salkoff, L., K. Baker, A. Butler, M. Covarrubias, M.D. Pak, and A. Wei. 1992. An essential 'set' of K+ channels conserved in flies, mice and humans. *Trends in Neurosciences*. 15:161–166. doi:https://doi.org/10.1016/0166-2236(92)90165-5.

Salkoff, L., and T. Jegla. 1997. Chapter 11 Molecular genetics and evolution of voltage-gated ion channels. *In* Membranes and Cell Signaling. E.E. Bittar and N.B.T.-P. of M.B. Bittar, editors. Elsevier. 229–244.

Sanders, L. 2018a. Just 21 People Are Known to Have This Rare Genetic Condition. Can You Help Us Find More? *New York Times Magazine*.

Sanders, L. 2018b. A Diagnosis Update: New Information on a Young Girl's Rare Genetic Condition. *New York Times Magazine*.

Sanders, L. 2019. Looking for a Village.

Sansom, M.S.P., I.H. Shrivastava, J.N. Bright, J. Tate, C.E. Capener, and P.C. Biggin. 2002. Potassium channels: structures, models, simulations. *Biochimica et Biophysica Acta (BBA) - Biomembranes*. 1565:294–307. doi:https://doi.org/10.1016/S0005-2736(02)00576-X.

Santos, R., O. Ursu, A. Gaulton, A.P. Bento, R.S. Donadi, C.G. Bologa, A. Karlsson, B. Al-Lazikani, A. Hersey, T.I. Oprea, and J.P. Overington. 2016. A comprehensive map of molecular drug targets. *Nature Reviews Drug Discovery*. 16:19–34. doi:10.1038/nrd.2016.230.

Sausbier, M., H. Hu, C. Arntz, S. Feil, S. Kamm, H. Adelsberger, U. Sausbier, C.A. Sailer, R. Feil, F. Hofmann, M. Korth, M.J. Shipston, H.-G. Knaus, D.P. Wolfer, C.M. Pedroarena, J.F. Storm, and P. Ruth. 2004. Cerebellar ataxia and Purkinje cell dysfunction caused by Ca2+-activated K+ channel deficiency. *Proceedings of the National Academy of Sciences of the United States of America*. 101:9474 LP – 9478.

doi:10.1073/pnas.0401702101.

Schewe, M., E. Nematian-Ardestani, H. Sun, M. Musinszki, S. Cordeiro, G. Bucci, B.L. De Groot, S.J. Tucker, M. Rapedius, and T. Baukrowitz. 2016. A Non-canonical Voltage-Sensing Mechanism Controls Gating in K2P K+Channels. *Cell*. 164:937–949. doi:10.1016/j.cell.2016.02.002.

Schewe, M., H. Sun, Ü. Mert, A. Mackenzie, A.C.W. Pike, F. Schulz, C. Constantin, K.S. Vowinkel, L.J. Conrad, A.K. Kiper, W. Gonzalez, M. Musinszki, M. Tegtmeier, D.C. Pryde, H. Belabed, M. Nazare, B.L. de Groot, N. Decher, B. Fakler, E.P. Carpenter, S.J. Tucker, and T. Baukrowitz. 2019. A pharmacological master key mechanism that unlocks the selectivity filter gate in K+ channels. *Science*. 363:875 LP – 880. doi:10.1126/science.aav0569.

Schreiber, M., and L. Salkoff. 1997. A novel calcium-sensing domain in the BK channel. *Biophysical journal*. 73:1355–1363. doi:10.1016/S0006-3495(97)78168-2.

Schumacher, M.A., A.F. Rivard, H.P. Bächinger, and J.P. Adelman. 2001. Structure of the gating domain of a Ca2+-activated K+ channel complexed with Ca2+/calmodulin. *Nature*. 410:1120–1124. doi:10.1038/35074145.

Schwartzkroin, P.A., and D.A. Prince. 1980. Effects of tea on hippocampal neurons. *Brain Research*. 185:169–181. doi:https://doi.org/10.1016/0006-8993(80)90680-0.

Semper, A. 1910. Über die Wirkungen der Kamala und ihrer Bestandteile. *Archiv für experimentelle Pathologie und Pharmakologie*. 63:10–37. doi:10.1007/BF01840850.

Shao, L.-R., R. Halvorsrud, L. Borg-Graham, and J.F. Storm. 1999. The role of BK-type Ca2+-dependent K+ channels in spike broadening during repetitive firing in rat hippocampal pyramidal cells. *The Journal of Physiology*. 521:135–146. doi:10.1111/j.1469-7793.1999.00135.x.

Sheehan, J.J., B.L. Benedetti, and A.L. Barth. 2009. Anticonvulsant effects of the BK-channel antagonist paxilline. *Epilepsia*. 50:711–720. doi:10.1111/j.1528-1167.2008.01888.x.

Shelley, C., X. Niu, Y. Geng, and K.L. Magleby. 2010. Coupling and cooperativity in voltage activation of a limited-state BK channel gating in saturating Ca²⁺ *The Journal of General Physiology*. 135:461 LP – 480. doi:10.1085/jgp.200910331.

Shelley, C., J.P. Whitt, J.R. Montgomery, and A.L. Meredith. 2013. Phosphorylation of a constitutive serine inhibits BK channel variants containing the alternate exon “SRKR.” *Journal of General Physiology*. 142:585–598. doi:10.1085/jgp.201311072.

Shen, K.Z.-Z., A. Lagrutta, N.W. Davies, N.B. Standen, J.P. Adelman, and R.A. North. 1994. Tetraethylammonium block of Slowpoke calcium-activated potassium channels expressed in Xenopus oocytes: Evidence for tetrameric channel formation. *Pflügers Archiv*. 426:440–445. doi:10.1007/BF00388308.

Shi, J., and J. Cui. 2001. Intracellular Mg2+ Enhances the Function of Bk-Type Ca2+-Activated K+ Channels . *The Journal of General Physiology*. 118:589–606. doi:10.1085/jgp.118.5.589.

Shipston, M.J. 2001. Alternative splicing of potassium channels: A dynamic switch of cellular excitability. *Trends in Cell Biology*. 11:353–358. doi:10.1016/S0962-8924(01)02068-2.

Shruti, S., J. Urban-Ciecko, J.A. Fitzpatrick, R. Brenner, M.P. Bruchez, and A.L. Barth. 2012. The brain-specific Beta4 subunit downregulates BK channel cell surface expression. *PloS one*. 7.

Sigworth, F.J., and S.M. Sine. 1987. Data transformations for improved display and fitting of single-channel dwell time histograms. *Biophysical Journal*. 52:1047–1054. doi:https://doi.org/10.1016/S0006-3495(87)83298-8.

Silverman, J.A., M.L. Hayes, B.J. Luft, and K.A. Joiner. 1997. Characterization of anti-Toxoplasma activity of SDZ 215-918, a cyclosporin derivative lacking immunosuppressive and peptidyl-prolyl-isomerase-inhibiting activity: possible role of a P glycoprotein in Toxoplasma physiology. *Antimicrobial Agents and Chemotherapy*. 41:1859 LP – 1866. doi:10.1128/AAC.41.9.1859.

Singer, S.J., and G.L. Nicolson. 1972. The Fluid Mosaic Model of the Structure of Cell Membranes. *Science*. 175:720 LP – 731. doi:10.1126/science.175.4023.720.

Sofer, S., and M. Gueron. 1988. Respiratory failure in children following envenomation by the scorpion Leiurus quinquestriatus: Hemodynamic and neurological aspects. *Toxicon*. 26:931–939. doi:https://doi.org/10.1016/0041-0101(88)90258-9.

Stockand, J.D., and S.C. Sansom. 1994. Large Ca(2+)-activated K+ channels responsive to angiotensin II in cultured human mesangial cells. *American Journal of Physiology-Cell Physiology*. 267:C1080–C1086. doi:10.1152/ajpcell.1994.267.4.C1080.

Sun, J., and R. MacKinnon. 2017. Cryo-EM structure of a KCNQ1/CaM complex reveals insights into congenital long QT syndrome. *Cell*. 169:1042–1050. doi:10.1016/j.cell.2017.05.019.

Sweet, T.-B., and D.H. Cox. 2009. Measuring the influence of the BKCa β1 subunit on Ca2+ binding to the BKCa channel. *Journal of General Physiology*. 133:139–150.

Swenson, R.P., and C.M. Armstrong. 1981. K+ channels close more slowly in the presence of external K+ and Rb+. *Nature*. 291:427–429. doi:10.1038/291427a0.

Tang, Q.Y., Z. Zhang, X.Y. Meng, M. Cui, and D.E. Logothetis. 2014. Structural determinants of phosphatidylinositol 4,5-bisphosphate (PIP2) regulation of BK channel activity through the RCK1 Ca2+ coordination site. *Journal of Biological Chemistry*. 289:18860–18872. doi:10.1074/jbc.M113.538033.

Tao, X., R.K. Hite, R. Mackinnon, and K. Ca. 2016. Cryo-EM structure of the open high-. *Nature*. 1–14. doi:10.1038/nature20608.

Tasaki, I., and S. Hagiwara. 1957. Capacity of Muscle Fiber Membrane. *American Journal of Physiology-Legacy Content*. 188:423–429. doi:10.1152/ajplegacy.1957.188.3.423.

Tempel, B.L., D.M. Papazian, T.L. Schwarz, Y.N. Jan, and L.Y. Jan. 1987. Sequence of a probable potassium channel component encoded at Shaker locus of Drosophila. *Science*. 237:770–775.

Thompson, J., and T. Begenisich. 2012. Selectivity filter gating in large-conductance Ca2+-activated K+ channels. *The Journal of General Physiology*. 139:235 LP – 244. doi:10.1085/jgp.201110748.

Tian, L., R.R. Duncan, M.S.L. Hammond, L.S. Coghill, H. Wen, R. Rusinova, A.G. Clark, I.B. Levitan, and M.J. Shipston. 2001. Alternative Splicing Switches Potassium

Channel Sensitivity to Protein Phosphorylation. *Journal of Biological Chemistry* . 276:7717–7720. doi:10.1074/jbc.C000741200.

Tian, Y., S.H. Heinemann, and T. Hoshi. 2019. Large-conductance Ca²⁺- and voltage-gated K⁺ channels form and break interactions with membrane lipids during each gating cycle. *Proceedings of the National Academy of Sciences*. 116:8591 LP – 8596. doi:10.1073/pnas.1901381116.

Tseng-Crank, J., C.D. Foster, J.D. Krause, R. Mertz, N. Godinot, T.J. DiChiara, and P.H. Reinhart. 1994. Cloning, expression, and distribution of functionally distinct Ca(2+)-activated K+ channel isoforms from human brain. *Neuron*. 13:1315–1330. doi:10.1016/0896-6273(94)90418-9.

Vaithianathan, T., A. Bukiya, J. Liu, P. Liu, M. Asuncion-Chin, Z. Fan, and A. Dopico. 2008. Direct Regulation of BK Channels by Phosphatidylinositol 4,5-Bisphosphate as a Novel Signaling Pathway . *The Journal of General Physiology*. 132:13–28. doi:10.1085/jgp.200709913.

Valdivia, H.H., J.S. Smith, B.M. Martin, R. Coronado, and L.D. Possani. 1988. Charybdotoxin and noxiustoxin, two homologous peptide inhibitors of the K+(Ca2+) channel. *FEBS Letters*. 226:280–284. doi:https://doi.org/10.1016/0014-5793(88)81439-X.

Vandenberg, C.A., and R. Horn. 1984. Inactivation viewed through single sodium channels. *Journal of General Physiology*. 84:535–564. doi:10.1085/jgp.84.4.535.

Vang, A., J. Mazer, B. Casserly, and G. Choudhary. 2010. Activation of endothelial BKCa channels causes pulmonary vasodilation. *Vascular Pharmacology*. 53:122–129. doi:10.1016/j.vph.2010.05.001.

Vergara, C., R. Latorre, N. V Marrion, and J.P. Adelman. 1998. Calcium-activated potassium channels. *Current Opinion in Neurobiology*. 8:321–329. doi:https://doi.org/10.1016/S0959-4388(98)80056-1.

Vogalis, F., T. Vincent, I. Qureshi, F. Schmalz, M.W. Ward, K.M. Sanders, and B. Horowitz. 1996. Cloning and expression of the large-conductance Ca(2+)-activated K+ channel from colonic smooth muscle. *American Journal of Physiology-Gastrointestinal and Liver Physiology*. 271:G629–G639. doi:10.1152/ajpgi.1996.271.4.G629.

Vouga, A.G., M.E. Rockman, M.A. Jacobson, and B.S. Rothberg. 2019. Mechanism of BK Channel Inhibition by the Opioid Agonist Loperamide. *Biophysical Journal*. 116:543a.

Wang, B., B.S. Rothberg, and R. Brenner. 2006. Mechanism of β4 Subunit Modulation of BK Channels. *The Journal of General Physiology*. 127:449 LP – 465. doi:10.1085/jgp.200509436.

Wang, B., B.S. Rothberg, and R. Brenner. 2009. Mechanism of increased BK channel activation from a channel mutation that causes epilepsy. *The Journal of general physiology*. 133:283–294. doi:10.1085/jgp.200810141.

Wang, W., and R. MacKinnon. 2017. Cryo-EM structure of the open human ether-à-go-go-related K+ channel hERG. *Cell*. 169:422-430.e10. doi:10.1016/j.cell.2017.03.048.

Wang, X., S.R.A. Burke, R.J. Talmadge, A.A. Voss, and M.M. Rich. 2020. Depressed neuromuscular transmission causes weakness in mice lacking BK potassium

channels. *Journal of General Physiology*. 152. doi:10.1085/jgp.201912526.

Webb, T.I., A.S. Kshatri, R.J. Large, A.M. Akande, S. Roy, G.P. Sergeant, N.G. McHale, K.D. Thornbury, and M.A. Hollywood. 2015. Molecular mechanisms underlying the effect of the novel BK channel opener GoSlo: Involvement of the S4/S5 linker and the S6 segment. *Proceedings of the National Academy of Sciences*. 112:2064 LP – 2069. doi:10.1073/pnas.1400555112.

Weiss, D.S., and K.L. Magleby. 1992. Voltage-dependent gating mechanism for single fast chloride channels from rat skeletal muscle. *The Journal of Physiology*. 453:279–306. doi:10.1113/jphysiol.1992.sp019229.

Whitmire, L.E., L. Ling, V. Bugay, C.M. Carver, S. Timilsina, H.-H. Chuang, D.B. Jaffe, M.S. Shapiro, J.E. Cavazos, and R. Brenner. 2017. Downregulation of KCNMB4 expression and changes in BK channel subtype in hippocampal granule neurons following seizure activity. *PloS one*. 12.

Wickman, K.D., and D.E. Clapham. 1995. G-protein regulation of ion channels. *Current opinion in neurobiology*. 5:278–285.

Wouters, O.J., M. McKee, and J. Luyten. 2020. Estimated Research and Development Investment Needed to Bring a New Medicine to Market, 2009-2018. *JAMA - Journal of the American Medical Association*. 323:844–853. doi:10.1001/jama.2020.1166.

Wu, S.-N., H.-F. Li, C.-R. Jan, and A.-Y. Shen. 1999. Inhibition of Ca2+-activated K+ current by clotrimazole in rat anterior pituitary GH3 cells. *Neuropharmacology*. 38:979–989. doi:https://doi.org/10.1016/S0028-3908(99)00027-1.

Wu, Y., Y. Yang, S. Ye, and Y. Jiang. 2010. Structure of the gating ring from the human large-conductance Ca(2+)-gated K(+) channel. *Nature*. 466:393–7. doi:10.1038/nature09252.

Xia, X.-M., X. Zeng, and C.J. Lingle. 2002. Multiple regulatory sites in large-conductance calcium-activated potassium channels. *Nature*. 418:880–884. doi:10.1038/nature00956.

Xie, J., and D.P. McCobb. 1998. Control of alternative splicing of potassium channels by stress hormones [In Process Citation]. *Science*. 280:443–446. doi:10.1126/science.280.5362.443.

Yan, J., and R.W. Aldrich. 2010. LRRC26 auxiliary protein allows BK channel activation at resting voltage without calcium. *Nature*. 466:513–516. doi:10.1038/nature09162.

Yan, J., and R.W. Aldrich. 2012. BK potassium channel modulation by leucine-rich repeat-containing proteins. *Proceedings of the National Academy of Sciences*. 109:7917 LP – 7922. doi:10.1073/pnas.1205435109.

Yan, Z., P. Chi, J. A. Bibb, T. A. Ryan, and P. Greengard. 2002. Roscovitine: a novel regulator of P/Q-type calcium channels and transmitter release in central neurons. *The Journal of Physiology*. 540:761–770. doi:10.1113/jphysiol.2001.013376.

Yang, C., V. Gonzalez-Perez, T. Mukaibo, J.E. Melvin, X.-M. Xia, and C.J. Lingle. 2017. Knockout of the LRRC26 subunit reveals a primary role of LRRC26-containing BK channels in secretory epithelial cells. *Proceedings of the National Academy of Sciences of the United States of America*. 114:E3739–E3747. doi:10.1073/pnas.1703081114.

Yang, H., L. Hu, J. Shi, K. Delaloye, F.T. Horrigan, and J. Cui. 2007. Mg²⁺ mediates interaction between the voltage sensor and cytosolic domain to activate BK channels. *Proceedings of the National Academy of Sciences*. 104:18270 LP – 18275. doi:10.1073/pnas.0705873104.

Yang, H., J. Shi, G. Zhang, J. Yang, K. Delaloye, and J. Cui. 2008. Activation of Slo1 BK channels by Mg2+ coordinated between the voltage sensor and RCK1 domains. *Nature Structural & Molecular Biology*. 15:1152–1159. doi:10.1038/nsmb.1507.

Yang, J., G. Krishnamoorthy, A. Saxena, G. Zhang, J. Shi, H. Yang, K. Delaloye, D. Sept, and J. Cui. 2010. An Epilepsy/Dyskinesia-Associated Mutation Enhances BK Channel Activation by Potentiating Ca2+ Sensing. *Neuron*. 66:871–883. doi:10.1016/j.neuron.2010.05.009.

Yao, J., B. Liu, and F. Qin. 2011. Modular thermal sensors in temperature-gated transient receptor potential (TRP) channels. *Proceedings of the National Academy of Sciences*. 108:11109–11114.

Ye, S., Y. Li, L. Chen, and Y. Jiang. 2006. Crystal Structures of a Ligand-free MthK Gating Ring: Insights into the Ligand Gating Mechanism of K+ Channels. *Cell*. 126:1161–1173. doi:10.1016/j.cell.2006.08.029.

Yellen, G. 1984a. Ionic permeation and blockade in Ca2+-activated K+ channels of bovine chromaffin cells. *Journal of General Physiology*. 84:157–186. doi:10.1085/jgp.84.2.157.

Yellen, G. 1984b. Relief of Na+ block of Ca2+-activated K+ channels by external cations. *Journal of General Physiology*. 84:187–199. doi:10.1085/jgp.84.2.187.

Yellen, G. 1987. Permeation in potassium channels: implications for channel structure. *Annual review of biophysics and biophysical chemistry*. 16:227–246.

Yellen, G., M.E. Jurman, T. Abramson, and R. MacKinnon. 1991. Mutations affecting internal TEA blockade identify the probable pore-forming region of a K+ channel. *Science*. 251:939–942.

Yoshihara, K., A. Tajima, S. Adachi, J. Quan, M. Sekine, H. Kase, T. Yahata, I. Inoue, and K. Tanaka. 2011. Germline copy number variations in BRCA1-associated ovarian cancer patients. *Genes, Chromosomes and Cancer*. 50:167–177. doi:10.1002/gcc.20841.

Zack, M.M., and R. Kobau. 2017. National and state estimates of the numbers of adults and children with active epilepsy—United States, 2015. *MMWR. Morbidity and mortality weekly report*. 66:821.

Zagotta, W.N., T. Hoshi, and R.W. Aldrich. 1994. Shaker potassium channel gating. III: Evaluation of kinetic models for activation. *The Journal of General Physiology*. 103:321–362. doi:10.1085/jgp.103.2.321.

Zakharov, S.I., J.P. Morrow, G. Liu, L. Yang, and S.O. Marx. 2005. Activation of the BK (SLO1) Potassium Channel by Mallotoxin. *Journal of Biological Chemistry* . 280:30882–30887. doi:10.1074/jbc.M505302200.

Zeng, X.-H., J.-P. Ding, X.-M. Xia, and C.J. Lingle. 2001. Gating Properties Conferred on Bk Channels by the β3b Auxiliary Subunit in the Absence of Its Nh2- and Cooh Termini . *The Journal of General Physiology*. 117:607–628. doi:10.1085/jgp.117.6.607.

Zeng, X.-H., X.-M. Xia, and C.J. Lingle. 2005. Divalent cation sensitivity of BK channel activation supports the existence of three distinct binding sites. *The Journal of general physiology*. 125:273–286. doi:10.1085/jgp.200409239.

Zhang, G., Y. Geng, Y. Jin, J. Shi, K. McFarland, K.L. Magleby, L. Salkoff, and J. Cui. 2017. Deletion of cytosolic gating ring decreases gate and voltage sensor coupling in BK channels. *Journal of General Physiology*. 149:373–387. doi:10.1085/jgp.201611646.

Zheng, J., and M.C. Trudeau. 2015. Handbook of ion channels. CRC Press.

Zhou, Y., and C.J. Lingle. 2014. Paxilline inhibits BK channels by an almost exclusively closed-channel block mechanism. *The Journal of General Physiology*. 144:415 LP – 440. doi:10.1085/jgp.201411259.

Zhou, Y., Q.-Y. Tang, X.-M. Xia, and C.J. Lingle. 2010. Glycine311, a determinant of paxilline block in BK channels: a novel bend in the BK S6 helix. *The Journal of General Physiology*. 135:481–494. doi:10.1085/jgp.201010403.

Zhou, Y., X.-M. Xia, and C.J. Lingle. 2018. BK channel inhibition by strong extracellular acidification. *eLife*. 7:e38060. doi:10.7554/eLife.38060.

Zhou, Y., X.-M. Xia, and C.J. Lingle. 2020. The functionally relevant site for paxilline inhibition of BK channels. *Proceedings of the National Academy of Sciences*. 117:1021 LP – 1026. doi:10.1073/pnas.1912623117.

Zhou, Y.Y., X.-H. Zeng, and C.J. Lingle. 2012. Barium ions selectively activate BK channels via the Ca2+-bowl site. *Proceedings of the National Academy of Sciences*. 109:11413–11418. doi:10.1073/pnas.1204444109.